

Developmental Programming of Blood Pressure and Renal Function Through the Life Course

Valérie A. Luyckx | Karen M. Moritz | John F. Bertram

CHAPTER OUTLINE

DEVELOPMENTAL PROGRAMMING, 668

DEVELOPMENTAL PROGRAMMING IN THE KIDNEY, 668

EVIDENCE FOR PROGRAMMING IN THE KIDNEY, 671

PROGRAMMING OF RENAL FUNCTION AND DISEASE, 681

PROPOSED MECHANISMS OF DEVELOPMENTAL PROGRAMMING IN THE KIDNEY, 692

MATERNAL HEALTH AND INTERGENERATIONAL EFFECTS OF PROGRAMMING, 703

IMPACT OF NEPHRON ENDOWMENT IN TRANSPLANTATION, 704

GLOBAL HEALTH RELEVANCE OF RENAL PROGRAMMING, 707

CONCLUSION, 707

The large variation in individual susceptibility to kidney disease and other chronic diseases is not easily explained.¹ Genetic factors are important determinants of development and function of major organ systems as well as of susceptibility to disease. Rare genetic and congenital abnormalities leading to abnormal kidney development are associated with the occurrence of subsequent renal dysfunction, often manifest very early in life.^{2,3} Most renal disease in the general population, however, is not ascribable to genetic mutations, with the most common etiologic associations with end-stage kidney disease (ESKD) worldwide being the polygenic disorders of diabetes and hypertension. Of note here is the association between mutations in the apolipoprotein-1 (*APOLI*) gene in people of African descent with increased predisposition to the development of HIV-associated nephropathy and focal and segmental glomerulosclerosis (FSGS). However, searches for other specific gene polymorphisms or mutations have not implicated specific genes but instead point to a likely complex interplay between polygenic predisposition and environmental factors in the development of hypertension, diabetes, and renal disease.⁴⁻⁹ Hypertension and renal disease prevalence vary between populations from different ethnic backgrounds, with very high rates observed among Aboriginal Australians, Native Americans, and people of African descent.¹⁰⁻¹² It is well established that lifestyle and socioeconomic factors

pose significant risk for the development and persistence of hypertension and diabetes in the general population, with obesity becoming an increasing concern, especially in the developing world.^{7,13} However, more and more evidence is pointing also to the far-reaching effects of the intrauterine environment and early postnatal growth on organ development, organ function, and subsequent susceptibility to adult disease.¹⁴⁻¹⁶ Stresses experienced during fetal life (for which low birth weight, being small for gestational age, preterm birth, or high birth weight may be surrogate markers), such as maternal malnutrition, ill health, preeclampsia, or gestational diabetes, or those experienced in early childhood, such as poor early nutrition, infections, and environmental exposures, may “program” long-term organ function and may be the first in a succession of challenges or “hits” that ultimately manifest in overt disease. This chapter outlines the effects of fetal and early-life programming on renal development (particularly nephrogenesis), nephron endowment, and the risks of hypertension and kidney disease throughout the life course. Major congenital renal anomalies are discussed elsewhere in this book (see Chapter 72). In addition, it must be borne in mind that low birth weight and preterm birth also predict later-life diabetes, cardiovascular disease, metabolic syndrome, and preeclampsia; therefore, renal function may be additionally impacted through developmental programming

of these disorders and in turn may impact outcomes of these diseases, the discussion of which is beyond the scope of the current chapter.^{16–20}

DEVELOPMENTAL PROGRAMMING

The process through which an environmental insult experienced early in life can predispose to adult disease is known as “developmental programming,” which refers to the observation that an environmental stimulus experienced during a critical period of development in utero or early postbirth can induce long-term structural and functional effects in the organism.^{15,21} This phenomenon, often termed “developmental origins of health and disease,” can have far-reaching implications in that the effects can be perpetuated across generations.^{22,23} The association between adverse intrauterine events and subsequent cardiovascular disease has long been recognized.^{15,21,24,25} In early studies, adults of low birth weight were found to have higher cardiovascular morbidity and mortality than those of normal birth weight.²⁶ Subsequently, evidence from diverse populations has confirmed these findings and expanded them to include other conditions, such as hypertension, impaired glucose tolerance, type 2 diabetes, obesity, preeclampsia, and chronic kidney disease (CKD).^{21,27–33} Of these, the associations between low birth weight and preterm birth and subsequent hypertension have been the most studied.^{34–37} Until recently, attention has largely focused on preterm birth and low birth weight as markers for developmental programming of hypertension and renal disease, but being small for gestational age or having a high birth weight, often as a result of a diabetic pregnancy or maternal obesity, are also emerging as risk factors.^{38–41} Currently, birth weight and preterm birth are the best available surrogates for an adverse intrauterine environment, but some intrauterine stresses may not manifest as such and therefore may not be recognized. Ongoing work is required to develop more sensitive measures of developmental stress. Table 21.1 outlines the definitions of birth weight and gestational age categories, which are referred to throughout this chapter. Globally, the respective incidences of low birth weight and preterm birth are around 15%–20% and 11%.^{42,43} Importantly, as shown in Fig. 21.1, a large proportion of babies born small for gestational age have a birth weight above 2.5 kg, highlighting the need to identify all small-for-gestational-age infants who are also at risk of programming effects.⁴⁴ The global incidence of high birth weight is rising, ranging from 5%–20% in high-income and 0.5%–15% in lower-income countries.⁴⁰ A significant number of infants born yearly therefore likely undergo developmental programming and are at risk for chronic disease later in life.

DEVELOPMENTAL PROGRAMMING IN THE KIDNEY

The kidney is the organ central to the development of hypertension. The relationship between renal sodium handling, intravascular volume homeostasis, and hypertension is well accepted.^{45,46} That factors intrinsic to the kidney itself affect blood pressure has been demonstrated clinically in kidney transplantation in which the blood pressure in the

Table 21.1 Definitions of Birth Weight and Prematurity Categories

Category	Definition
Birth Weight Categories	
Normal birth weight	>2500 g and <4000 g (usually)
Large for gestational age	>2 standard deviations above the mean birth weight for gestational age
Low birth weight	<2500 g
Very low birth weight	<1500 g
Appropriate for gestational age	Within ± 2 standard deviations of the mean birth weight for gestational age
Small for gestational age	>2 standard deviations below the mean birth weight for gestational age
Intrauterine growth restriction	Evidence of fetal malnutrition and growth restriction at any time during gestation
Gestational Categories	
Extremely preterm	<28 weeks of gestation
Very preterm	<32 and >28 weeks of gestation
Moderately preterm	<34 and >32 weeks of gestation
Late preterm	<37 and >34 weeks of gestation
Full term	>37 weeks of gestation

Taken from Abitbol CL, Rodriguez MM. The long-term renal and cardiovascular consequences of prematurity. *Nat Rev Nephrol.* 2012;8:265–274.

recipient after transplantation has been shown to be related to the blood pressure or hypertension risk factors of the donor: that is, hypertension “follows” the kidney.⁴⁷ In 1988, Brenner and colleagues⁴⁸ proposed that congenital (programmed) low nephron number may explain why some individuals are susceptible to hypertension and renal injury, whereas others may seem relatively resistant under similar circumstances (e.g., sodium excess or diabetes mellitus). Low nephron number and low whole-kidney glomerular surface area would result in reduced sodium excretory capacity, enhancing susceptibility to hypertension, and a relatively reduced renal reserve capacity, limiting compensation for renal injury. This hypothesis was attractive in that an association between low nephron number and low birth weight, for example, could explain differences in hypertension and renal disease prevalence observed in populations of different ethnicity, among whom those who tend to have lower birth weights often have a greater prevalence of hypertension and renal disease.^{49–52}

PLAUSIBILITY OF THE NEPHRON NUMBER HYPOTHESIS

An obstacle to investigation of the nephron number hypothesis has been the difficulty of accurately counting or estimating the total number of nephrons in a kidney.⁵³ Review of early studies shows that humans were believed to have an average of approximately 1 million nephrons per kidney.⁵⁴ This was the nephron number published in textbooks for decades, and implied that there is little variability in human nephron number and therefore likely little relationship with adult disease risk. Such studies, however, were performed using

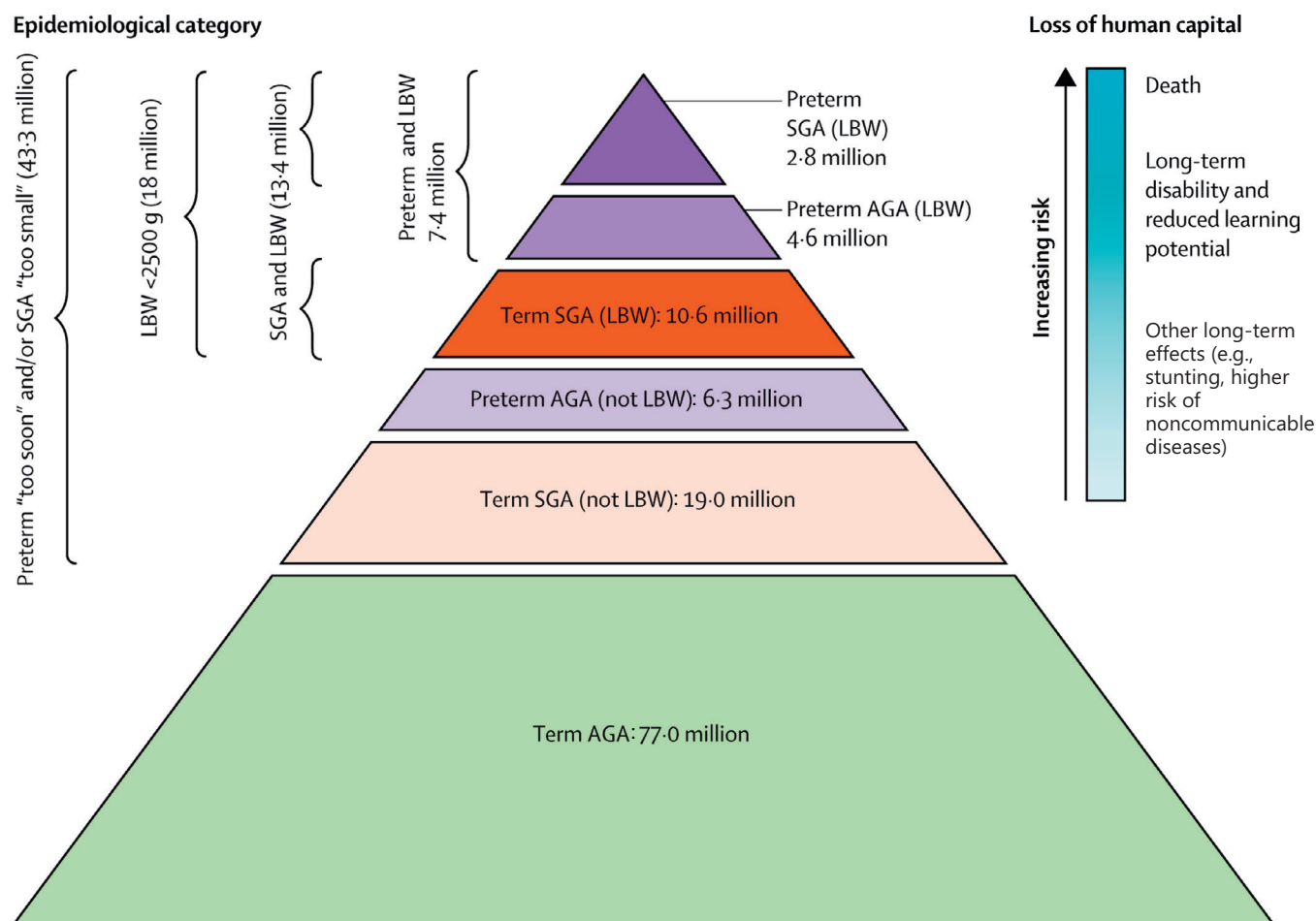


Fig. 21.1 Public health implications of the burden of preterm and SGA births for 120 million births in countries of low and middle income in 2010. AGA, Appropriate for gestational age; LBW, low birth weight; SGA, small for gestational age. (From Lee AC, Katz J, Blencowe H, et al. National and regional estimates of term and preterm babies born small for gestational age in 138 low-income and middle-income countries in 2010. *Lancet Glob Health*.)

techniques such as acid maceration or traditional model-based stereologic approaches, which are prone to bias because of required assumptions, extrapolations, and operator sensitivity.^{53–55} Moreover, many of the early studies analyzed only a handful of kidneys, and therefore, population variability in nephron number was not observed.

Over the past 25 years, the design-based (often termed “unbiased”) dissector/fractionator method has emerged as the gold standard method for estimating total glomerular, and thereby nephron, number because it generates accurate (no bias) and precise (low-variance) estimates.^{53–55} With this approach, the dissector principle⁵⁶ is used to sample glomeruli in representative histologic samples of the three-dimensional renal cortex with equal probability (regardless of their size, shape, and location), and then these sampled glomeruli are counted. Glomeruli are counted in a “known” fraction of kidney tissue, allowing simple algebraic estimation of the total number of glomeruli in the whole kidney.

As the dissector/fractionator method requires sampling from a whole kidney, all studies to date with this technique have been performed on autopsy samples. For 37 normal Danish adults, the average glomerular (nephron) number was reported to be 617,000 per kidney (range: 331,000–1,424,000).⁵⁴ A positive correlation was noted

between glomerular number and kidney weight, but this has not been the case in all studies.⁵⁷ The large variation in nephron number in Danes has subsequently been reported in several other populations (Table 21.2), with the largest range to date being a 13-fold variation in the largest sample yet investigated (African Americans; 159 subjects; 210,332 to 2,702,079 per kidney).⁵⁸ This previously unappreciated variability in total nephron number in subjects without kidney disease, apart from age-associated changes, may influence susceptibility to hypertension and kidney disease.^{54,58,59}

In general, numbers of viable glomeruli are reduced in kidneys from older subjects, owing to age-related glomerulosclerosis and obsolescence.^{54,60} Recently, Denic et al.⁶¹ estimated glomerular number in 1638 living donor kidneys using a method combining computed tomographic (CT) estimation of renal cortical volume (prior to donation) with stereologic estimation of glomerular numerical density (number per unit volume of cortex) obtained on biopsies at the time of donation. Multiplication of these two estimates provided an estimate of total glomerular number. Separate estimates of nonsclerotic and sclerotic glomeruli were obtained. A striking decrease in numbers of nonsclerotic glomeruli was observed between 18–29-year-olds and donors aged 70–75 years. Moreover, the total number of sclerotic and nonsclerotic

Table 21.2 Variability of Nephron Number Per Kidney in Humans

Population	Sample Size	Mean	Range	Fold ^a	Reference
Danish	37	617,000	331,000–1,424,000	4.3	54
French	28	1,107,000	655,000–1,554,000	2.4	534
German (all)	20	1,074,414	531,140–1,959,914	3.7	59
German hypertensive	10	702,379	531,140–954,893	1.8	
German normotensive	10	1,429,200	884,458–1,959,914	2.2	
African-American	105	884,938	210,332–2,026,541	9.6	292
White American	84	843,106	227,327–1,660,232	7.3	
Australian non-Aborigines	24	861,541	380,517–1,493,665	3.9	220
Australian Aborigines	19	713,209	364,161–1,129,223	3.1	
Senegalese	47	992,353	536,171–1,764,421	3.3	291, 535
African and white Americans, Australian Aborigines and non-Aborigines and Senegalese	420	901,902	210,332–2,702,079	12.8	536
Japanese males (all)	18	544,819	306,092–960,756	3.1	221
Japanese Hypertensive (males)	9	423,498	306,092–550,222	1.8	
Japanese Normotensive (males)	9	666,140	419,282–960,756	2.3	
Kidney donors	1638	873,696			61

^aFold change from lowest to highest nephron number.

Adapted from Puelles VG, Hoy WE, Hughson MD, et al. Glomerular number and size variability and risk for kidney disease. *Curr Opin Nephrol Hypertens*. 2011;20:7–15.

glomeruli was much lower in the older donors, suggesting that many glomeruli had disappeared without trace following global glomerulosclerosis, and/or older donors were born with fewer nephrons.

In support of the nephron number hypothesis, it is known that persons born with severe nephron deficits, for example, unilateral renal agenesis, bilateral renal hypoplasia, and oligomeganephronia, develop progressive proteinuria, glomerulosclerosis, and renal dysfunction with time.^{3,62} Analogously, therefore, people born with nephron numbers at or below the median level may be more susceptible to superimposed postnatal factors that act as subsequent “hits”; thus, a significant proportion of the population may be at risk for the development of hypertension and renal disease, given that some 30% of the world’s adult population is hypertensive.^{7,14}

The counterargument to the nephron number hypothesis is that in experimental animals and in humans, removal of one kidney (presumed reduction in nephron number by 50%) under varying circumstances may be associated with higher blood pressures or low-grade proteinuria but does not always lead to hypertension and renal disease.^{55,63,64} Of interest, however, uninephrectomy on postnatal day 1 in rats or fetal uninephrectomy in sheep, that is, loss of nephrons at a time when nephrogenesis is not yet completed, does lead to adult hypertension prior to any evidence of renal injury.^{65–67} These data support the hypothesis that congenitally acquired low nephron number may be associated with different compensatory mechanisms in the developing and growing kidney, or with a reduced compensatory capacity compared with later nephron loss, resulting in increased risk of hypertension. Consistent with this, kidneys from rats that underwent unilateral nephrectomy at 3 days of age showed a similar total glomerular number but a significantly reduced number of mature glomeruli in the remaining kidney compared with those that underwent nephrectomy at 120 days of age.⁶⁸ Furthermore, the mean glomerular volume in the remaining kidney of rats following unilateral nephrectomy

in the neonatal period increased by 59% as compared with 20% in adult rats, likely indicating a greater burden of compensatory hypertrophy and hyperfiltration in response to neonatal nephrectomy. Similarly, rats spontaneously born with a solitary kidney are highly susceptible to the induction of hypertension (via deoxycorticosterone acetate (DOCA) salt), as well as proteinuria and reduced glomerular filtration rate (GFR) compared with rats that underwent a uninephrectomy 8 weeks after birth.⁶⁹ Nephrons in mice born with low nephron endowment were found to be less capable of coping with a stress such as unilateral ureteral obstruction than mice with the normal nephron complement.⁷⁰

In potential contrast, however, a study of 97 subjects with a radiologically normal single kidney, aged 2.5–25 years, found that renal function declined faster over time in those with acquired single kidneys (surgical removal of the other kidney) versus congenital single kidneys, although blood pressures and proteinuria were not different.⁷¹ These findings may be confounded by indication for nephrectomy, however, as approximately 25% of the nephrectomies were performed for obstruction, which may impact nephron development in the contralateral kidney as will be discussed later.^{72,73} In addition, unilateral in utero nephrectomy in sheep was associated with significant renal hypertrophy and a 45% increase in nephron number in the remaining kidney, indicating that a congenital solitary kidney may have a higher than normal nephron endowment and therefore be relatively protected compared with an acquired single kidney.^{74,75} Thus, timing of nephron loss is likely a crucial factor in determining the compensatory capacity of remaining nephrons.

ESTIMATING NEPHRON NUMBER

Until recently, most estimates of glomerular number were obtained using acid maceration of kidneys obtained at autopsy or model-based or design-based stereological analysis of histological samples obtained at autopsy. While useful, reliance on autopsy samples precludes studies in living animals and

Table 21.3 Glomerular Characteristics by Birth Weight in Humans

Birth Weight Mean (Range), kg	N	Number of Glomeruli ^a	Mean Glomerular Tuft Volume ($\mu\text{m}^3 \times 10^6$)	Total Glomerular Tuft Volume (cm^3)
2.65 (1.81–3.12)	29	770,860 (658,757–882,963)	9.2	6.7
3.27 (3.18–3.38)	28	965,729 (885,714–1,075,744)	7.2	6.8
3.93 (3.41–4.94)	30	1,005,356 (900,094–1,110,599)	6.9	6.6

^aAdjusted for age, gender, race, body surface area.

From Hoy WE, Hughson MD, Bertram JF, et al. Nephron number, hypertension, renal disease, and renal failure. *J Am Soc Nephrol.* 2005;16:2557–2264.

patients, as well as longitudinal studies. In recent years, we have seen the development of several new approaches for imaging, counting, and sizing glomeruli, some of which do not require autopsy tissue. For example, the methods of Denic et al.^{61,76} and of Fulladosa et al.,⁷⁷ which use CT or magnetic resonance imaging (MRI) estimation of cortical volume and estimation of glomerular numerical density in biopsies, do not require autopsy tissue. The number of functional (nonsclerotic) glomeruli has also been estimated by dividing the whole kidney ultrafiltration coefficient (K_f ; calculated from renal plasma flow from para-amino-hippurate clearance) by single-nephron K_f estimated from electron microscopic measurements of the glomerular filtration barrier using kidney biopsies.^{78,79}

An MRI technique using cationic ferritin to label glomeruli shows promise,^{80,81} but to date, studies have been ex vivo. Cationic ferritin is injected intravenously into animals or into ex vivo human kidneys via the renal artery and binds to anionic charges on the glomerular basement membrane. With this approach, all glomeruli in whole rat,^{82,83} mouse,⁸⁴ and human⁸⁰ kidneys are imaged, counted, and sized, providing, for the first time, the glomerular volume distribution for whole kidneys. New tissue-clearing approaches also facilitate imaging (using light sheet microscopy), counting, and sizing of all glomeruli in ex vivo rodent kidneys,⁸⁵ but whether modifications to this approach can be safely used in vivo appears unlikely at this time. With these recent advances in glomerular imaging and counting, we can hope that in vivo glomerular imaging and quantitation will be possible in the not-too-distant future.

NEPHRON NUMBER AND GLOMERULAR VOLUME

Human glomeruli have been reported to increase in size up to sevenfold from infancy to adulthood.^{86–88} This can be considered normal physiological growth. Glomerular size also increases in adulthood in people without overt renal disease and is associated with increasing age, increasing body size, and lower birth weight.⁸⁹ Mean glomerular volume has also been consistently noted to vary inversely with total glomerular number, although the correlation appears stronger among Caucasians and Australian Aborigines than in people of African origin.^{59,90–92} This relationship suggests that larger glomeruli may reflect compensatory hyperfiltration and hypertrophy in subjects with fewer nephrons and may therefore be a surrogate marker for a reduced nephron number.^{60,91} In fact, Hoy and coworkers⁹⁰ found that, although mean glomerular volume was increased in subjects with

reduced nephron number, total renal glomerular tuft volume (a surrogate for total renal filtration surface area) was not different among groups with different nephron numbers (Table 21.3). This suggests that total renal filtration surface area may initially be maintained in the setting of a low or reduced nephron number but at the expense of glomerular hypertension and hypertrophy, which are maladaptive and predictors of poor outcomes.^{93–95} Consistent with this possibility, glomerulomegaly is common in renal biopsies from Australian Aborigines, a population with high rates of low birth weight and renal disease, and has also been associated with faster rate of decline of GFR in Pima Indians.^{96–98} Furthermore, in a study of donor kidneys, maximal planar area of glomeruli was found to be higher in kidneys from African Americans compared with whites and a predictor of poorer transplant function.⁹⁴ Among 111 adult males from four ethnic groups, mean glomerular volume and variability were highest among African Americans and Aboriginal Australians, likely associated with susceptibility to hypertension and renal disease.⁹⁹ In populations at high risk of kidney failure, therefore, large glomeruli are a common finding at early stages of renal disease and may reflect programmed reductions in nephron number in these populations in which access to prenatal and subsequent health care is often suboptimal.^{100–102} An increase in glomerular volume, however, is not always associated with lower glomerular number, and individual glomerular volume varies considerably within kidneys.⁹² Overall, however, glomerulomegaly and greater intrasubject variability in glomerular volume were associated with older age, fewer nephrons, lower birth weight, hypertension, obesity, and severity of cardiovascular disease.⁹⁹

EVIDENCE FOR PROGRAMMING IN THE KIDNEY

DEVELOPMENTAL PROGRAMMING OF NEPHRON ENDOWMENT

EXPERIMENTAL EVIDENCE FOR PROGRAMMING OF NEPHRON ENDOWMENT

Developmental programming of nephron number has been the most rigorously studied link to later-life hypertension and kidney disease so far. Numerous animal models have demonstrated the association between low birth weight (induced by gestational exposure to low-protein or low-calorie diets, uterine ischemia, dexamethasone, vitamin A deprivation, alcohol) with subsequent hypertension.^{103–110} The link between

Table 21.4 Associations of Nephron Number and Renal Size With Blood Pressure and Renal Function**Experimental Evidence****Reduction in Nephron Number**

<i>Experimental Model</i>	<i>Animal</i>	<i>Glomerular Number (% change)</i>	<i>Birth Weight</i>	<i>Blood Pressure</i>	<i>Renal Function</i>
Maternal calorie restriction ^{174,537,538}	Rat	↓ 20–40	↓	↑	↓ GFR proteinuria
Uterine artery ligation ^{112,302,402}	Rat	↓ 20–30	↓	↑	Impaired proteinuria
Low-protein diet during gestation ^{108,115,539–541}	Rat	↓ 25 ↓ 17 ↓ 16	↓/↔	↑	↓ GFR proteinuria ↓ Longevity
Postnatal nutrient restriction ⁴⁰⁶	Rat	↓ 27	Normal	↑	NA
Iron deficiency ³²²	Rat	↓ 22	↓	↑	NA
Vitamin A deficiency ^{115,326}	Rat	↓ 20	↔	NA	NA
Zinc deficiency ³²⁴	Rat	↓ 25	NA	↑	↓ GFR proteinuria
Ethanol ^{379,381}	Sheep	↓ 11	↔	NA	NA
	Rat		↓	↑	↓ GFR (females) ↑ GFR (males)
Hypoxia ⁵⁴²	Rats	↓ 26–52	↓	NA	NA
Cigarette smoke ⁵⁴³	Mouse	NA	↓	NA	↓ Kidney mass
Ureteral obstruction, neonatal ⁷³	Rat	↓ 50	NA	↑	↓ GFR ↓ Renal growth postrelief of obstruction
Prematurity ¹¹³	Mouse	↓ 17–24	↓	↑	↓ GFR ↑ Albuminuria
Glucocorticoids ^{103,107,356,544}	Rat	↓ 20	↓/↔	↑	Glomerulosclerosis
	Sheep	↓ 38	↔	↑	↑ Collagen deposition
Maternal diabetes ^{148,341}	Rat	↓ 10–35	↔	↑	Salt sensitivity
Gentamicin ^{370,372}	Rat	↓ 10–20	↓	NA	NA
β-Lactams ³⁷⁴	Rat	↓ 5–10	↔	NA	Tubular dilatation Interstitial inflammation
Cyclosporine ^{176,545}	Rabbits	↓ 25–33	↓/↔	↑	↓ GFR ↑ RVR proteinuria
Dahl salt sensitive ⁴⁸	Rat	↓ 15		↑ with Na intake	Accelerated FSGS
Munich-Wistar-Frömter ^{48,546}	Rat	↓ 40		↑ with age	↑ SNGFR FSGS
Milan hypertensive ⁴⁸	Rat	↓ 17		↑	NA
PVG/c ⁴⁸	Rat	↑ 122		Resistant	Resistant to FSGS
PAX2 mutations ^{306,396,400}	Mouse	↓ 22		NA	Renal coloboma syndrome in humans, small kidneys
GDNF heterozygote ^{392,547}	Human				
	Mouse	↓ 30	↔	↑	Normal GFR Enlarged glomeruli
c-ret null mutant ²⁹⁷	Mouse	↓	NA	NA	Severe renal dysplasia
hIGFBP-1 overexpression ³³⁹	Mouse	↓ 18–25	↓	NA	Glomerulosclerosis
Bcl-2 deficiency ³⁹⁷	Mouse	↓	NA	NA	↑ BUN and creatinine
p53 transgenic ⁴⁰¹	Mouse	↓ 50	NA	NA	Glomerular hypertrophy Renal failure
COX2 null mutant ⁵⁴⁸	Mouse	NA	↔	↔	↓ GFR
Fibroblast growth factor (FGF) 7 null mutant ¹¹⁵	Mouse	↓ 30	NA	NA	Reduced ureteric branching

Augmentation of Nephron Number

Vitamin A supplementation (with low-protein diet) ⁴¹⁹	Rat	Normalized	NA	NA	NA
Amino acid (glycine, urea, or alanine) supplementation to maternal low protein diet ⁵⁴⁹	Rat	Normalized	NA	Normalized with glycine only	NA
Restoration of postnatal nutrition postintrauterine growth restriction ¹¹²	Rat	Normalized	↓	Normalized	NA

Table 21.4 Associations of Nephron Number and Renal Size With Blood Pressure and Renal Function (Cont'd)

<i>Experimental Model</i>	<i>Animal</i>	<i>Glomerular Number (% change)</i>	<i>Birth Weight</i>	<i>Blood Pressure</i>	<i>Renal Function</i>
Iron supplementation to iron-deficient mothers ³²³	Rat	Partial rescue	NA	NA	NA
Ouabain administration (with low-protein diet) ⁴¹⁵	Mouse	Prevented ↓	NA	NA	NA
Maternal uninephrectomy prior to gestation ⁴²⁰	Rat	↑	NA	NA	NA
Postnatal overfeeding, normal birth weight ¹³⁷	Rat	↑ 20	↔	↑	Glomerulosclerosis
Maternal high-fat diet prior to and during pregnancy and lactation ³⁵⁰	Mouse	↑ 20–25	↔	NA	NA
Human Evidence					
<i>Clinical Circumstance</i>	<i>Population/ Age</i>	<i>Glomerular Number/ Kidney Volume^a (% change)</i>	<i>Birth Weight</i>	<i>Blood Pressure</i>	<i>Renal Function</i>
Low birth weight ^{91,124}	Human 0–1 year	↓ 13–35	↓	NA	NA
Preterm birth ¹²²	Human	↓ Correlated with gestational age	↓	NA	NA
Females vs. males ²⁸⁵	Human	↓ 12	NA	Variable	Variable
Hypertensive vs. normotensive Caucasian ^{59,120}	Human 35–59 years	↓ 19–50	NA	↑	No ↑ Glomerulosclerosis
Hypertensive vs. normotensive African American ¹²⁰	Human 35–59 years	NS ↓	NA	↑	No ↑ Glomerulosclerosis
Aboriginal Australians vs. Caucasian Australians ⁵⁵	Human 0–85 years	↓ 23	↓	NA	
Senegalese Africans ^{535,550}	Human 5–70 years	NA	NA	NA	↑ Variability of glomerular size with ↓ glomerular numbers
Maternal vitamin A deficiency ⁴⁶⁰	Indian vs. Canadian newborns	↓ Newborn renal volume	NA	NA	NA
Genetic polymorphisms:					
<i>RET</i> (1476A) polymorphism ¹²¹	Newborns	↓ 10 ^a	NA	NA	NA
<i>PAX2</i> AAA haplotype ³⁰⁶	Newborns	↓ 10 ^a	NA	NA	NA
Combined <i>RET</i> (1476A) polymorphism and <i>PAX2</i> AAA haplotype ¹²¹	Newborns	↓ 23 ^a	NA	NA	NA
<i>I/D ACE</i> polymorphism ⁵⁵¹	Newborns	↓ 8 ^a	NA	NA	NA
<i>BMPRI1A</i> ^{rs7922846} polymorphism ³⁰⁷	Newborns	↓ 13 ^a	NA	NA	NA
<i>OSR1</i> ^{rs12329305(T)} polymorphism ³¹⁰	Newborns	↓ 12 ^a	NA	NA	NA
Combined <i>OSR1</i> and <i>RET</i> polymorphisms ³¹⁰	Newborns	↓ 22 ^a	NA	NA	NA
Combined <i>OSR1</i> and <i>PAX2</i> polymorphisms ³¹⁰	Newborns	↓ 27 ^a	NA	NA	NA
<i>ALDH1A2</i> ^{rs7169289(G)} polymorphism ³⁰⁸	Newborns	↑ 22 ^a	NA	NA	NA

^aNumbers relate to kidney volume and not to glomerular number.

BUN, Blood urea nitrogen; FSGS, focal segmental glomerulosclerosis; GDNF, glial cell line-derived neurotrophic factor; GFR, glomerular filtration rate; NA, not assessed; NS, nonsignificant; RVR, renal vascular resistance; SNGFR, single-nephron GFR.

Adapted from Luyckx VA, Bertram JF, Brenner BM, et al. Effect of fetal and child health on kidney development and long-term risk of hypertension and kidney disease. *Lancet*. 2013;382:273–283; Brenner BM, Garcia DL, Anderson S. Glomeruli and blood pressure. Less of one, more the other? *Am J Hypertens*. 1988;1:335–347; Kett MM, Bertram JF. Nephron endowment and blood pressure: what do we really know? *Curr Hypertens Rep*. 2004;6:133–139; Clark AT, Bertram JF. Molecular regulation of nephron endowment. *Am J Physiol*. 1999;276:F485–F497; Moritz KM, Wintour EM, Black MJ, et al. Factors influencing mammalian kidney development: implications for health in adult life. *Adv Anat Embryol Cell Biol*. 2008;196:1–78.

adult hypertension and low birth weight in these animal models appears to be mediated, at least in part, by an associated congenital nephron deficit.^{103,107,108} Corresponding blood pressures and nephron numbers associated with various programming models are outlined in Table 21.4. As shown, the association between birth weight, nephron numbers, and blood pressures varies between models, as discussed in detail later, underscoring the complexity of developmental programming and the need for better markers than birth weight.

Vehaskari and colleagues¹⁰⁸ demonstrated an almost 30% lower glomerular number in offspring of pregnant rats fed a low-protein diet compared with a normal-protein diet during pregnancy (see Fig. 21.2A). As shown in Fig. 21.2B, the low-protein offspring had tail-cuff systolic blood pressures that were around 40 mm Hg higher by 8 weeks of age.¹⁰⁸ Similarly, prenatal administration of dexamethasone was associated with low birth weight and fewer glomeruli than controls.¹⁰³ In these nephron-deficient rats, GFR was reduced, albuminuria was increased, and urinary sodium excretion was lower than those with a normal nephron complement.¹⁰³ Uteroplacental insufficiency, induced by maternal uterine artery ligation late in gestation, also results in low nephron number and was

associated with increased profibrotic renal gene expression with age, although hypertension developed only in males.^{111,112}

Conversely, adequate postnatal nutrition, by cross-fostering onto normal lactating females at birth, restored nephron number and prevented subsequent hypertension in growth-restricted male rats.¹¹² In a study of the impact of preterm birth, mice delivered 1–2 days early (normal mouse gestation, 21 days) had lower nephron numbers, lower GFR, and higher blood pressures and albuminuria than mice born at term.¹¹³ Interestingly, nephron numbers were lower in mice delivered 2 days compared with 1 day early, suggesting the degree of preterm birth is important in determining final nephron endowment, even though nephrogenesis continues for about 5 days after birth in the normal mouse. Not surprisingly, in animal studies, timing of the gestational insult has been found to be crucial in renal programming, with the greatest impact on nephron number occurring during periods of most active nephrogenesis.¹¹⁴

The complexity of the effects of perturbations to the fetomaternal environment and genetic variants on kidney development was recently highlighted in two studies. Sampogna et al.¹¹⁵ compared the effects of vitamin A deficiency,

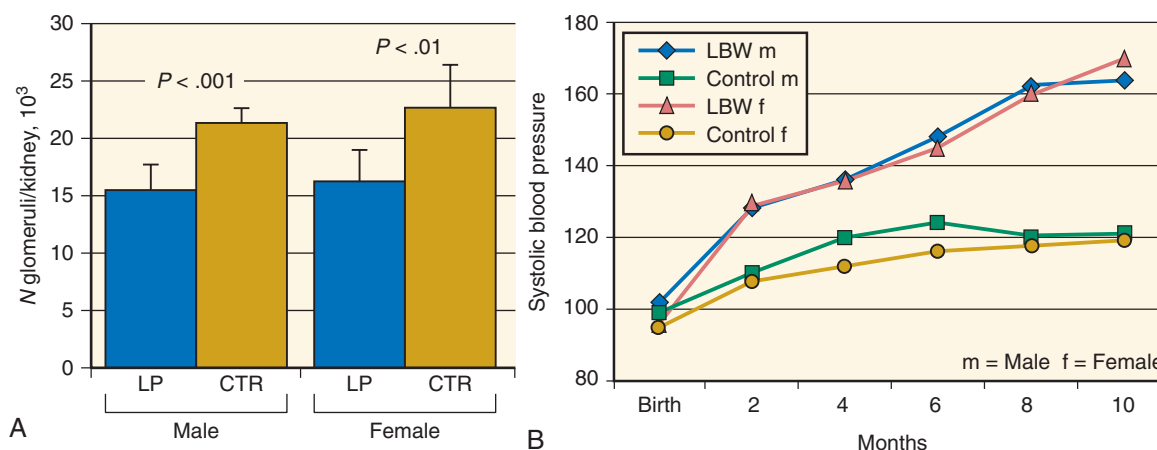


Fig. 21.2 Fetal programming of hypertension in low-birth-weight rats (induced by maternal low-protein diet). (A) Total number of glomeruli per kidney at 8 weeks of age, and (B) systolic blood pressure throughout life, in male ($n = 7$) and female ($n = 6$) offspring from low-protein pregnancies and in male ($n = 7$) and female ($n = 7$) control rats (Control). LBW, Low birth weight. (Adapted from Vehaskari VM, Aviles DH, Manning J. Prenatal programming of adult hypertension in the rat. *Kidney Int.* 2001;59:238–245.)



Fig. 21.3 Architecture of wild type (WT), FGF7 null mutant, vitamin A-deficient, and protein-deficient mouse kidneys. Embryonic day 15.5 ureteric tree tracings for each condition (to scale). N is the maximum branching generation number, and $Glom$ represents the mean glomerular count (minimum of three kidneys studied per condition). In each case, glomerular count is reduced compared with the wild type. The number of branching generations is specific to each condition. (From Sampogna RV, Schneider L, Al-Awqati Q. Developmental programming of branching morphogenesis in the kidney. *J Am Soc Nephrol.* 2015;26(10):2414–2422.)

protein deficiency, and FGF7 deletion on kidney development and nephron endowment in mice (Fig. 21.3). Perturbations to kidney development included developmental delay, defects in nephron induction, changes in the growth axis, and alterations in ureteric branching morphogenesis. These produced up to a threefold decrease in glomerular number and a twofold decrease in total branching events. Boubred et al.¹¹⁶ carefully compared the effects of two perturbations to the fetomaternal environment on nephron number and adult renal physiology in rats. Maternal gestational low-protein diet and betamethasone both led to a similar level of growth restriction, but nephron numbers were lower in betamethasone than the low-protein-diet offspring. Betamethasone offspring had impaired GFR, increased blood pressure, and glomerulosclerosis, whereas renal function and structure were unaltered in low-protein-diet offspring at 22 months of age. These findings suggest that the degree of reduction in nephron number is a risk factor for cardiovascular and renal disease. In summary, the available evidence suggests that the effect of a suboptimal fetomaternal environment on offspring nephron endowment and adult health may be influenced by species, the nature, timing, duration and severity of the perturbation, gender, and the postnatal nutritional environment (including lactation, infant nutrition, and subsequent growth).

PROGRAMMING OF NEPHRON NUMBER IN HUMANS

As noted earlier, total nephron number varies widely in the normal human population (see Table 21.2).^{58,117} A significant proportion of the interindividual variability in nephron number appears to be already present perinatally, demonstrating a strong developmental effect.^{118,119} Overall, the data support a direct relationship between nephron number and birth weight and an inverse relationship between nephron number and glomerular volume.^{87,91,120,121} Hughson and colleagues⁸⁷ reported a linear relationship between glomerular number and birth weight, and calculated a regression coefficient predicting an increase of 257,426 glomeruli per increase in birth weight, although the generalizability of the regression coefficient to populations in which the distribution of nephron number appears bimodal, such as among African Americans, may not be valid.

In recent years, glomerular loss associated with aging in healthy human kidneys has come to be better appreciated. Rates of loss of glomeruli per year have been variously reported as approximately 6750 glomeruli per year after the age of 18 years,⁶⁰ 4500 glomeruli per year,⁹⁰ and 6200 glomeruli per year.⁶¹ Specifically, the number of nonsclerotic functional glomeruli per kidney was found to decrease by almost 50% from young adulthood (aged 18–29 years: 990,661 glomeruli) to old age (aged 70–75 years: 520,410 glomeruli).⁶¹ Glomerular number tends to be lower in women than in men.^{60,61} A kidney starting with a lower nephron number, therefore, would conceivably reach a critical reduction of nephron mass, either with age or in response to a renal insult, earlier than a kidney with a greater nephron complement, predisposing to hypertension and/or renal dysfunction.

Nephrogenesis in humans begins during the 9th week of gestation and continues until the 34th to 36th week.⁹⁰ Nephron number at birth is therefore largely dependent on the intrauterine environment and gestational age. It is generally believed that no new nephrons are formed in humans

after term birth. To investigate whether glomerulogenesis does continue postnatally in preterm infants, Rodriguez and colleagues¹²² studied kidneys at autopsy from 56 extremely preterm infants compared with 10 full-term infants as controls. The radial glomerular counts (an estimate of glomerular number based on the number of layers of glomeruli in the cortex) were lower in preterm versus full-term infants and correlated with gestational age. Furthermore, evidence of active glomerulogenesis, indicated by the presence of S-shaped bodies immediately under the renal capsule, was seen in preterm infants who died before 40 days, but was absent in those who died after 40 days of life, suggesting that nephrogenesis may continue for up to 40 days after preterm birth. These authors also stratified their cases by presence or absence of renal failure.¹²² Among infants surviving longer than 40 days, those with renal failure (serum creatinine >2.0 mg/dL) had significantly fewer glomeruli than those without renal failure. This cross-sectional observation may suggest that renal failure inhibited glomerulogenesis or, conversely, that fewer glomeruli lowered the threshold to develop renal failure in these infants. Those preterm infants surviving longer than 40 days without renal failure exhibited glomerulomegaly, which may reflect, at least in the short term, a compensatory renoprotective response. Faa and colleagues¹¹⁸ also reported evidence of active glomerulogenesis in kidneys of preterm infants and two term infants who died at birth, but not in a child who died at age 3 months, suggesting that glomerular maturation may continue even after term birth for a short period.

In contrast, Hinchliffe and associates^{123,124} studied nephron number in preterm or full-term stillbirths or infants who died at 1 year of age and who were born with either appropriate weight for gestational age or small-for-gestational age. At both time points, growth-restricted infants had fewer nephrons than controls. In addition, the number of nephrons in growth-restricted infants dying at 1 year of age had not increased compared with the growth-restricted stillbirths, demonstrating a lack of postnatal nephrogenesis (Fig. 21.4A). Manalich and coworkers⁹¹ examined the kidneys of neonates dying within 2 weeks of birth in relation to their birth weights (see Fig. 21.4B). A significant direct correlation was found between glomerular number and birth weight, and a strong inverse correlation between glomerular volume and glomerular number, independent of sex and race. These studies all support the hypothesis that an adverse intrauterine environment, which may manifest as low birth weight, small-for-gestational age, or preterm birth, is associated with a congenital reduction in nephron endowment and an early, compensatory increase in glomerular volume.

In a population of 140 adults aged 18 to 65 years who died of various causes, a significant correlation was also observed between birth weight and glomerular number.¹²⁰ Glomerular volume was inversely correlated with glomerular number. Total glomerular number did not differ statistically between African-American and white subjects, although the distribution among African Americans appeared bimodal. The range of nephron number was greatest in African Americans. Significantly, however, none of the subjects in this study had been of low birth weight; therefore, no conclusion can be drawn as to whether an association with low birth weight and nephron number existed in either population group.¹²⁰ It may be argued that as low birth weight is more

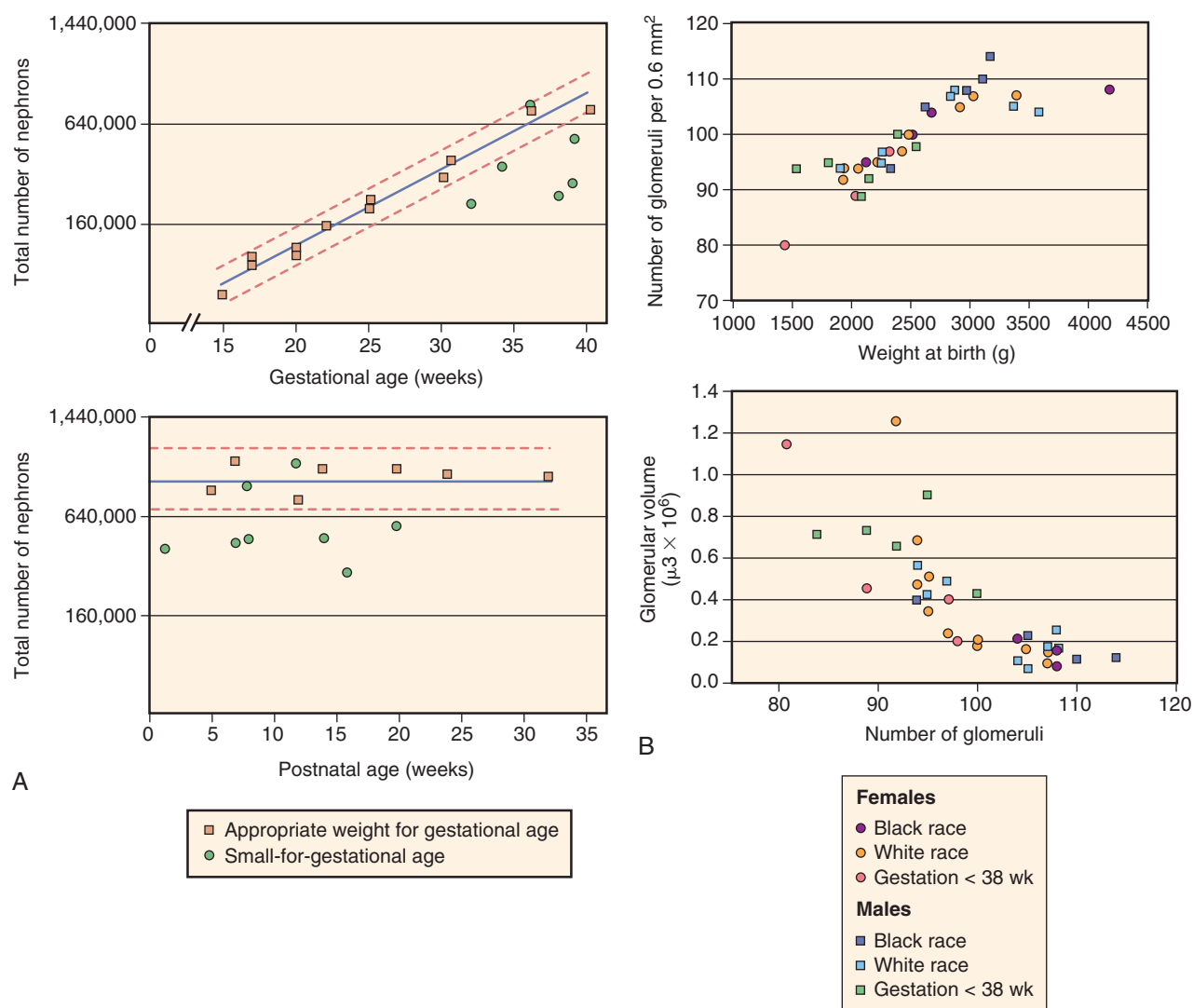


Fig. 21.4 Effect of intrauterine growth restriction (IUGR) on nephron number in humans. (A) Nephron number at birth in relation to gestational age (*upper panel*), and lack of postnatal catch-up in nephron number (*lower panel*). (From Hinchliffe SA, Lynch MR, Sargent PH, et al. The effect of intrauterine growth retardation on the development of renal nephrons. *Br J Obstet Gynaecol.* 1992;99:296–301.) (B) Relationship between birth weight and glomerular number (*upper panel*), and between glomerular number and glomerular volume (*lower panel*) in neonates. (From Manalich R, Reyes L, Herrera M, et al. Relationship between weight at birth and the number and size of renal glomeruli in humans: a histomorphometric study. *Kidney Int.* 2000;58:770–773.)

prevalent among African Americans, this cohort was more representative of the general white population than the general black population, having included only subjects of normal birth weight.¹²⁵ In a European study, among 26 subjects with non-insulin-dependent diabetes compared with 19 age-matched nondiabetic controls, no difference in glomerular number was found, but again, all subjects had birth weights above 3 kg; therefore, the impact of low birth weight on nephron number could not be assessed.¹²⁶

Kidney Size as a Correlate for Nephron Number

Analysis of the relationship between kidney weight and nephron endowment in infants younger than 3 months of age (a time at which compensatory hypertrophy has likely not yet occurred) revealed a direct relationship (Fig. 21.5).¹²¹ Regression analysis predicted an increase of 23,459 nephrons per gram of kidney weight.¹²¹ Renal mass is therefore proportional to nephron number in the early postnatal period,

and renal volume is proportional to renal mass; therefore, renal volume has been used as a surrogate for nephron endowment in infants in vivo.⁵⁴ Ultrasound evaluation of fetal renal function in utero revealed a reduction in hourly urine volume, higher prevalence of oligohydramnios, reduced renal perfusion, and reduced renal volume in growth-restricted fetuses.^{127–129} These findings may represent reduced fetal perfusion in situations of uterine compromise, however, and do not necessarily reflect altered renal development. Similarly, among preterm infants, kidney volume at a corrected age of 38 weeks was significantly lower than in term infants and was associated with a significantly lower GFR estimated from serum cystatin C.¹³⁰ Analysis of kidney size and postnatal growth measured by ultrasound in 178 children born preterm or small for gestational age compared with 717 mature children with appropriate weight for gestational age at 0, 3, and 18 months found that weight for gestational age was positively associated with kidney volume at all three time points.¹³¹ Slight

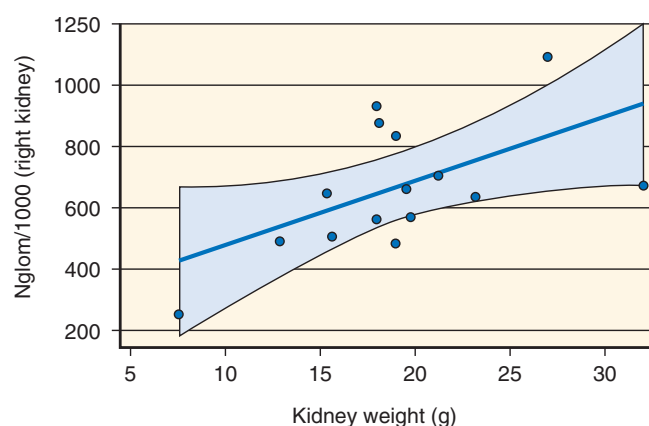


Fig. 21.5 Relationship between nephron number and mass of the right kidney in Caucasian infants aged ≤ 3 months who died within the first 3 months of life. *Nglom/1000*, Number of glomeruli per kidney in 1000s. (From Zhang Z, Quinlan J, Hoy WE, et al. A common RET variant is associated with reduced newborn kidney size and function. *J Am Soc Nephrol.* 2008;19:2027–2034.)

catch-up in kidney growth was observed in growth-restricted infants but not in preterm infants. Similarly, in a South Indian population, kidney volumes were lower in low-birth-weight and small-for-gestational-age neonates compared with those of normal and appropriate-for-gestational-age birth weights.¹³² Kidneys in the low-birth-weight and small-for-gestational-age infants remained small and grew more slowly than kidneys of normal-birth-weight or appropriate birth-weight-for-gestational-age infants during the first 2 years of life. In a study of multiethnic children in the Netherlands, lower fetal weight gain and lower early infant weight gain led to smaller kidneys at 6 years of age.¹³³ However, only lower fetal weight gain, not postnatal weight gain, was associated with lower estimated GFR (eGFR). These findings suggest that suboptimal early growth affects kidney function in later life.

Among term neonates, renal parenchymal thickness, proposed as a more accurate screening tool than renal volume estimation, was significantly reduced in those with low compared with normal birth weights.¹³⁴ In Australian Aboriginal children, low birth weight was also found to be associated with lower renal volumes on ultrasound.¹³⁵ Comparison of renal volume between children aged 9–12 years born preterm, at either small or appropriate weight for gestational age, and controls, found that kidneys were smallest in those who had been preterm and small for gestational age, but when adjusted for body surface area (BSA), there were no significant differences between the groups.¹³⁶ A smaller kidney size, therefore, may be a surrogate marker for a reduced nephron endowment, but importantly, growth in kidney size on ultrasound cannot distinguish between normal growth with age and renal hypertrophy. In contrast to the situation in infants and children, in adults, kidney size is not a reliable marker of nephron number tautology. Findings from six studies in adults in which kidney size was reported and nephron number had been estimated using the dissector/fractionator combination were analyzed.⁵⁷ Only data from subjects aged 18–60 years without renal disease were included. Although an association between renal size and nephron number was found, only about 5% of the variation in nephron numbers was explained by differences in renal size.

Table 21.5 Developmentally Programmed Changes Observed in the Kidney

- Nephron number
- Glomerular volume
- Accelerated maturation of glomeruli
- Tubular sodium transporter expression and/or activity
- Altered renal vascular reactivity
- Alterations in renin-angiotensin system
- Alterations in sympathetic nervous system activity
- Accelerated senescence, especially after catch-up growth
- Predisposition to inflammation and fibrosis
- Kidney size

EVIDENCE OF ADDITIONAL PROGRAMMING EFFECTS IN THE KIDNEY

Taken together, there is no doubt that nephron endowment varies widely in human kidneys and that a range of perturbations to the feto-maternal environment in animal models can dramatically influence nephron endowment. Once nephrogenesis ends, no new nephrons can form, and therefore, any deficit in nephron number is permanent. Nephron endowment is likely an independent factor determining susceptibility to essential hypertension and subsequent renal injury. However, low nephron number alone does not account for all observed programmed hypertension (see Table 21.4). Supplementation of a low-protein diet during gestation with glycine, urea, or alanine resulted in a normalization of nephron number in rat offspring, but blood pressure only normalized in those supplemented with glycine.³⁴ Likewise, augmentation of nephron number by postnatal hypernutrition resulted in a 20% increase in nephron number but also obesity, hypertension, and glomerulosclerosis with age.¹³⁷ These findings suggest that additional factors also contribute to developmental programming of hypertension. Recent evidence has shown alterations in renal tubular sodium handling and vascular function in developmentally programmed kidneys that likely also contribute to later-life blood pressure and renal functional changes, as outlined in Table 21.5.^{114,138}

Altered Sodium Handling by the Kidney

The pressure–natriuresis curve is shifted to the right in most forms of hypertension. A low total renal filtration surface area associated with a low nephron number is one plausible hypothesis to explain the associated higher blood pressures. Consistent with this, salt sensitivity has been reported in several animal models associated with low birth weight and low nephron number.^{35,139–143} Some authors have reported the presence of salt-sensitive hypertension in rats in which low birth weight was induced by maternal uterine artery ligation, whereas others report no salt sensitivity in rats in which low birth weight was induced by maternal protein restriction, although timing of dietary intervention and age at study appear to play a role.^{140,144,145} Elevations in blood pressure in response to a high-salt diet have been more consistently observed in aging rather than young rats, suggesting an early adaptive mechanism that may decline with age or worsening salt sensitivity as nephron number declines with age.¹⁴⁶ In

young rats, however, despite no change in blood pressure, an increase in plasma volume was observed, consistent with sodium retention.¹⁴⁷ Similar salt sensitivity was observed in adult rat offspring exposed to maternal gestational diabetes, a model also associated with low nephron number.¹⁴⁸

The effect of nephron “dose” and total renal filtration surface area on salt sensitivity was examined in glial cell line–derived neurotrophic factor (GDNF) heterozygous mice that have either 1 kidney (HET1K) and 65% few nephrons than wild type littermates, or two kidneys (HET2K) and 25% fewer nephrons than GDNF wild type mice.¹⁴⁹ Given the accompanying glomerular hypertrophy, total glomerular surface area was normal in HET2K but remained reduced in HET1K mice at 48–52 weeks of age. At baseline, the mice did not have elevated blood pressures; however, both heterozygous groups became hypertensive in response to high-sodium feeding. A gradient of increasing blood pressure was observed from GDNF wild type to HET2K to HET1K, suggesting dependence on nephron number and filtration surface area, given that no change in expression of tubular sodium transporters was observed.¹⁴⁹ In contrast, transforming growth factor- β 2 (TGF- β 2) heterozygous mice, which have 30% more nephrons than TGF- β 2 wild type mice, were relatively protected against developing high blood pressures on a chronic high-salt diet compared with wild type mice.¹⁵⁰ Surprisingly, however, these mice did develop increased blood pressures in response to an acute sodium load, suggesting that the benefit conferred by the higher nephron number requires time for adaptation to occur. Early change in sodium diet in itself has been found to have a long-term impact on programming of hypertension in low-birth-weight rats. Short-term feeding of a low-salt diet from weaning to 6 weeks of age abrogated, whereas high-salt feeding exacerbated hypertension at 10 and 51 weeks despite reinstitution of normal salt diet at 6 weeks.^{139,151} The role of sodium intake on long-term renal programming requires further study.

Salt sensitivity, therefore, does appear to be developmentally programmed. From the GDNF mouse data, total renal filtration surface area may be crucial in determining salt sensitivity, but as discussed earlier, it is often not reduced in the setting of low nephron number due to concomitant glomerular hypertrophy. Expression and activity of renal tubule sodium transporters have therefore been investigated. Expression of the Na-K-2Cl (NKCC2) and Na-Cl (NCC) transporters were significantly increased in prehypertensive offspring of rats fed a protein-restricted diet during gestation compared with controls, although expression of the sodium–hydrogen exchanger type III (NHE3) and epithelial sodium channel (ENaC) were not changed (Fig. 21.6A).¹⁵² Increased activity of NKCC2 was shown by increases in chloride transport and lumen positive transepithelial potential difference in the medullary thick ascending limb in offspring of protein-restricted or dexamethasone-treated mothers (Fig. 21.6B).¹⁵³ Furthermore, after development of hypertension, furosemide administration reduced blood pressure, supporting increased NKCC2 activity as a mediator of hypertension in the protein restriction model.¹⁵³ Expression of the glucocorticoid receptor and the glucocorticoid responsive α 1- and β 1-subunits of Na-K-ATPase were increased in offspring of pregnant rats fed a low-protein diet.¹⁵⁴ In rats suckled by low protein–fed mothers during lactation, expression of Na-K-ATPase was increased by 40%, but Na-K-ATPase activity was increased by

300%, demonstrating that expression levels may not fully reflect activity levels.¹⁵⁵ Prenatal dexamethasone administration was associated with increased expression of proximal tubular NHE3, as well as the more distal NKCC2 and NCC, but no change in ENaC expression.¹⁵⁶ Interestingly, renal denervation reduced systolic blood pressure and sodium transporter expression in this model, suggesting indirect regulation of these genes via sympathetic nerve activity.¹⁵⁶ In rats subjected to maternal diabetes, baseline expression of β and γ ENaC, but not α ENaC, as well as Na-K-ATPase, were significantly increased compared with controls.¹⁴⁸ Despite several studies showing no change in ENaC expression in programmed animals, an enhanced natriuretic response to the ENaC inhibitor benzamil demonstrated increased ENaC activity in offspring of mothers fed a low-protein diet (Fig. 21.6C).¹⁵⁷ Taken together, despite differences among models, the data suggest increased sodium transport in all segments of the renal tubule. Whether a reduced nephron number contributes indirectly to increased sodium transport through increased single-nephron GFR (SNGFR), necessitating glomerulotubular balance, or sodium transporter activity is independently programmed, has not yet been elucidated.

Renin–Angiotensin System

All components of the renin–angiotensin–aldosterone system (RAAS) are expressed in the developing kidney.¹⁵⁸ Alterations in the renal RAAS have been studied in various programming models, and a common, though not universal, finding has been inhibition of the system during the period of active nephrogenesis with an upregulation in adulthood, often associated with changes in blood pressure.¹⁵⁹ For example, expression of angiotensinogen and renin mRNA was decreased in neonatal kidneys of rats subjected to uterine ischemia but increased in mouse offspring of diabetic mothers.^{160,161} Outcomes likely reflect species differences, differences in timing of intervention, timing of study, and so on, as summarized in Table 21.6.⁷⁵ The importance of angiotensin II in nephrogenesis was demonstrated by the administration of the angiotensin II subtype 1 receptor (AT1R) blocker, losartan, to normal rats during the first 12 days of life (while nephrogenesis is proceeding), which resulted in a low final nephron number and subsequent development of hypertension.^{162,163} Angiotensin II can stimulate the expression of Pax-2 (an antiapoptotic factor) through AT2R.¹⁶⁴ AT2R expression, therefore, is likely to affect nephrogenesis and kidney development, but its role in programming is still unclear. Administration of an angiotensin-converting enzyme inhibitor (ACEI), captopril, or losartan to low-birth-weight rats from 2 to 4 weeks of age abrogated the development of adult hypertension.^{21,112,160,165} Similarly, administration of angiotensin II or ACEI to adult rats subjected to a low-protein diet in utero resulted in a more exaggerated hypertensive or hypotensive response, respectively, than in control rats.^{21,166–168} However, a recent study in aged offspring born growth restricted demonstrated that AT1 receptor blockade could not prevent hypertension, suggesting lesser dependence on the RAAS with age.¹⁶⁹ Differential regulation of the RAAS by sex hormones during development is thought to contribute to the observation that the effects of developmental programming are often less severe in young females.^{159,170} Overall, programmed suppression of the intrarenal RAAS during nephrogenesis is likely to contribute to low nephron number

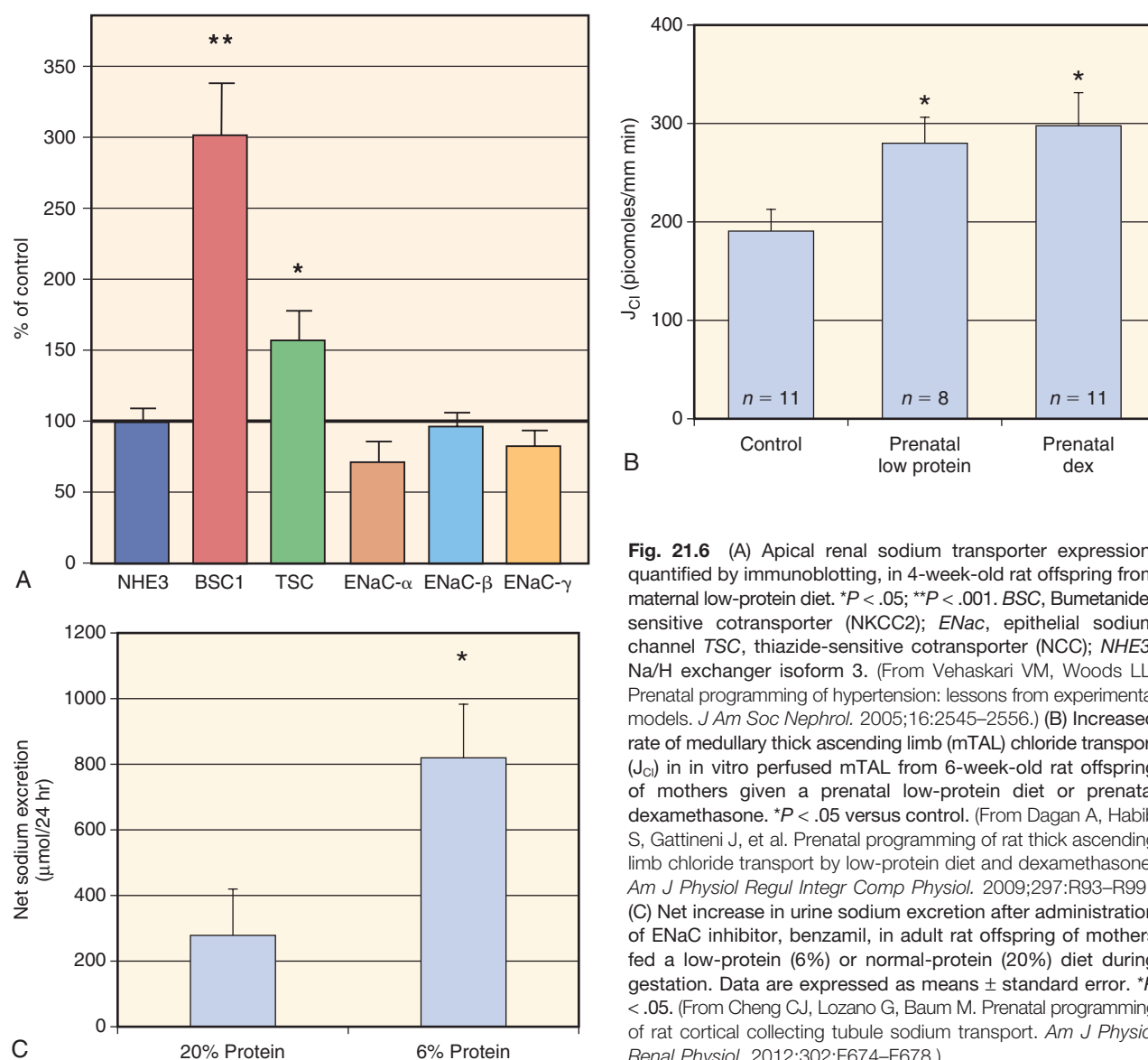


Fig. 21.6 (A) Apical renal sodium transporter expression, quantified by immunoblotting, in 4-week-old rat offspring from maternal low-protein diet. * $P < .05$; ** $P < .001$. BSC, Bumetanide-sensitive cotransporter (NKCC2); ENaC, epithelial sodium channel TSC, thiazide-sensitive cotransporter (NCC); NHE3, Na/H exchanger isoform 3. (From Vehaskari VM, Woods LL. Prenatal programming of hypertension: lessons from experimental models. *J Am Soc Nephrol*. 2005;16:2545–2556.) (B) Increased rate of medullary thick ascending limb (mTAL) chloride transport (J_{Cl}) in in vitro perfused mTAL from 6-week-old rat offspring of mothers given a prenatal low-protein diet or prenatal dexamethasone. * $P < .05$ versus control. (From Dagan A, Habib S, Gattineni J, et al. Prenatal programming of rat thick ascending limb chloride transport by low-protein diet and dexamethasone. *Am J Physiol Regul Integr Comp Physiol*. 2009;297:R93–R99.) (C) Net increase in urine sodium excretion after administration of ENaC inhibitor, benzamil, in adult rat offspring of mothers fed a low-protein (6%) or normal-protein (20%) diet during gestation. Data are expressed as means $\pm$ standard error. * $P < .05$. (From Cheng CJ, Lozano G, Baum M. Prenatal programming of rat cortical collecting tubule sodium transport. *Am J Physiol Renal Physiol*. 2012;302:F674–F678.)

under adverse circumstances, and postnatal upregulation of the AT1R, possibly mediated by an increase in glucocorticoid activity or sensitivity, may contribute to the subsequent development of hypertension and glomerular hyperfiltration as reviewed in detail elsewhere.^{75,159}

Underscoring the relevance of the RAAS in developmental programming of blood pressure, angiotensin-converting enzyme (ACE) activity was significantly elevated in children of low birth weight compared with normal birth weight, and there was a greater frequency of the ACE gene DD genotype among low-birth-weight children with highest blood pressures, suggesting that the programming effect of blood pressure may be in part modulated by ACE gene polymorphisms.¹⁷¹

The Sympathetic Nervous System and Renal Vascular Reactivity

Within the kidney, the sympathetic nervous system modulates activity of the RAAS, sodium transport, and vascular function

and thereby contributes to blood pressure through regulation of vascular tone and volume status.⁷⁵ Development of the renal sympathetic nervous system and how this may be programmed during nephrogenesis, and modulated by the RAAS, is expertly reviewed by Kett and Denton.⁷⁵ Renal denervation has been shown to abrogate development of adult hypertension and alter sodium transporter expression in the prenatal dexamethasone and uterine ischemia programming models, as well as the age-associated hypertension that develops in growth-restricted female rats.^{156,172,173} Consistent with the whole animal findings, within the kidney, an increase in baseline renal vascular resistance has been described in several programming models.^{174–176} For example, renal arterial responses to β -adrenergic stimulation and sensitivity to adenylyl cyclase were increased in 21-day-old growth-restricted offspring subjected to placental insufficiency.¹⁷⁷ Although the renal expression of β 2-adrenoreceptor mRNA was increased in these pups, there was also evidence of adaptations

Table 21.6 Programming Effects on the Renin–Angiotensin System

Model	Species	Timing of Insult	Age and Sex of Offspring at Study	mRNA or Protein Expression	Physiologic Response	Reference(s)
Glucocorticoids	Sheep	Early gestation	40 mo ♀	↔ Plasma renin, ANG II, or Aogen	↑ Basal MAP, females only	116
	Rat		6–7 mo ♂♀	↔ Renal Aogen ↑ PRA, plasma Aogen in females only	↑ Basal TBP, females only	108
	Rat	Mid- to late gestation	6 mo	↑ Renal ACE and renin in males and females	ND	159
	Rat	Mid- to late gestation	4 and 8 wk ♂	↔ PRA, plasma ANG II, and renal ANG II levels ↑ Urine ANG II at 4 and 8 wk	↑ Basal TBP at 8 but not 4 wk of age	27
Maternal nutrient restriction or low-protein diet	Sheep	Early- to midgestation	9 mo old	↑ Renal cortical ACE protein ↔ AT1R in the renal cortex and medulla ↔ Renal cortical but ↑ renal medulla AT2R	↑ Basal MAP	47
	Rat	Mid- to late gestation	4–12 wk ♂♀	↑ PRA	↑ Basal TBP from 8 wk of age	11
	Rat	Throughout gestation	1–5 days and 22 wk ♂	↓ Renal renin mRNA and ANG II levels at 1 to 5 days of age	↑ Basal MAP at 22 wk of age	154, 155
	Rat	Throughout gestation	16 wk ♂	↓ Renal AT1R and AT2R protein	↔ No change GFR or RBF ↑ Basal TBP, ↓ sodium excretion, ↔ GFR	99
	Rat	Throughout gestation	4 wk ♂	↑ Renal AT1R protein ↓ Renal AT2R protein	↑ Basal MAP (anesthetized) ↑ Basal renal vascular resistance	125, 126
				↑ Ang II receptor binding ↔ Renal renin and Ang II tissue levels	↔ No change GFR or RBF	
	Rat	Mid- to late gestation	1 to 11 mo ♂♀: 1–2 mo	↓ PRA ↓ Renal AT1R protein and mRNA	↑ Basal TBP at 8 wk of age Salt-sensitive TBP TBP normalized by ACE inhibition and low-salt diet	94, 95, 146, 147
			6 to 11 mo	↓ Renal AT2R protein, ↑ AT2R mRNA ↑ PRA ↔ Plasma or renal ANG I and ANG II ↑ AT1R protein and mRNA ↑ AT2R protein, ↔ AT2R mRNA	Urinary protein/creatinine ratio increased in males only	
Placental insufficiency	Rat	Late gestation	0 to 16 wk ♂ Newborn:	↓ renal renin and aogen	↑ Basal MAP that was abolished by ACEi treatment	53, 110
			16 wk:	↑ Renal renin and aogen mRNA, ↑ ACE activity ↔ Renal AT ₁ R and ANG II, ↔ PRA and plasma ACE	↑ Pressor response to ANG II in presence of ACEi ↓ GFR	
Maternal renal hypertension	Rabbit	Throughout gestation	10–45 wk, ♂ ♀	↓ PRA–5 and 10 wk ↔ PRA 30 and 45 wk	↑ Basal MAP at 30 and 45 wk	31, 32, 92

ACE, Angiotensin-converting enzyme; ACEi, angiotensin-converting enzyme inhibition; Aogen, angiotensinogen; AT1R, angiotensin receptor type 1; GFR, glomerular filtration rate; MAP, mean arterial pressure; ND, not done; PRA, plasma renin activity; RBF, renal blood flow; TBP, tail artery pressure.

Taken from Kett MM, Denton KM. Renal programming: cause for concern? *Am J Physiol Regul Integr Comp Physiol*. 2011;300:R791–R803.

to the signal transduction pathway contributing to the β -adrenergic hyperresponsiveness.¹⁷⁷ Intriguingly, these findings were much more marked in the right compared with the left kidney, an observation that remains unexplained but that is not without precedent: asymmetry of renal blood flow was found in 51% of a cohort of hypertensives without renovascular disease.^{177,178} In this study, the growth-restricted rats had low glomerular number, glomerular hyperfiltration, and hyperperfusion, and had significantly increased proteinuria compared with controls, suggesting alteration in glomerular pressures likely mediated by renal vasoreactivity. Interestingly, in a cohort of white and black U.S. subjects, the effect of birth weight on subsequent blood pressures was significantly modified by β -adrenergic receptor genotype, further underscoring a relationship between birth weight, sympathetic activity, and blood pressure.¹⁷⁹

PROGRAMMING OF RENAL FUNCTION AND DISEASE

In contrast to preterm infants or those of low birth weight in whom nephron numbers have been shown to be reduced, in adults, there are no data on nephron number, specifically in those born with low birth weight. The association between nephron number and birth weight and preterm birth, however, is a consistent finding in infants, so it seems reasonable to extrapolate that nephron numbers would remain reduced in adults of low birth weight.⁸⁷ The determination of nephron number *in vivo* is not yet possible; therefore, the most used *in vivo* surrogate markers at present are birth weight and preterm birth. Importantly, however, in some animal models, low nephron numbers have also been observed in the setting of normal birth weight (see Table 21.4). Therefore, in humans, if birth weight is the only surrogate marker used, the impact of renal programming on any outcome is likely to be underestimated.¹⁸⁰ Other clinical surrogates for an adverse intrauterine environment and low nephron numbers are outlined in Table 21.7.

EXPERIMENTAL EVIDENCE

Glomerulomegaly is consistently observed in the setting of low nephron number (see Fig. 21.4B). In rats in which low birth weight was induced by maternal protein restriction, GFR was reduced by 10%, although nephron number was reduced by 25%, implying some degree of compensatory

hyperfunction per nephron.³⁵ Although this may be a compensatory mechanism to restore filtration surface area, it is conceivable that renal reserve in these kidneys is reduced.⁹⁰ If this is the case, these kidneys may be expected to be less able to compensate further in the setting of additional renal insults and to begin to manifest signs of renal dysfunction (i.e., proteinuria, elevations in serum creatinine, and hypertension). To investigate this hypothesis, diabetes was induced by streptozotocin injection in subgroups of low-birth-weight (induced by maternal protein restriction) and normal-birth-weight rats.¹⁸¹ Low-birth-weight rats had reduced nephron numbers and higher blood pressures compared with those of normal birth weight. Among those rendered diabetic, there was a greater proportional increase in renal size and glomerular hypertrophy in the low-birth-weight rats compared with normal-birth-weight controls after 1 week (Fig. 21.7).¹⁸¹ This study demonstrates that the renal response to injury in the setting of a reduced nephron number may be exaggerated and could lead to accelerated loss of renal function.

Subsequently, the same authors published outcomes in low-birth-weight versus normal-birth-weight diabetic rats at 40 weeks.¹⁸² Histologically, the podocyte density was reduced and the average area covered by each podocyte was greater in the low-birth-weight diabetic rats than in the normal-birth-weight controls. These findings correlated with urine albumin excretion rate, which was higher in low-birth-weight diabetic rats, although this did not reach statistical significance. In support of the role of altered podocyte physiology in renal disease progression, similar findings were observed in the Munich Wistar-Frömter rat, a strain that has congenitally reduced nephron numbers and develops spontaneous renal disease.¹⁸³ Whether these podocyte changes are secondary to an increase in glomerular pressure in the setting of reduced nephron numbers or a primary programmed structural change leading to glomerular injury is not yet known. The podocyte depletion hypothesis has emerged as a potentially unifying concept in glomerular pathology in recent years, with podocyte depletion defined as either loss of podocytes, decreased podocyte density due to glomerular hypertrophy, or a change in podocyte phenotype.¹⁸⁴ Given that podocytes have little or no capacity to increase their numbers in adulthood, it will be important to determine if those fetomaternal environmental factors that result in low nephron endowment also produce low podocyte endowment. Humans and animals born with both a low nephron endowment and low podocyte endowment could be expected to be at further risk of developing CKD.

Of interest, in contrast, in low-birth-weight rats exposed to prenatal dexamethasone and subsequently fed a high-protein diet, GFR was similar to that in normal-birth-weight controls.¹⁸⁵ Nephron numbers were reduced by 13% in only the male low-birth-weight rats. This finding suggests that there is a threshold reduction in nephron number, above which compensation is adequate or that the high-protein diet-induced supranormal GFRs in both groups, masking subtle differences in baseline GFR. Another study that measured GFR in low-birth-weight rats, induced by placental insufficiency, also failed to demonstrate lower GFRs in low-birth-weight rats, but the low-birth-weight offspring were hypertensive compared with normal-birth-weight controls.¹¹⁰ Conceivably, in this study, the higher intraglomerular pressure due to elevated blood pressure and reduced nephron mass in low-birth-weight rats

Table 21.7 Clinical Surrogates for Programmed Low Nephron Number in Humans

- Low birth weight^{91,122,124}
- Preterm birth^{122,124}
- Low kidney mass^{54,121}
- Reduced kidney volume^{131,135}
- Glomerulomegaly^{87,91,124}
- Female sex²⁸⁵
- Ethnicity: Australian Aboriginal²⁸⁵
- Older age⁶¹

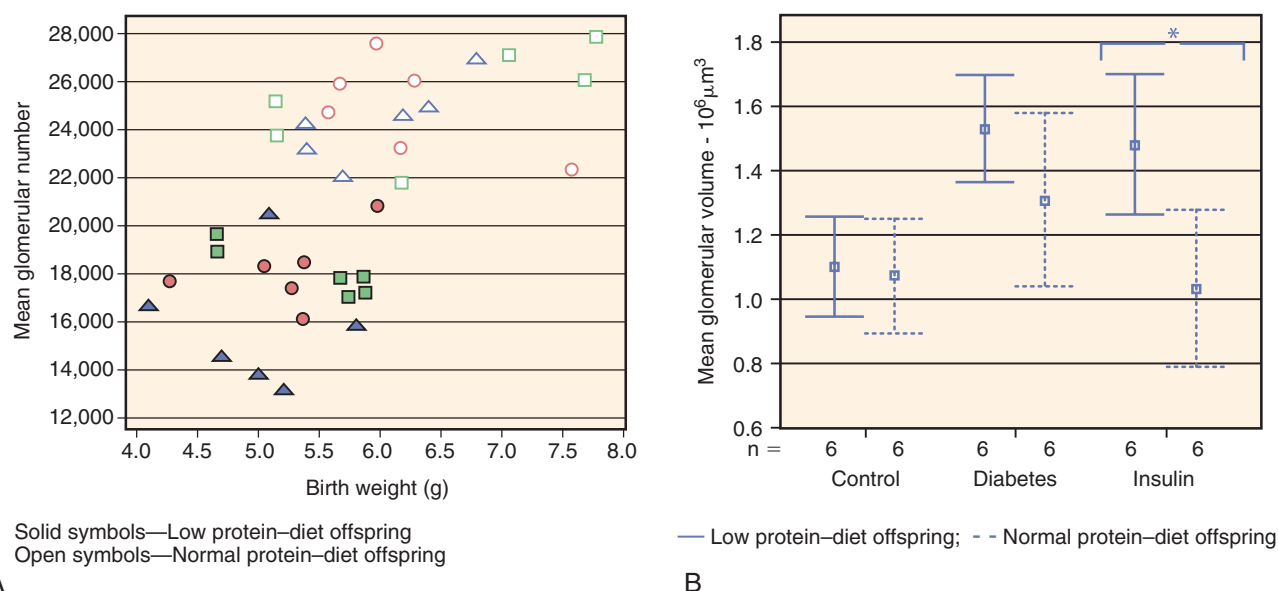


Fig. 21.7 Influence of glomerular number on adaptation to diabetes in rats. (A) Scatterplot of mean glomerular number and birth weight in rats on low-protein diet (solid symbols) and normal protein diet (open symbols); control groups (triangles); diabetes (circles); diabetes treated with insulin (squares). (B) Plot of glomerular volume in rats on low-protein diet (solid lines) and normal-protein diet (dotted lines). Error bars represent 95% confidence intervals. * $P = .015$. (From Jones SE, Bilous RW, Flyvbjerg A, et al. Intra-uterine environment influences glomerular number and the acute renal adaptation to experimental diabetes. *Diabetologia* 2001;44:721–728.)

may have led to a compensatory increase in SNGFR and, thus, normalization of whole-kidney GFR.

The definitive pathophysiologic impact of a reduction in nephron number in the development of renal dysfunction is difficult to elucidate from the existing literature comprising very varied experimental conditions. Overall, however, it is possible that, although whole-kidney GFR may not change, SNGFR is likely to be increased in the setting of a reduced nephron number and exacerbated in the face of renal injury. Interestingly, SNGFR was found to be significantly elevated in the Munich-Wistar-Frömter rat, which has reduced nephron numbers and is known to develop spontaneous progressive glomerular injury, compared with the control Wistar rat strain.¹⁸³ Renal dysfunction may also result from a programmed predisposition to inflammation and scarring, which may be independent of glomerular pressures. In a glomerulonephritis model, anti-Thy-1 antibody injection in low-birth-weight rats resulted in significant upregulation of inflammatory markers and development of sclerotic lesions by day 14, but with no difference in blood pressure or proteinuria compared with normal-birth-weight controls.¹⁸⁶

HUMAN EVIDENCE

Most human data rely on the surrogates of birth weight and preterm birth and kidney size to reflect likelihood of renal programming. Although there is no direct proven relationship between renal risk and nephron number in humans, the consistency of the data is strongly suggestive of a programming effect.

BIRTH WEIGHT, PRETERM BIRTH, AND BLOOD PRESSURE

Two meta-analyses and systematic reviews have shown consistent associations with lower birth weight and preterm birth

and higher blood pressures in later life.^{36,37} Meta-analysis of 27 studies investigating the relationship between birth weight and blood pressure found that systolic blood pressures were 2.28 mm Hg (95% confidence interval 1.24–3.33 mm Hg) higher in subjects with birth weights below, compared with higher 2.5 kg (Fig. 21.8A).³⁷ Many studies do not discriminate between low birth weight occurring as a result of growth restriction (a marker of intrauterine stress) at any gestational age or as a result of preterm birth with an appropriate (low) weight for gestational age. Therefore the relative impact of growth restriction and preterm birth on subsequent blood pressures is not always easy to dissect.¹⁸⁷ To investigate this question, a study of 50-year-old subjects all born at term, but with or without growth restriction, reported an odds ratio (OR) of 1.9 (95% confidence interval [CI]: 1.1 to 3.3) for hypertension among those who had experienced growth restriction compared with those who had normal birth weights.¹⁸⁸ Growth restriction before birth per se, therefore, is associated with subsequent higher blood pressure.

A systematic review of 10 studies comparing preterm or very low-birth-weight subjects versus those born at term found that the preterm subjects, having a mean gestational age of 30.2 weeks and a mean birth weight of 1280 g, had 2.5 mm Hg higher (95% CI: 1.7 to 3.3 mm Hg) systolic blood pressures in later life compared with those born at term (see Fig. 21.8B).³⁶ Preterm birth, therefore, is also independently associated with higher blood pressure, which, in some studies, meets the definition of hypertension by 1–2 years of age.^{189–191} Whether the risk of higher blood pressure is greater among preterm subjects who were born small for gestational age (growth restricted) versus appropriate for gestational age, however, is not yet clear, with some studies suggesting an additional effect of growth restriction, whereas others do not.^{192–195} Ultimately, the importance of dissecting the risk from low birth weight “versus” preterm birth may lie in the

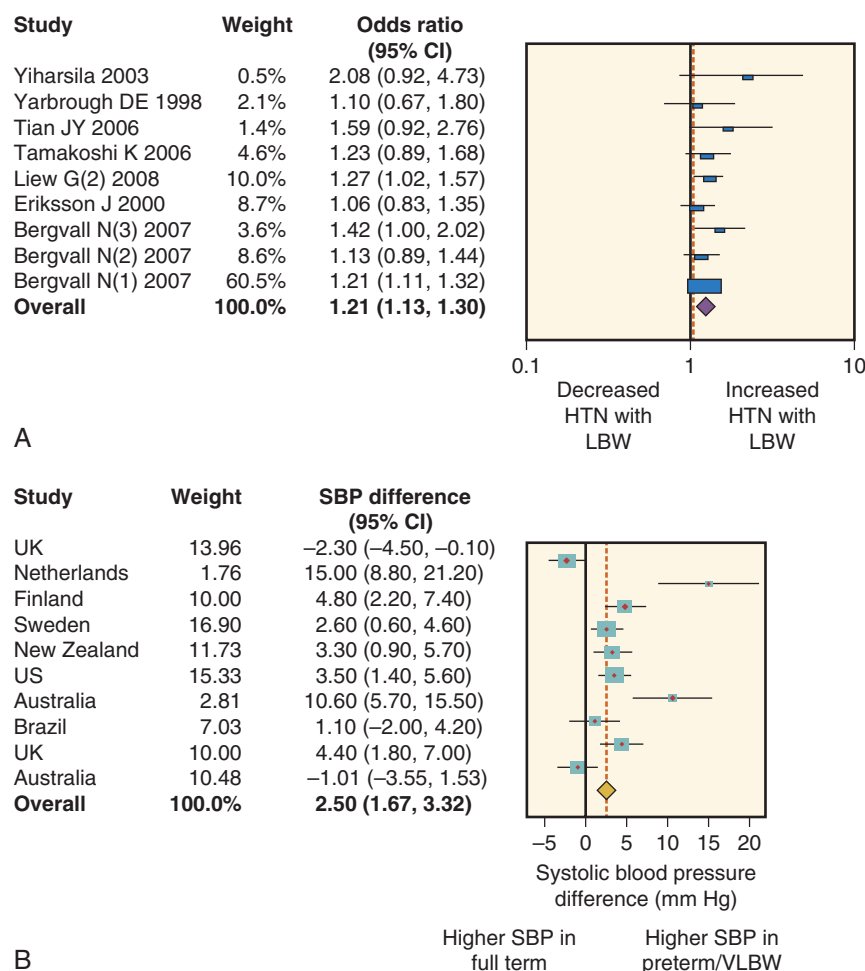


Fig. 21.8 Relationship between birth weight, preterm birth, and blood pressure. (A) Meta-analysis of odds for hypertension (HTN) in individuals with birth weights below 2500 g (low birth weight [LBW]) compared with birth weights above 2500 g. The pooled odds ratio is shown as a *diamond*. (From Mu M, Wang SF, Sheng J, et al. Birth weight and subsequent blood pressure: a meta-analysis. *Arch Cardiovasc Dis*. 2012;105:99–113; see original paper for full references). (B) Meta-analysis of difference in systolic blood pressure (SBP) between individuals born preterm or very low birth weight (VLBW) compared with full term. Pooled SBP difference is indicated by the *diamond* and *dashed vertical line*. (From de Jong F, Monuteaux MC, van Elburg RM, et al. Systematic review and metaanalysis of preterm birth and later systolic blood pressure. *Hypertension*. 2012;59:226–234; see original paper for full references).

future potential for prevention, but, given that effect estimates for risk of higher blood pressures were similar in the meta-analyses and systematic reviews cited earlier, at present, both events must be deemed important risk factors for subsequent high blood pressure.

Importantly, blood pressures of low- and normal-birth-weight subjects, although different, may still be within the normal range in childhood, but differences become amplified with age, such that adults who had been of low birth weight often develop overt hypertension, which increases with age.¹⁹⁶ Although the majority of studies have been conducted in Caucasian populations, generally consistent data are accumulating in other populations.¹⁹⁷ An association of higher blood pressure with lower birth weight in African-American children has been reported in some studies, but not all, suggesting additional factors may contribute to the greater severity of blood pressure in those of African origin.^{197–202} An important effect modifier of the association with low birth weight or preterm birth and blood pressure, noted in diverse populations, is current body mass index, which may override an effect of birth weight,

especially in children at different stages of growth.^{197,203} Furthermore, in most populations, blood pressures are highest in those born preterm or of low birth weight who “catch up” fastest in postnatal weight (i.e., rapid upward crossing of weight centiles), highlighting the importance of early postnatal nutrition in developmental programming.^{204–208}

Elevated blood pressure associated with low birth weight was recently reported in a multiethnic cohort of adolescents.²⁰⁹ From the National Health and Nutrition Examination Survey (NHANES) study, 5352 participants aged 12–15 years were classified as low birth weight, very low birth weight, or normal birth weight by parental/proxy recall. Low/very low-birth-weight adolescents had a greater odds of having a blood pressure ≥ 95 th percentile for age, height, and sex [Low birth weight: OR: 2.90; 95% CI: 1.48 to 5.71; very low birth weight: OR 5.23; 95% CI: 1.11 to 24.74]. Whether preterm birth and/or small for gestational age contributed to the low birth weight in the population was not known.

Associations between blood pressure and other markers of potential developmental stresses have also been reported.

A meta-analysis of 31 studies found that blood pressures were also higher in children who had high birth weights; however, blood pressures tended to be lower in high-birth-weight adults, suggesting that age may modify this risk differently compared with that in low-birth-weight subjects in whom it increases with age (Table 21.8).^{39,210} Additionally, a systematic review and meta-analysis investigating the impact of a diabetic pregnancy on blood pressure found an overall association with higher blood pressure in offspring aged 2–20 years, but this effect was only seen in males (Table 21.8).²¹¹ The authors did not discuss the potential impact of birth weight in this study, however, and whether these effects may have been modified by offspring of high birth weight was not reported. Another potential risk factor for higher offspring blood pressure is maternal gestational hypertension or preeclampsia.^{212,213} Indeed, in a systematic review of 18 studies among children and young adults, systolic blood pressures were 2.39 mm Hg (95% CI: 1.74 to 3.05) higher in young adults who had been exposed to preeclampsia compared with those not exposed (Table 21.8).²¹⁴ Whether the effect is mediated by the often accompanying fetal growth restriction or preterm birth or may be associated with circulating antiangiogenic factors or other humoral changes in preeclampsia requires further investigation. In a recent study of young adults (males and females) born preterm, soluble endoglin and soluble fms-like tyrosine kinase-1 (sflt-1) levels were significantly elevated ($P < .001$ for both) compared with controls born at term, and proportional to current systolic blood pressures, suggesting a programming effect of preterm birth on angiogenesis and blood pressure.²¹⁵ Interestingly, sflt-1 levels in this study were even further elevated in those who had been preterm and exposed to a hypertensive pregnancy ($P < .002$), suggesting an additional impact of preeclampsia. Having been born small for gestational age or preterm, in turn, are risk factors for a woman subsequently developing preeclampsia in her own pregnancies, emphasizing the intergenerational effects of developmental programming.^{33,190} Whether the programmed risk for preeclampsia is mediated by elevated circulating antiangiogenic factors and/or the programmed risk of higher blood pressures and renal dysfunction resulting from a mother's own low birth weight or preterm birth status is not known.

Gender differences in programming effects on blood pressure have been inconsistently reported. In some studies, programming effects appear more pronounced in males, and in others, the differential effects of gender are modified by age, ethnicity, and body mass index.¹⁹⁷ In a meta-regression of 20 Nordic cohorts, including 183,026 males and 14,928 women, a linear inverse association between birth weight and systolic blood pressure was present across all birth weights in males, which strengthened with age, whereas the relationship was U-shaped in women, with increasing risk also observed with birth weights above 4 kg.²¹⁶ Potential mechanisms whereby developmental programming may be expressed differently in males and females will be discussed later and are reviewed in detail elsewhere.^{159,170}

The relative importance of genetics and environmental factors in programming of blood pressure has been studied in twins.^{217–219} In a large Swedish cohort of 16,265 twins, the overall adjusted OR for hypertension was 1.42 (95% CI: 1.25 to 1.62) for each 500-g decrease in birth weight. Within like-sexed twin pairs, the ORs were 1.43 (95% CI: 1.07 to

1.69) and 1.74 (95% CI: 1.13 to 2.70) for dizygotic and monozygotic pairs, respectively, suggesting that environmental factors that contributed to differences in birth weight had a greater impact than genetics in this cohort, consistent with a developmental programming effect.²¹⁷

Nephron Number and Blood Pressure

In support of the potential association between nephron number and hypertension, a study of Caucasians aged 35 to 59 years who died in accidents found that in 10 subjects with a history of essential hypertension, the number of glomeruli per kidney was significantly lower and glomerular volume significantly higher than in 10 normotensive-matched controls (Fig. 21.9).⁵⁹ Birth weights were not reported in this study, but the authors concluded that a reduced nephron number is associated with susceptibility to essential hypertension. Similarly, kidneys of Australian Aborigines with a history of hypertension contained approximately 30% fewer nephrons than Aborigines with no history of hypertension.²²⁰ Although the sample size was small and birth weights were not available, this is a population with high rates of socioeconomic disadvantage and low birth weight. Kanzaki et al.²²¹ also recently reported that kidneys from Japanese men with a history of hypertension contained approximately 40% fewer nephrons than age-matched normotensive men. The data therefore seem consistent across several populations that higher blood

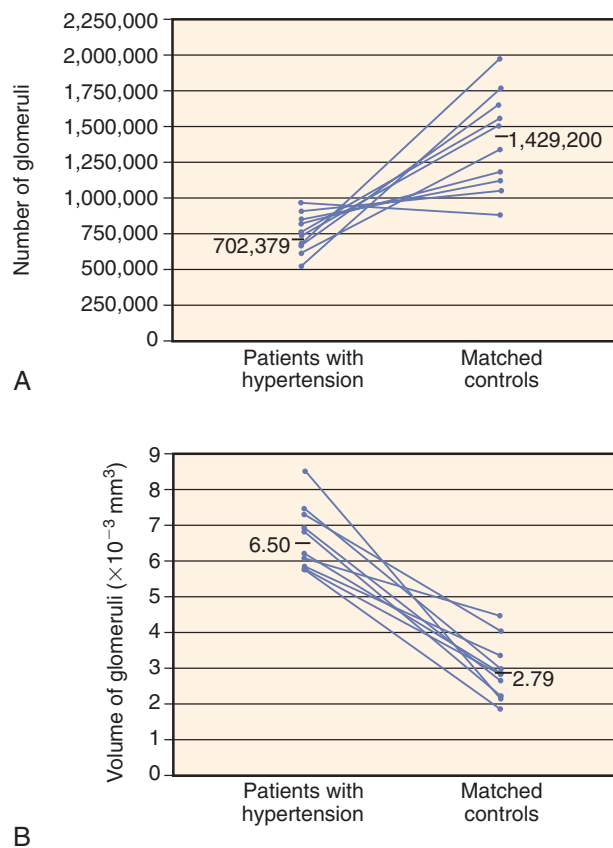


Fig. 21.9 (A) Nephron number, and (B) glomerular volume in Caucasian subjects with primary hypertension compared with controls. (From Keller G, Zimmer G, Mall G, et al. Nephron number in patients with primary hypertension. *N Engl J Med.* 2003;348:101–108.)

Table 21.8 Systematic Reviews, Metaanalyses, and Population-Based Studies of Developmentally Programmed Associations With Blood Pressure and Kidney Disease

Reference	Condition	Age	Number Included in the Analysis	Study Type	Outcome	Risk (95% CI)
Programmed Associations With Blood Pressure						
36	Preterm birth ^a	17.8 years	3080 subjects	Systematic review and meta-analysis	SBP	Increase by 2.5 mm Hg (1.7–3.3 mm Hg) vs. term birth
37	Low birth weight	4–84 years	20 studies	Meta-analysis	Hypertension SBP	OR 1.21 (1.13–1.3) Increase by 2.28 mm Hg (1.24–3.33 mm Hg) for BW <2500 g vs. BW >2500 g
	High birth weight				SBP	Decrease by 2.08 mm Hg (–2.98 to –1.17 mm Hg) for BW >4000 g vs. BW <4000 g
39	High birth weight	4–83 years	31 studies	Meta-analysis	Hypertension, children Hypertension, adult Overall	RR 1.18 (95% CI 1.05–1.32) RR 0.97 (95% CI 0.86–0.97) RR 1.0 (95% CI 0.93–1.06)
552	Birth weight	0–84 years	444,000 subjects	Systematic review	SBP	Decrease by 2 mm Hg per kg increase in BW
	Catch-up growth				SBP	Increase with catch-up growth Highest blood pressure values in LBW after catch-up
214	Offspring of pregnancy complicated by preeclampsia	Children, young adult	45,249 subjects	Systematic review	SBP DBP BMI SBP	Increase by 2.39 mm Hg (1.74–3.05 mm Hg) Increase by 1.35 mm Hg (0.9–1.8 mm Hg) Increase by 0.62 kg/m ²
211	Pregnancy complicated by diabetes	2–17 years	61,852 individuals	Systematic review and meta-analysis	SBP	Increase (0.47–3.28 mm Hg) in ODM (only statistically significant in males when stratified by sex)
217	Genes vs. environment	Birth years 1926–1958	16,265 subjects 595 subjects 250 subjects	Same-sex twin study, all Dizygotic pairs ^b Monozygotic pairs ^b	Hypertension Hypertension Hypertension	OR 1.42 (1.25–1.69) per 500-g decrease OR 1.34 (1.07–1.69) per 500-g decrease OR 1.74 (1.13–2.70) per 500-g decrease
293	Fetal and/or infant exposure to famine	37–43 years	1,339 subjects	Cohort, Biafra famine, Nigeria (1967–1970)	SBP Hypertension	Increase in exposure OR 2.87 (1.9–4.34) in exposure
295		59 years	971 subjects	Cohort, Dutch famine (1944–1945)	SBP DBP Hypertension	Increase (0.25–5.30 mm Hg) after ≥10 weeks' exposure Not significant OR 1.44 (1.04–2.00) after ≥10 weeks' exposure
238		48–53 years	724 subjects	Cohort, Dutch famine (1944–1945)	Albuminuria	OR 3.2 (1.4–7.7) for exposure to famine in midgestation
553		52–53 years	549 subjects	Cohort, siege of Leningrad (1941–1944)	SBP, DBP SBP Albuminuria	Not significant Not significant Not significant

Continued on following page

Table 21.8 Systematic Reviews, Metaanalyses, and Population-Based Studies of Developmentally Programmed Associations With Blood Pressure and Kidney Disease (Cont'd)

Reference	Condition	Age	Number Included in the Analysis	Study Type	Outcome	Risk (95% CI)
Programmed Associations With Kidney Disease						
32	Low birth weight	12–75 years	46,249 subjects	Systematic review	Chronic kidney disease	OR 1.73 (1.44–2.08)
					End-stage renal disease	OR 1.58 (1.33–1.88)
					Albuminuria	OR 1.81 (1.19–2.77)
					Reduced glomerular filtration rate	OR 1.79 (1.31–2.45)
274	Low birth weight	<21 years	1994 cases	Case-control study	Childhood chronic kidney disease ^c	OR 2.88 (2.28–3.63)
	Maternal GDM		20,032 controls			OR 1.54 (1.13–2.09)
	Maternal overweight					OR 1.24 (1.05–1.48)
	Maternal obesity					OR 1.26 (1.05–1.52)
	High birth weight					Not significant

^aMean gestational age 30.2 weeks, mean birth weight 1280 g.
^bTwin pairs discordant for hypertension.
^cChronic kidney disease definition included reduced renal function, renal dysplasia and/or aplasia, and obstructive uropathy.
BMI, Body mass index; CI, confidence interval; DBP, diastolic blood pressure; GDM, gestational diabetes mellitus; ODM, offspring of diabetic mother; OR, odds ratio; RR, relative risk; SBP, systolic blood pressure.
From Luyckx VA, Brenner BM. Birth weight, malnutrition and kidney-associated outcomes—a global concern. *Nat Rev Nephrol.* 2015;11:135–149.

pressures are associated with lower nephron numbers. These studies attempted to exclude loss of nephrons due to hypertension as a potential confounder of the association in the absence of birth weight data.

Interestingly, among a subset of 63 subjects in whom mean arterial pressures and birth weights were available, Hughson and coworkers¹²⁰ reported a significant correlation between birth weight and glomerular number, mean arterial pressure, and glomerular number as well as mean arterial pressure and birth weight among Caucasian but not African American subjects. Among African Americans having nephron numbers below the mean, however, twice as many were hypertensive as normotensive, suggesting a possible contribution of lower nephron number in this group as well.¹²⁰ The relationship of low birth weight and nephron number was similar in a cohort of black and white Cuban neonates; therefore, it is expected that a similar relationship between low birth weight and low nephron number exists in the black population.⁹¹ Glomerular volumes were found to be higher among the hypertensive African American subjects than the hypertensive whites.¹²⁰ The consistent finding of larger glomeruli among African Americans may suggest a greater prevalence of low nephron number in this population as a result of the known higher prevalence of low birth weight, or may reflect independent or additional programming of glomerular size. This topic warrants further research.

Consistent with an association between nephron number and blood pressure in humans, salt sensitivity has been found to correlate inversely with birth weight and inversely with kidney size in adults and children (Fig. 21.10).^{142,143} In both studies, salt-induced changes were independent of GFR, excluding confounding by renal function. Among 1512 subjects aged 62 years, investigators found an inverse associa-

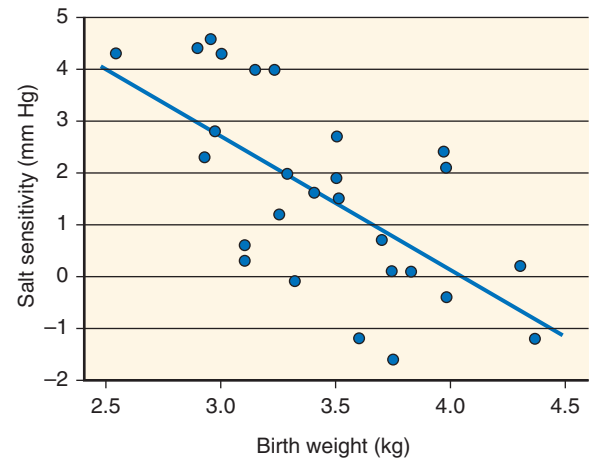


Fig. 21.10 Correlation between birth weight and salt sensitivity in 27 normotensive adults. Salt sensitivity of blood pressure was defined as the difference of mean arterial pressure on high-salt diet (200 mmol/d) compared with low-salt diet (60 mmol/d). $R = -0.06$; $P = .002$. (From de Boer MP, Ijzerman RP, de Jongh RT, et al. Birth weight relates to salt sensitivity of blood pressure in healthy adults. *Hypertension* 51:928–932, 2008.)

tion between birth weight and blood pressure among those with birth weights below 3050 g, with a progressive 2.48-mm Hg (95% CI: 0.4 to 4.52 mm Hg) increase in blood pressure for every 1-g rise in daily salt intake until 10 g/d.²²² Interestingly, above this threshold birth weight, there was no association between blood pressure and salt intake, potentially suggesting protection against salt sensitivity with higher birth weights.

BIRTH WEIGHT, PRETERM BIRTH, AND KIDNEY FUNCTION

As with blood pressure, programmed changes in renal function that occur, at least in the early stages, may not be outside of normal limits. With time or exposure to additional insults, however, these changes may manifest as kidney disease.

Glomerular Filtration Rate

The GFR, in the absence of compensatory hyperfiltration, should reflect the filtration surface area and, therefore, nephron number. Compensatory hyperfiltration is thought not to occur in the immediate neonatal period, and therefore, measurement of neonatal GFR may be a good proxy for nephron endowment. Consistent with this, amikacin clearance on day 1 of life, measured as a correlate for neonatal GFR, was found to be significantly lower in low-birth-weight and preterm neonates than in term controls.²²³ Similarly, in a cohort of very preterm children aged 7.6 years, although still within the normal range, GFR measured by inulin clearance was significantly lower among those who had been growth restricted compared with non-growth restricted perinatally.²²⁴ Importantly, the GFR was lower among children who had been growth restricted before (in utero) or in the first weeks after birth (in intensive care), pointing also to the role of postnatal nutrition in renal programming. Several studies in children found similar associations of decreased GFR in low-birth-weight or preterm children; however, studies using creatinine-based formulas may underestimate the impact of birth weight on GFR, suggesting a need to validate measures of renal function in low-birth-weight and preterm subjects, whose body composition may remain different over time.^{223,225–231} The relationship between birth weight and eGFR appears to track with time, suggesting consistency of the associations. Among 5352 adolescents from the NHANES study, eGFRs were lower in low-birth-weight (OR: 1.49; 95% CI: 1.06 to 2.10) and very low-birth-weight (OR: 2.49; 95% CI: 1.20 to 5.18) subjects compared with those with normal birth weight.²⁰⁹ As discussed earlier, this study is the first population-based assessment among adolescents of multiple ethnicities permitting an estimation of the population-attributable fraction of birth weight to blood pressures and eGFR: 1 in 13 low-birth-weight or 1 in 5 very low-birth-weight adolescents had systolic blood pressures $\geq$ 95th percentile and/or eGFR < 90 mL/min/1.73 m².²⁰⁹

Overall, a recent metaanalysis found an OR of 1.79 (95% CI: 1.31 to 2.45) for a reduced GFR with low birth weight (see Table 21.8).³² Linear regression analysis in a cohort of 2192 British adults aged 60–64 years revealed that for each 1-kg decrease in birth weight, adult GFR estimated using cystatin C was reduced by 2.25 mL/min per 1.73 m² (95% CI: 0.69 to 3.58 mL/min per 1.73 m²).²³² Taken together, these findings are consistent with low birth weight and preterm birth being risk factors for a reduced GFR. The relative contributions of genetics and the fetal environment on programming of renal function were investigated in 653 twins.²³³ Creatinine clearance was significantly lower in low-birth-weight compared with normal-birth-weight twins. Furthermore, intrapair birth weight differences were positively correlated with GFR in both monozygotic and dizygotic twin pairs, suggesting that feto-placental factors have a greater impact than genetic factors on adult renal function.

To approach an understanding of the mechanism of the reduced GFR in low-birth-weight and preterm individuals, renal functional reserve, determined by measuring GFR and effective renal plasma flow (ERPF) before and after low-dose dopamine infusion or an oral amino acid load, was studied in 20-year-old subjects who had been preterm and either small or appropriate for gestational age, versus controls, born full term and normal birth weight.¹⁹² After renal stimulation, the relative increase in GFR tended to be lower in small-for-gestational-age compared with appropriate-for-gestational-age and control subjects, and ERPF was lower in both groups of preterm subjects, although statistical significance was not reached, likely because of small numbers. Reduced renal functional reserve was also observed among young adults with type 1 diabetic mothers, who had been exposed to diabetes during gestation, but not those with diabetic fathers, again suggesting a programming rather than genetic effect.⁷² A reduced renal reserve capacity in these subjects may be consistent with a programmed reduction in nephron number.

Proteinuria

One of the earliest signs of hyperfiltration, which would be expected in the setting of a reduced nephron number and filtration surface area, is microalbuminuria, which may progress to overt proteinuria with ongoing renal injury and worsening hyperfiltration. Consistent with this hypothesis, Aboriginal Australians who had low birth weights evidenced an OR for macroalbuminuria of 2.8 (95% CI: 1.26 to 6.31) compared with those who had normal birth weights, and this increased with age.²³⁴ Importantly, proteinuria was also associated with a higher rate of cardiovascular and renal deaths, underscoring its clinical relevance.⁹⁷ A meta-analysis including eight additional studies reported an OR of 1.81 (95% CI: 1.19 to 2.77) for albuminuria with low birth weight (see Table 21.8).³² More recently, the Australian group reported that low birth weight, childhood poststreptococcal glomerulonephritis, and current body mass were all independent predictors of albumin to creatinine levels in young Aboriginal adults.^{235,236} These findings are compatible with the “multihit” model of CKD, of which nephron endowment at birth may be the first “hit” increasing susceptibility to kidney disease throughout the life course.

As with blood pressure, whether preterm birth modifies the association of birth weight with proteinuria is not always easy to dissect, although studies of preterm children and adolescents show consistent findings. Among children aged 4 years who had been preterm, albuminuria was higher in both boys and girls who had reached normal height (presumably caught up in growth), and among 19-year-olds who had been very preterm, albuminuria was higher among those who had been growth restricted, again highlighting the interplay between preterm birth, growth restriction, and catch-up growth on later risk of disease.^{230,237} In contrast, however, among 12- to 15-year-old NHANES participants, despite there being significant differences in blood pressure and eGFRs, there was no difference in the albumin/creatinine ratio between adolescents born with low birth weight, very low birth weight, or normal birth weight.²⁰⁹ This unexpected finding might be due to the fact that albuminuria was determined on a single random, nonsupine sample, which may lack the specificity needed to distinguish patients with true kidney disease.

Analysis of 724 subjects aged 48 to 53 years who had been subjected to malnutrition in midgestation during the Dutch famine revealed an increased prevalence of microalbuminuria (12%) when compared with those subjected to malnutrition during early gestation (9%), late gestation (7%), or not exposed to famine (4%–8%) (see Table 21.8).²³⁸ Size at birth was not associated with the observed increase in microalbuminuria, however, suggesting that renal development may have been irreversibly affected in midgestation, although in later gestation, whole-body growth was able to catch up with restoration of more normal nutrition. This observation again emphasizes the need for surrogate markers in addition to birth weight in order to identify individuals at risk for renal programming.

A U-shaped association between birth weight and proteinuria was described among Pima Indians, showing that the risk increased for birth weights below 2.5 kg and above 4.5 kg.¹² The strongest predictor of proteinuria among high-birth-weight subjects in this study was exposure to gestational diabetes, raising the question of whether gestational diabetes exposure, rather than birth weight per se, was the predominant programming risk factor.¹² In a Canadian study, urine albumin/creatinine ratios were lower in infants of diabetic mothers compared with nondiabetic mothers at 1 year of age, but were higher at 3 years, although independent of birth weight.²³⁹ The authors interpret these findings to reflect abnormal renal programming in offspring of diabetic mothers; however, the effects of gestational diabetes and high birth weight on renal programming require much more study.

Neonatal Acute Kidney Injury

Preterm birth is an important risk factor for acute kidney injury (AKI) in neonates, estimated to occur in between 12.5% and 71%, depending on the population studied.^{240,241} In turn, AKI in very low-birth-weight infants is an independent predictor of longer hospital stays, mortality, and subsequent CKD.^{187,242} A retrospective analysis of preoperative renal volume in neonates undergoing congenital heart surgery found higher peak postoperative creatinine values and, therefore, potentially increased risk of AKI in infants with a renal volume ≤ 17 cm³.²⁴³ High creatinine values were associated with lower gestational age and lower birth weight z-score. Neonates in intensive care are particularly susceptible to renal dysfunction, not only because of potentially programmed risk and low nephron number but also because of critical illness and frequent nephrotoxin exposure such as aminoglycosides and nonsteroidal antiinflammatory drugs.^{122,241,244–248} Medications administered during pregnancy or prior to delivery, such as tocolytics and antibiotics, may also impact fetal nephrogenesis and increase the risk of neonatal AKI.^{249–251} These factors all likely adversely impact any postnatal nephrogenesis, which may occur under optimal circumstances for several weeks following preterm birth.¹²²

The actual risk of AKI with exposure to nephrotoxins among preterm infants is not well described. Among 269 infants perinatally exposed to potential nephrotoxic medication (i.e., in late pregnancy or within the first 7 days postbirth), ibuprofen exposure was associated with an OR of 2.6 (95% CI: 1.2 to 5.3) for a reduced GFR on day 7 of life, which persisted over the first month.²⁵⁰ Other authors found that exposure to aminoglycosides was associated with higher serum creatinine in preterm infants born small for

gestational age at 2 months.²⁵² Neonatologists are working to raise awareness of the renal risk; however, the definition of renal failure in this population remains a challenge. Creatinine cutoff values have been proposed according to gestational age.²²⁷ Serum creatinine in the neonate reflects maternal creatinine early on, however; therefore, urinary biomarkers and cystatin C have been proposed as superior markers to detect AKI early.^{244,253} More recently, the Neonatal Kidney Disease: Improving Global Outcomes Classification has been proposed for more consistency in the definition.²⁴⁴ The risk and significant adverse consequences of AKI associated with preterm birth therefore highlight the need for long-term follow-up of preterm infants.²⁴⁰ Studies are emerging linking perinatal AKI to the long-term risk of CKD, which is significantly increased among those who subsequently become obese.^{187,254–257} Prevention of perinatal AKI may therefore be important to reduce the risk of later-life CKD, and long-term follow-up of infants who develop AKI is required to appropriately modify this risk.^{242,258}

Chronic Kidney Disease and End-Stage Kidney Disease

A case series of six patients, aged 15–52 years, who had been born preterm with very low birth weight, described findings consistent with secondary FSGS, associated with glomerulomegaly in all biopsies.²⁵⁹ The authors suggest a susceptibility to hyperfiltration and glomerulosclerosis associated with preterm birth and low birth weight. Similar histologic findings have been reported in several Japanese individuals who had very low birth weights and developed early renal dysfunction or proteinuria.^{260,261} These individuals responded well to ACEI therapy, supporting the pathophysiologic role of hyperfiltration in developmental programming of renal disease. Consistent with these findings, a variety of generally small studies have reported a greater severity of renal disease and more rapid progression of diverse renal diseases, including immunoglobulin A (IgA) nephropathy, membranous nephropathy, minimal change disease, chronic pyelonephritis, Alport syndrome, and polycystic kidney disease among children and adults who had been of low birth weight.^{41,262–270} A handful of studies have examined the relationship between birth weight and diabetic nephropathy and found an increased susceptibility among subjects who had been growth restricted, although not invariably.^{12,265,271,272}

Strong associations have recently been reported between both low birth weight (defined as <2.5 kg) and gestational age with the development of CKD in Japanese and North American children attending pediatric CKD clinics.^{273,274} In the U.S. population, low birth weight, maternal gestational diabetes, and maternal overweight and obesity were all significantly associated with an increased risk of childhood CKD (see Table 21.8).²⁷⁴ In addition, a significant association was found between low birth weight and maternal pregestational diabetes and renal dysplasia/aplasia, and between low birth weight, maternal gestational diabetes, and maternal overweight or obesity and congenital obstruction. In the Japanese population, the association remained highly significant, even after exclusion of children with congenital anomalies of the kidney and urinary tract (CAKUT), with an estimated population attributable fraction of pediatric CKD being 21.1% (95% CI: 16.0% to 26.1%) for low birth weight and 18.2% (95% CI: 16.5 to 25.6%) for preterm birth.²⁷³

Table 21.9 Risk of End-Stage Renal Disease According to Birth Weight and Gestational Age Category^a

	LBW (BW <10th percentile)	LBW (<2.5 kg)	SGA (<37 weeks)	Preterm (<37 weeks)	Term LBW	Preterm LBW	Term SGA	Preterm AGA	Preterm SGA
All ages	1.63 (1.29–2.06)	2.25 (1.59–3.19)	1.67 (1.3–2.07)	1.36 (0.94–1.99)	1.56 (1.18–2.07)	1.89 (1.25–2.86)	1.54 (1.2–1.96)	1.09 (0.69–1.73)	4.03 (2.08–7.80)
1–18 y	2.72 (1.88–3.92)		1.93 (1.28–2.91)						
18–42 y	1.23 (0.9–1.68)		1.53 (1.15–2.03)			1.42 (0.82–2.48)	1.41 (1.05–1.90)		4.02 (1.79–9.03)

^aAll comparisons for term birth, LBW term, AGA term as relevant. Data expressed as hazard ratios (95% confidence intervals).

AGA, Appropriate for gestational age; BW, birth weight; LBW, low birth weight; SGA, small for gestational age.

Reproduced from Low Birth Weight and Nephron Number Working Group. The impact of kidney development on the life course: a consensus document for action. *Nephron*. 2017;136:3–49.

Derived from Ruggajo P, Skrunes R, Svarstad E, et al. Familial factors, Low birth weight, and development of ESRD: a nationwide registry study. *Am J Kidney Dis*. 2016;67(4):601–608.

Taken together, most observations suggest that low birth weight and preterm birth are risk factors for renal disease. Consistent with this notion, a meta-analysis of 18 studies reported an OR for CKD of 1.73 (95% CI: 1.44 to 2.08) with low birth weight (see Table 21.8).³² Similarly, in retrospective analysis of a cohort of over 2 million white children, the relative risk of developing ESKD was 1.7 (95% CI: 1.4 to 2.2) in those with birth weights <10th percentile.³¹ In a follow-up study, these investigators showed a stronger relationship between low birth weight and ESKD in individuals under 18 years (OR: 2.72; 95% CI: 1.88 to 3.92) compared with 18–42 years of age (OR: 1.23; 95% CI: 1.15 to 2.03), again demonstrating the likely impact of CAKUT in the younger population (Table 21.9).⁴¹ Overall in this study, both low birth weight and being small for gestational age were associated with a progressively increasing risk of ESKD (Fig. 21.11). There was no evidence for modulation of these associations by familial factors, again emphasizing the strong impact of environmental exposures in renal programming.⁴¹ In a dialysis-based study, low birth weight was also associated with increased risk of ESKD, but the OR for diabetic ESKD was also increased among those having birth weights greater than 4000 g (OR 2.4; 95% CI: 1.3 to 4.2).⁵⁰ The relevance of high birth weight was also highlighted by a U-shaped association of renal disease with birth weight found in two large population-based studies, although effects differed between males and females in both studies, again reflecting potential effect modification by gender under conditions as yet not fully elucidated.^{30,31} Whether high birth weight in these studies was associated with intrauterine exposure to diabetes is not known.

Among 1850 Pima Indians younger than 45 years with diabetes, the incidence rate ratio of developing ESKD was 4.12 (95% CI: 1.54 to 11.02) among those who had been exposed to diabetes in utero compared with those without exposure.²⁷⁵ Interestingly, this effect disappeared after controlling for duration of diabetes in this relatively young cohort, suggesting the ESKD risk may be mediated predominantly by a programmed risk for earlier-onset diabetes among those exposed to diabetes in utero. This study emphasizes that programming of renal risk may be indirect as a consequence of programming of other disorders such as diabetes, under-

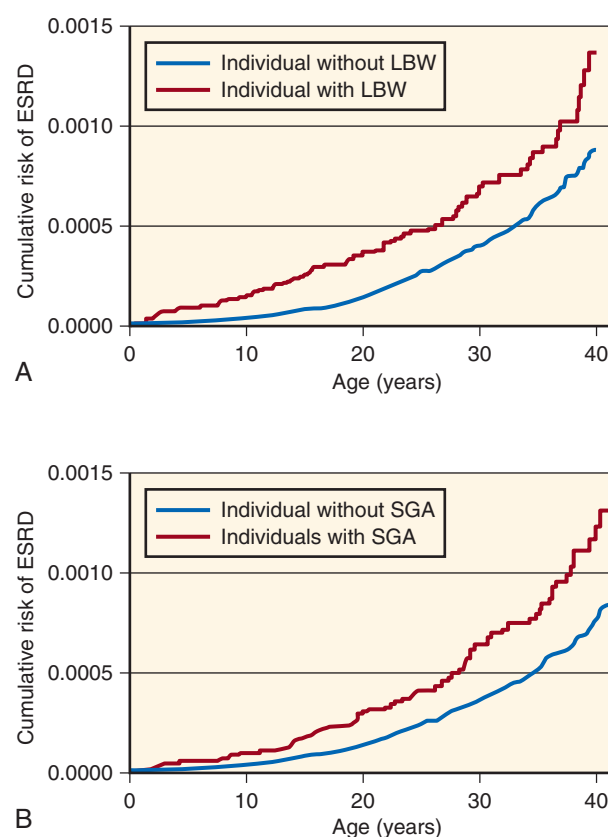


Fig. 21.11 Cumulative risk for end-stage renal disease (ESRD) in individuals with age, according to whether the individual was (A) low birth weight (LBW) or (B) small for gestational age (SGA). From Ruggajo P, Skrunes R, Svarstad E, et al. Low birth weight, and development of ESRD: a nationwide registry study. *Am J Kidney Dis*. 2016;67:601–608.

scoring the need for a holistic and life-course approach to understand and tackle programming of renal disease.

It is likely too simplistic to assume that altered kidney development associated with low birth weight, preterm birth, or other developmental stresses itself is enough to cause renal disease, but exposure to additional “hits” (e.g., nephrotoxin exposure, AKI, glomerulonephritis), or superimposed

developmental programming of conditions that are themselves risk factors for kidney disease (i.e., diabetes, cardiovascular disease, metabolic syndrome, obesity), all likely exacerbate renal risk.^{16,19,187,235,275,276} Overall, combined meta-analysis of 31 studies, including over 2 million subjects, concluded that individuals of low birth weight have a 70% increased risk of developing CKD, including albuminuria, reduced GFR, and renal failure in later life.³² Clinical associations of developmental programming in the kidney are outlined in Table 21.10.

RELATIVE IMPACT OF BEING BORN SMALL OR PRETERM ON RENAL PROGRAMMING

As illustrated in Fig. 21.1, a low birth weight may be the result of preterm birth and/or growth restriction. Most studies have not differentiated between these two occurrences, so the relative impact of each has remained unclear. A recent Norwegian population study including data from >1.8 million subjects under age 40 years has examined these effects, using the following definitions: low birth weight <10th percentile for Norway (<2800 g), small for gestational age <10th percentile for gestational age (determined by ultrasonography in gestational weeks 17–20), and preterm birth <37 weeks' gestation (see Table 21.9).⁴¹ Overall, the hazard ratio (HR) for ESKD was 1.56 (95% CI: 1.18 to 2.07) for low-birth-weight term births, 1.89 (95% CI: 1.25 to 2.86) for low-birth-weight preterm births, 1.54 (95% CI: 1.20 to 1.96) for term small-for-gestational-age births, and 4.03 (95% CI: 2.08 to 7.80) for preterm small-for-gestational-age births (Fig. 21.11).⁴¹ Preterm birth with appropriate weight for gestational age was not associated with an increased risk of ESKD in this study. Being born small for gestational age was therefore the strongest predictor of ESKD risk. Subsequently, the same authors reported that being small for gestational age and low birth weight were associated with an increased risk of ESKD among a cohort of young adults with IgA nephropathy.²⁷⁷ The IgA subgroup was chosen because it is a common disease and unlikely to be directly impacted by programming (in contrast to CAKUT or diabetic nephropathy). Low birth weight and small for gestational age were associated with increased risk of progression to ESKD in IgA nephropathy.⁴¹ This study again points to developmental programming being a first “hit” leading to accelerated progression of a primary renal disease. Again, in this study, the highest risk for ESKD was conferred by a small birth weight for gestational age (HR: 2.2; 95% CI: 1.1 to 4.2); however, this effect was only significant in males. Patients who were both low birth weight and small for gestational age had the highest risk (HR: 3.2; 95% CI: 1.5 to 6.8), suggesting a greater impact at lower birth weights, likely reflecting more adverse intrauterine exposures.

POTENTIAL IMPACT OF ETHNICITY ON RENAL PROGRAMMING

The nephron number hypothesis was initially put forward as a potential explanation for the disproportionately higher risks of hypertension and kidney disease among disadvantaged populations.⁴⁸ Hypertension, CKD, and ESKD prevalences tend to be highest among Aboriginal Australians, Native Americans, people of African origin, and ethnic minorities in developed countries.^{10,12,50,278} The incidence of birth circumstances and maternal risk factors for renal programming are high across low- and middle-income countries, as highlighted in Table

21.11, suggesting a potentially significant population impact. Most studies regarding programming of blood pressure and kidney disease have thus far been conducted in western Caucasian populations and may not be generalizable to all populations. In the Norwegian population, for example, only 3.3% of subjects had birth weights below 2.5 kg, whereas around 13%–15% of babies in sub-Saharan Africa have low birth weights.^{41,279} Furthermore, the causes of low birth weight or preterm birth differ across regions, with maternal nutrition, poverty, and infections being important in lower-income regions and advanced maternal age, multiple gestations, and use of assisted reproduction technology predominating in high-income regions.^{280–282} How such differences may modulate programming risk across populations is not known.^{283–286}

Kidney size, as measured by ultrasound at birth, among 715 South Asians (Pakistani, Indian, and Bangladeshi origin) from the Born in Bradford (United Kingdom) cohort study was smaller than that among 872 white babies, and the difference persisted after adjustment for potential confounders, including birth weight.²⁸⁷ Total kidney volume was also found to be lower in low-birth-weight and preterm Australian Aborigines at 32 and 38 weeks of gestation and at term compared within nonindigenous Australians, suggesting a lower nephron number in the indigenous children.²⁸⁸ However, in this study, calculated GFR (serum cystatin C) was similar in Aboriginal and nonindigenous neonates, suggesting SNGFR was increased in the Aborigines. In a study comparing eGFR in 152 African and white American children at a mean age of 1.5 years, the eGFR was found to be significantly lower among African American children (82 vs. 95 mL/min/1.73 m²).²⁸⁹ Birth weight was significantly and positively associated with eGFR in African American but not white children, suggesting that some of the racial disparities in adult CKD may have origins in the prenatal period.

Nephron numbers have thus far been counted only in small cohorts of Caucasians, African Americans, Australian Aborigines, Senegalese Africans, and Japanese males, in many of whom birth weights were not available. Therefore, the relationship between birth weight and nephron number remains unknown in most populations. A recent study on nephron number and hypertension among older Japanese males found the nephron count in the normotensive subjects was one of the lowest reported to date (~640,000 nephrons), suggesting that Japanese people may be especially vulnerable to adult hypertension and CKD, particularly if a “second hit” occurs.²⁹¹ This hypothesis would be consistent with the fact that Japan has the second-highest incidence of ESKD in the world.²⁹⁰ Low birth weight is strongly associated with reduced nephron numbers and susceptibility to kidney disease among Australian Aborigines, a population that is also significantly disadvantaged compared with their white counterparts.^{96,97} The association between nephron number and birth weight does appear to be consistent among subjects of African origin as in Caucasian subjects.^{87,91,291,292} The relationship between low nephron numbers and higher blood pressures seen in Aboriginal Australians and Caucasians, however, was not as consistent among African Americans.²⁸⁵ Similarly, the association of low birth weight with blood pressure is also less consistently reported among African-American compared with Caucasian children; therefore, additional factors likely augment blood pressure risk in the African-origin population.^{198,289}

Evidence for developmental programming of blood pressure in an African population has, however, been shown among

Table 21.10 Clinical Associations of Renal Programming

	Low Birth Weight	Small for Gestational Age	Preterm Birth	High Birth Weight	Gestational Diabetes Exposure	Exposure to Preeclampsia/Eclampsia	Maternal Overweight/Obesity	Maternal Vitamin A Deficiency	Rapid Catch-Up Growth/Overweight	Low Nephron Number	Smaller Kidney Size/Weight	Increased Glomerular Volume
Increased blood pressure	✓		✓	✓	✓	✓	✓		✓	✓	✓	✓
Salt sensitivity	✓				a					a	✓	
Reduced GFR	✓	✓	✓		a				✓	a	✓	
Reduced renal functional reserve		✓	✓		✓							
Proteinuria	✓		✓	✓	✓				✓	a	✓	
Acute kidney injury (neonates)			✓							✓	✓	
Chronic kidney disease	✓	✓	✓		✓	✓	✓		✓	a	✓	✓
End-stage renal disease	✓			✓	✓							
Transplant outcomes	✓										✓	✓
Death	✓											
Increased glomerular volume	✓		✓		a				✓	✓		
Smaller kidney size/weight	✓	✓	✓					✓		✓		
Congenital obstruction	✓			NS	✓		✓			a		
Renal dysplasia/aplasia	✓			NS	✓		NS			✓		✓

✓, Evidence from human studies; a, evidence from animal studies only; NS, not significant. Modified from prior Table 23.8, readapted from Luyckx, 2015, 2002.

Table 21.11 Prevalence of LBW, Preterm Birth, Maternal Diabetes, and Obesity in Low- and Middle-Income Countries^a

	LBW (2010)	Preterm Birth (2010)	HBW (2004–2008)	Gestational Diabetes ^a (2013)	Maternal Overweight (2003–2009)	Maternal Obesity (2003–2009)	Chronic Hypertension/ Preeclampsia/ Eclampsia
Proportions in LMICs (country <i>n</i>)	15% (138)	11.3% (138)	0.5%–14.9% (23)	0.4%–24.3% (15)	13.7% (27 Sub-Saharan countries)	5.3% (27 sub-Saharan countries)	2.3% (17)

^aVariable rates in part related to differences in cut-off values for diagnosis.

HBW, High birth weight; LBW, low birth weight; LMIC, low- and middle-income countries.

Adapted from Low Birth Weight and Nephron Number Working Group. The impact of kidney development on the life course: a consensus document for action. *Nephron*. 2017;136:3–49. Compiled from Koyanagi A, Zhang J, Dagvadorj A, et al. Macrosomia in 23 developing countries: an analysis of a multicountry, facility-based, cross-sectional survey. *Lancet*. 2013;381:476–483; Lee AC, Katz J, Blencowe H, et al. National and regional estimates of term and preterm babies born small for gestational age in 138 low-income and middle-income countries in 2010. *Lancet Glob Health*. 2013;1:e26–e36; Cresswell JA, Campbell OM, De Silva MJ, et al. Effect of maternal obesity on neonatal death in sub-Saharan Africa: multivariable analysis of 27 national datasets. *Lancet*. 2012;380:1325–1330; (Kanguru L, Bezawada N, Hussein J, et al. The burden of diabetes mellitus during pregnancy in low- and middle-income countries: a systematic review. *Glob Health Action*. 2014;1;7:23987; and Abalos E, Cuesta C, Carroli G, et al. Pre-eclampsia, eclampsia and adverse maternal and perinatal outcomes: a secondary analysis of the world health organization multicountry survey on maternal and newborn health. *BJOG*. 2014;121(suppl 1):14–24.).

Nigerian adults exposed to the Biafran famine (1967–1970) during fetal life and early childhood.²⁹³ Consistent with observations among Caucasians exposed to the Dutch famine (1944–1945), blood pressures were elevated in adult Nigerians exposed to the famine.^{293–295} Interestingly, however, blood pressures in Biafran subjects were elevated 15 to 20 years earlier than among the Dutch subjects (see Table 21.8), potentially again suggesting aggravating factors present in African-origin subjects.^{293,295} In addition, in both populations, exposure to famine was also associated with increased risks of glucose intolerance and obesity, again underscoring the multisystem consequences of developmental programming.^{293,294}

Variants in the *APOL1* gene have been strongly linked to risk of CKD in African Americans and in West Africans.^{4,9} The relationship between *APOL1* genotype and nephron number was recently investigated.²⁹⁶ Glomerular number was not reduced, and glomerular size was not increased among African Americans with one or two *APOL1* variant alleles, but there was evidence of more accelerated loss of nephrons after the age of 38 years, which was exacerbated by obesity.²⁹⁶ A programming interaction with *APOL1* genotype, therefore, has not yet been established; however, age-related loss of glomeruli appears accelerated.

PROPOSED MECHANISMS OF DEVELOPMENTAL PROGRAMMING IN THE KIDNEY

Kidney development is a complex process involving tightly controlled expression of many genes and constant remodeling.^{111,297–301} The molecular regulation of kidney development is exhaustively reviewed elsewhere (see Chapter 1).^{298–301} Many experimental models, as discussed earlier and outlined in Table 21.4, have been shown to result in a reduced nephron number, and these have been used to gain insight into potential mechanisms contributing to programmed disease outcomes. In

many of the experimental models of programming, reduced nephron number has been shown to be associated with low birth weight and subsequent hypertension and renal injury. Interestingly, in normal rat litters, those pups with naturally occurring low birth weight (i.e., birth weights < –2 standard deviations [SDs] from the mean) were found to have a 13% reduction in nephron number, which was associated with glomerulomegaly and proteinuria.³⁰² Low birth weight in rodents, therefore, may be associated with a low nephron number, even under nonexperimental conditions. Maternal factors that affect birth weight and preterm birth in humans may also directly affect nephrogenesis, as illustrated in Table 21.12, and some of these factors may compound the effect of low birth weight or preterm birth.^{90,303} In humans, nephrogenesis begins in week 9 of gestation and continues until about 36 weeks. Approximately two-thirds of the nephrons develop during the last trimester, making this the window of greatest susceptibility to adverse effects, although earlier insults can also impact nephrogenesis.^{123,304} In rodents, nephrogenesis continues for up to 10 days after birth but is most active mid- to late gestation when studies show the most impact from manipulation of environmental factors.³⁰⁴

Our current understanding highlights three processes that are considered to play critical roles in determining nephron endowment at the conclusion of nephrogenesis: branching of the ureteric tree, condensation of metanephric mesenchymal cells at the ureteric branch tips, and conversion of these mesenchymal condensates into nephron epithelium.^{298–301} It has been estimated that a 2% decrease in ureteric tree branching efficiency would result in a 50% reduction in final nephron complement after 20 generations of branching.³⁰⁵ The specific molecular mechanisms whereby nephron numbers may be affected and/or function altered, however, are not yet completely understood. Perturbations to the fetomaternal environment that result in reduced nephron endowment are summarized in Table 21.13 and discussed later.

Table 21.12 Potential Renal Programming Effects of Risk Factors for Preterm Delivery

Maternal Risk Factor for Preterm Birth	Animal		Human		
	BP	Nephron Number/ Kidney Size	BP	Nephron Number/ Kidney Size ^a	Kidney Function
Vitamin D deficiency	↑	↑ Immature glomeruli	ND	ND	ND
Anemia/iron deficiency	↑	↓	ND	ND	ND
Smoking	ND	↓	↑	↓	↓
Alcohol intake	ND	↓	ND	ND	ND
Antibiotic use (UTI)	ND	↓	ND	ND	ND
Chorioamnionitis	ND	↓	ND	ND	ND
Steroids	↑	↓	Normal	Not different	Unknown
Cyclosporine/tacrolimus (maternal transplant)	↑	↓	ND	ND	Possibly normal
Maternal dialysis	ND	ND	ND	ND	Some ↓
Maternal diabetes	↑	↓	↑	ND	↓
Preeclampsia	ND	ND	↑	ND	ND

^aKidney size generally determined by ultrasound.

BP, Blood pressure; ND, not determined; UTI, urinary tract infection; ↑, increased; ↓, decreased.

Compiled from references 176, 246, 322, 330, 356, 362, 363, 374, 379, 465, 466, 543, and 555–559.

Reprinted with permission from Luyckx VA. Preterm birth and its impact on renal health. *Semin Nephrol.* 2017;37:311–319.

GENETIC VARIANTS ASSOCIATED WITH KIDNEY SIZE AND NEPHRON NUMBER IN HUMANS

Rare genetic and CAKUT contribute to about 40% of all childhood ESKD (NAPRTCS Report 2014). The common pathology linking these malformations involves a disturbance of the normal interaction between the ureteric bud and the pool of renal progenitor cells.³⁸³ Of the >25 mutant genes associated with monogenic forms of CAKUT, most encode transcription factors (e.g., *PAX2*, *GATA3*) and growth factor receptors (e.g., *RET*) expressed in ureteric bud/tree cells during branching morphogenesis or in intermediate mesoderm (e.g., *SIX2*, *EYA1*, *ROBO*), where they set the fate of renal progenitor cells and regulate interactions with the ureteric bud.

Completely dysfunctional alleles for crucial developmental genes are rare, because they produce malformations with a major disadvantage. However, there is some evidence that mild mutations of these same genes exert subtler effects on renal mesenchyme/ureteric bud interactions and may be fairly common in the normal population (see Table 21.4). For example, heterozygous null alleles of the *PAX2* gene cause the rare autosomal dominant renal coloboma syndrome, characterized by decreased ureteric branching during fetal life, sharply reduced nephron number at birth, and progressive renal failure in childhood. *PAX2* is highly expressed in the ureteric bud, where it suppresses apoptosis and optimizes the extent of branching. Interestingly, an intronic *PAX2* polymorphism, which reduces *PAX* transcript levels from the mutant allele by only 50%, is found in 18.5% of Canadians and is associated with a subtle (10%) reduction in newborn kidney size.³⁰⁶ Ureteric branching morphogenesis is also highly dependent on *GDNF* signaling from the metanephric mesenchyme to the ureteric bud via the *RET* tyrosine kinase receptor on ureteric tip cells. Although no common hypomorphic variants of the human *GDNF* gene have been shown to affect kidney size, a polymorphic variant of the *GDNF* receptor, *RET(1476A)*, was associated with a 10% reduction in kidney

volume at birth compared with the wild type (*RET 1476G*) allele.¹²¹ In this study, newborn kidney volume was shown to be proportional to nephron number, suggesting that the modest renal hypoplasia seen with somewhat dysfunctional *PAX2* and *RET* polymorphisms represents a reduction in congenital nephron number.¹²¹ Newborn kidney size was reduced by 13% among Polish babies with a common variant of the *BMPRIA* gene, encoding a bone morphogenetic protein receptor on ureteric epithelial cells.³⁰⁷ Conversely, 22% of Canadian newborns inherit a variant of the *ALH1A2* gene (*rs7169289*) associated with increased production of all-*trans* retinoic acid metabolism in fetal tissues; this retinoid is known to enhance *RET* expression in the ureteric epithelium, and newborns with the G allele were shown to have a 22% increase in newborn kidney size compared with the wild type allele.³⁰⁸

Although final nephron number is clearly affected by genes regulating the extent of ureteric branching, animal studies also indicate that the size of the renal progenitor cell pool may also be rate limiting. One of the earliest transcription factors marking the nephron progenitor cells of intermediate mesoderm is *OSR1*; *Osr1* knockout mice lack nephrogenic mesenchyme and are anephric at birth.³⁰⁹ A variant of the human *OSR1* gene that interferes with mRNA splicing was identified in about 6% of normal Caucasians.³¹⁰ This *OSR1rs12329305(T)* variant was associated with a 12% reduction in newborn kidney size. Taken together, these observations suggest that final nephron endowment may represent a complex polygenic trait determined by the additive effects of multiple genes regulating either the extent of ureteric branching or the renal progenitor cell pool during fetal life. As reviewed by Walker and Bertram,³⁰¹ full or partial deletion of more than 25 genes has been shown to result in kidney hypoplasia, and deletion of several of these genes results in low nephron endowment. Not all have been studied in humans; therefore, the impact of genetic variation on nephron endowment alone and in the context of developmental stresses and the risk of later-life hypertension and renal disease requires further study.

Table 21.13 Proposed Mechanisms of Developmental Programming in the Kidney

Experimental Model	Possible Mechanism of Nephron Number Reduction	References
Maternal low-protein diet	↑ Apoptosis in metanephros and postnatal kidney Altered gene expression in developing kidney Altered gene methylation ↓ Placental 11- β HSD2 expression → increased fetal exposure to glucocorticoids	313, 315, 320, 488
Maternal vitamin A restriction	↓ Branching of ureteric bud ? Maintenance of spatial orientation of vascular development ↓ c-ret expression	326
Maternal iron restriction	? Reduced oxygen delivery ? Altered glucocorticoid responsiveness ? Altered micronutrient availability ↑ Inflammation ? Tissue hypoxia	322, 323
Maternal zinc deficiency	↑ Apoptosis ↓ Antioxidant activity	324
Gestational glucocorticoid exposure	↑ Fetal glucocorticoid exposure ? Enhanced tissue maturation ↑ Glucocorticoid receptor expression ↑ Na-K-ATPase, α_1 and β_2 subunit expression ↓ Renal and adrenal 11- β HSD2 expression	154, 356, 361, 560
Uterine artery ligation/embolization	↑ Proapoptotic gene expression in developing kidney: caspase-3, Bax, p53 ↓ Antiapoptotic gene expression: PAX2, bcl-2 Altered gene methylation Altered renin-angiotensin gene expression	160, 402
Maternal diabetes/hyperglycemia	↓ IGF-2/mannose-6-phosphate receptor expression Altered IGF-2 activity/bioavailability Activation of NF- κ B Altered ureteric branching morphogenesis	161, 338, 342, 345
Gestational drug exposure	↓ Branching morphogenesis ↑ Mesenchymal apoptosis Arrest of nephron formation ? Via reduced vitamin A levels Affects prostaglandins	176, 372, 374, 376
Maternal hypoxia	? Affects expression of retinoid receptors ? Increased expression of glucocorticoid receptors ? Increased expression of angiopoietin-2 Accelerated aging	542
Ureteral obstruction: postnatal	↓ Cell proliferation ↑ Apoptosis of tubular cells Delayed maturation of interstitial fibroblasts → interstitial fibrosis ? Alteration of postinductive processes	73
Preterm birth	Abnormal maturation of glomeruli ? Factors associated with shift from intrauterine to extrauterine environment ? Loss of progenitor cell populations	113, 561

MATERNAL NUTRITION

UNDERNUTRITION

Suboptimal maternal nutrition during pregnancy has long been thought to be the basis of programmed deficits in offspring.³¹¹ In humans, maternal malnutrition, as measured by hemoglobin, triceps skinfold thickness, or lower weight gain during later pregnancy, were all associated with higher offspring blood pressures, suggesting a programming effect.³¹² Longitudinal study of people conceived or born during the Dutch Hunger Winter of 1944–1945 has demonstrated that periods of severe undernourishment can result in hypertension²⁹⁵ and albuminuria,²³⁸ even in the absence of overt fetal growth restriction. Experimental alterations in maternal dietary composition at different stages of gestation have been

shown to program kidney gene expression early in the course of gestation, which later affects nephron number (Table 21.13).³¹³ Maternal “protein and calorie restriction” during all or the later stages of pregnancy have been the most widely studied models of low birth weight and reduced nephron number since as early as 1968.³¹⁴ Not all low-protein diets have the same programming effects, however. It has been proposed that the source of carbohydrate and the relative deficiencies of specific amino acids—methionine or glycine, for example—may have a greater impact on organ development than total protein restriction per se, potentially through epigenetic modulation of gene expression.^{34,315} Effects are also dependent on the degree of protein restriction and the sex of the fetus, with a more severe restriction required to impair kidney development and program hypertension in

female rats than males.^{141,316} Fetal nutrient supply is also affected by alterations in placental development that affect uteroplacental blood flow and transfer of nutrients to the fetus. Placental insufficiency is the most common cause of fetal growth restriction in the Western world, and as recently reviewed,³¹⁷ the placental phenotype most likely underpins the developmental programming of chronic disease in the offspring. Similarly, in the rat, uterine ischemia in late pregnancy resulted in a nephron deficit and hypertension in male offspring, but interestingly, restoration of good fetal nutrition postnatally, during ongoing nephrogenesis, resulted in restoration of nephron number.¹¹²

Increased fetal exposure to glucocorticoids and alterations to the RAAS have been proposed as mechanisms whereby low-protein diet reduces nephron number, both of which are discussed later.³¹⁸ Other potential mechanisms include reduced renal angiogenesis, associated with reduced VEGF expression, observed in offspring of mothers exposed to 50% calorie restriction during gestation³¹⁹; global downregulation of gene expression, as seen in fetal kidneys of microswine exposed to low-protein diet in late gestation/early lactation³²⁰; and altered epigenetic DNA methylation, which may affect gene expression, described in livers of offspring of protein-restricted mothers.³²¹

In terms of micronutrients, maternal “iron restriction” during pregnancy also leads to a reduction in birth weight and nephron number and the development of hypertension in rat offspring.³²² In another study, offspring of iron-deficient dams had reduced radial glomerular counts and increased tubulointerstitial fibrosis, which were rescued with iron supplementation during gestation.³²³ Conceivably, fetal anemia may result in reduced tissue oxygen delivery, altered fetal kidney glucocorticoid sensitivity, or altered availability of other micronutrients that may affect nephrogenesis.³²² Similarly, pre- or postweaning “zinc deficiency” was also associated with decreased nephron number, reduced GFR, and higher blood pressures in rats, potentially mediated by reduction in the antioxidant antiapoptotic effects of zinc.³²⁴ As yet, there is no evidence for programmed renal outcomes in children of iron- or zinc-deficient mothers, although in a population with a high prevalence of anemia, iron supplementation resulted in an increase in birth weight.³²⁵

Maternal “vitamin A restriction” has also been associated with a reduction in nephron number in rat offspring.^{106,326} Severe vitamin A deficiency during pregnancy is associated with congenital malformations and renal defects in the offspring. Vitamin A and all-*trans* retinoic acid have been shown to stimulate nephrogenesis through modulation of branching capacity in ureteric epithelial cell culture and in maintenance of spatial organization of blood vessel development in cultured renal cortical explants.³²⁶ Analysis of 21-day-old fetal rats (just before birth) revealed a direct correlation between plasma retinol concentration and nephron number, as shown in Fig. 21.12.³²⁶ The reduction in nephron number in the setting of vitamin A deficiency is likely mediated, at least in part, by modulation of genes regulating branching morphogenesis.³²⁶ In vivo, a vitamin A-deficient diet sufficient to reduce circulating vitamin A levels by 50% in pregnant rats resulted in a 25% reduction in nephron endowment in the offspring, whereas supplementation of vitamin A increased nephron endowment.³²⁶ In contrast, supplemental retinoic acid was studied as a means

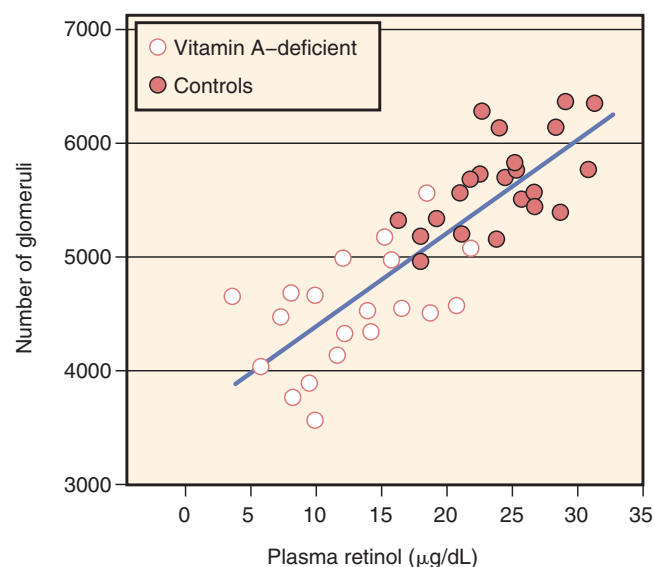


Fig. 21.12 Relationship between nephron number (number of glomeruli) and plasma retinol concentration in term rat fetuses. $P < .001$; $R = 0.829$. (From Merlet-Benichou C. Influence of fetal environment on kidney development. *Int J Dev Biol.* 1999;43:453–456.)

to stimulate nephrogenesis in postnatal preterm baboons, but no effect was observed, suggesting a more proximal window where vitamin A may be most critical.³²⁷

It is interesting to note that smoking and alcohol intake may be associated with reduced levels of circulating vitamin A, and both, as discussed later, are associated with low birth weight and programming of disease outcomes. Subtle differences in vitamin A level during pregnancy, therefore, may be a significant factor contributing to the wide distribution of nephron number in the general population.⁹⁰

Emerging evidence also suggests “vitamin D deficiency” during pregnancy can alter kidney development; in animal models, offspring from vitamin D-deficient dams had delayed glomerular development and altered renal function.^{328–330} In humans, severe 25-hydroxy Vitamin D (25OHD) deficiency is associated with low birth weight and gestational age at delivery, both of which are known to influence kidney development.³²⁹

Recognizing the importance of nutrition during pregnancy and after birth, maternal and neonatal replacement of micronutrients has been implemented as a public health policy in several countries (Table 21.14).³³¹ From a global review, the incidence of low birth weight was reduced by 19% with iron and folate supplementation and by 11%–13% with multimicronutrient supplements, and balanced energy supplementation increased birth weights by a mean of 73 g and reduced the risk of being small for gestational age by 34%.^{331,332} Vitamin A supplementation alone did not affect birth weight but did improve child mortality.^{331,333,334} Interestingly, children of mothers who received folate or a preparation containing folate + iron + zinc during pregnancy in addition to vitamin A supplementation in early postpartum in Nepal had a lower risk of low birth weight but no change in later blood pressure.^{334,335} Microalbuminuria was less frequent, however, suggesting a potential programming effect of these micronutrients.³³⁵ Very high doses of vitamin A have been

Table 21.14 Impact of Nutritional Interventions on Birth Weight and Preterm Birth and Programming of Blood Pressure and Kidney Disease

	LBW/SGA	Preterm Birth	Preeclampsia/ Eclampsia	HBW/LGA	Child Blood Pressure	Child GFR	Child Microalbuminuria
Iron and folate supplementation	↓	↓			↓	↑	
Micronutrient supplementation	↓				↑ or ↓ (No effect vs. iron/folate)		
Calcium supplementation		↓	↓		↓		
Protein supplementation	↓				No effect		
Vitamin A supplementation	No effect/↓				Possible ↓		
Folate supplementation							↓
Zinc supplementation		↓			No effect		
Iodine supplementation	↓						
Malaria prevention and Rx	↓						
Rx of genital infections	↓	↓					
Rx asymptomatic bacteriuria	↓						
Magnesium sulphate			↓				
Antiplatelet agents	↓	↓	↓				
Diabetes education				↓			
Smoking cessation	↓	↓					

GFR, Glomerular filtration rate; HBW, high birth weight; LBW, low birth weight, LGA, large for gestational age; SGA, small for gestational age.

Compiled from references 461, 457, 334, 335, 562, and 577–583.

Reproduced from Low Birth Weight and Nephron Number Working Group. The impact of kidney development on the life course: a consensus document for action. *Nephron*. 2017;136:3–49.

shown to be teratogenic and reduce nephrogenesis; therefore, vitamin A supplementation as a strategy to rescue nephron number should target normalization of vitamin A levels and avoid excess.³³⁶ Folate is not known to impact renal development; however, it does affect gene methylation, and deficiencies may program epigenetic effects.³²¹ Diastolic blood pressures appeared marginally lower (0.78 mm Hg; 95% CI: 0.16 to 1.28) in 4.5-year-old children whose mothers had received early prenatal food supplements in Bangladesh, but were marginally higher (0.87 mm Hg; 95% CI: 0.18 to 1.56) among those whose mothers received a multimicronutrient supplement.³³⁷ GFRs were 4.98 mL/min per 1.73 m² higher in children whose mothers had received higher-dose iron supplementation, however.³³⁷ Potential confounders or effect modifiers in these cohorts include baseline vitamin A supplementation, which may have an overriding renal programming effect in all subjects, as well as frequent persistent malnutrition and stunting. Long-term follow-up of children of mothers who received supplements during pregnancy is sparse and thus far does not consistently suggest a positive impact on renal programming. The effects should become clearer as cohorts of children age and more data emerges.^{331,337}

OVERNUTRITION: EXPOSURE TO MATERNAL DIABETES AND OBESITY

Focus on long-term programming effects of maternal diet has more recently moved to consider the effects of overnutrition, including the consequences for offspring born to diabetic women and obese women. As discussed earlier, in some populations high birth weight is associated with an increased susceptibility to proteinuria and renal disease.^{12,38,50} High birth weight is a complication of gestational hyperglycemia and diabetes and may therefore also be a surrogate marker of

abnormal intrauterine programming. Human offspring of diabetic pregnancies have a higher incidence of congenital malformations, resulting from defects in early organogenesis and have an increased risk of CKD (see Table 21.8).^{274,338} It is known that expression and bioavailability of the insulin-like growth factors (IGFs) are altered in diabetic pregnancies and that IGFs and their binding proteins are important regulators of fetal development.³³⁸ The impact of maternal diabetes on metanephric expression of IGFs and their receptors was studied in rats in which diabetes was induced by streptozotocin compared with gestational age-matched normal controls.³³⁸ There was no significant difference in IGF-1 or IGF-2 or insulin receptor expression at any stage, but in offspring of diabetic mothers, there was a significantly increased expression of the IGF-2/mannose-6-phosphate receptor. This receptor tightly regulates the action of IGF-2, and an increase in expression would lead to reduced IGF-2 bioavailability.³³⁸ IGF-2 is a critical player in renal development. The same investigators examined the role of IGF binding protein-1 (IGFBP-1) on nephrogenesis in genetically modified mice.³³⁹ Offspring of females overexpressing human IGFBP-1 were growth restricted and had an 18% to 25% reduction in nephron number, depending on whether human IGFBP-1 was overexpressed in the mother only, fetus only, or both. When metanephroi from these mice were cultured in the presence of IGF-I or IGF-II, only IGF-2 increased nephron numbers by 25% to 40% in a concentration-dependent manner.³³⁹ Interestingly, in a cohort of preterm infants, diastolic blood pressure at age 4 years was found to correlate positively with IGFBP-1 levels measured at postnatal weeks 32.6–34.6, potentially suggesting a programming effect of this pathway in humans.³⁴⁰

In offspring of rats rendered hyperglycemic during pregnancy either by inducing diabetes mellitus with streptozotocin

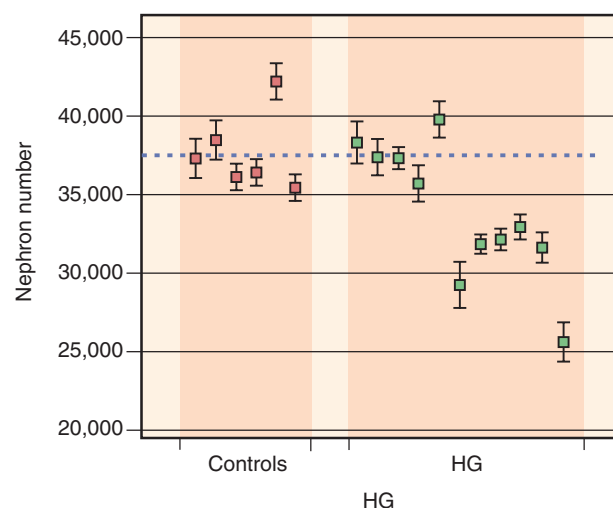


Fig. 21.13 Effects of maternal hyperglycemia on nephron number in rat offspring. Dotted line represents mean value in control group. HG, Hyperglycemia. (From Amri K, Freund N, Vilar J, et al. Adverse effects of hyperglycemia on kidney development in rats: in vivo and in vitro studies. *Diabetes* 1999;48:2240–2245.)

or by infusing glucose from gestational days 12 to 16, nephron numbers were reduced by 10% to 35%, correlating with the degree of maternal hyperglycemia (Fig. 21.13).³⁴¹ Furthermore, culture of metanephroi in varying glucose concentrations demonstrated that tight glucose control is necessary for optimal metanephric growth and differentiation. In mice, offspring of diabetic mothers had fewer nephrons, associated with increased evidence of apoptosis in tubules and podocytes, potentially mediated via increased renal angiotensinogen and renin mRNA expression and NF- κ B activation.¹⁶¹ Other authors suggest altered branching morphogenesis, increased asymmetric dimethylarginine, and reduced nitric oxide levels as potential mediators of reduced nephron numbers in offspring of diabetic mothers.^{342,343} These offspring developed higher blood pressures and renal hypertrophy and had greater tubulointerstitial injury compared with controls, and these changes were abrogated by normalization of asymmetric dimethyl arginine (ADMA) levels achieved by maternal supplementation with L-citrulline.³⁴³ Multiple pathways are therefore likely implicated in hyperglycemia-induced renal programming. Functionally, study of adult rat offspring of diabetic mothers revealed glomerular hypertrophy, reduced GFR and renal plasma flow, hypertension, and decreased endothelium-mediated vasodilation.^{344,345}

Maternal obesity and associated pregnancy complications, including gestational diabetes, maternal hypertension, and sleep apnea, are now thought to present the greatest risk for in utero programming of adult disease. Pregnancies in overweight and obese mothers can result in both low- and high-weight infants, suggesting that programming of offspring by maternal obesity may be multifactorial.³⁴⁶ Feeding animals a high-fat (+/- high carbohydrate) diet prior to and throughout pregnancy can result in programmed outcomes. Exposure to a high-fat diet for 10 days prior to pregnancy until weaning resulted in rat female offspring with hypertension but no change in nephron endowment, although renal renin and Na/K-ATPase activity were altered.³⁴⁷ Rat dams fed an

obesogenic diet for a more sustained period (a high-fat/high-fructose diet from 6 weeks prior to mating) resulted in offspring with albuminuria, which was further exacerbated by a postnatal high-fat diet.³⁴⁸ A similar model also caused increased renal norepinephrine and renin expression in offspring.³⁴⁹ Conversely, in mice, a maternal high-fat diet caused an increase in offspring nephron number, with no consequences for renal function.³⁵⁰ Dams in that study were glucose intolerant prior to mating but had normal glucose tolerance during pregnancy, suggesting complex interactions between maternal obesity and diabetes. These data suggest that it may not be the high-fat diet per se causing changes in renal development but rather the resultant maternal hyperglycemia. Of note, many calorie-dense foods are also deficient in essential micronutrients.²⁷⁹

MATERNAL AND FETAL EXPOSURE TO GLUCOCORTICOIDS

Elevated maternal glucocorticoids due to stress or glucocorticoid therapy may impact fetal growth and renal development. Under normal circumstances, the fetus is protected, at least in part, from exposure to excess endogenous maternal corticosteroids by the placental enzyme 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2), which metabolizes corticosterone to the inert 11-dehydrocorticosterone.²¹ However, this barrier is not complete, and administration of high physiologic concentrations of cortisol (in sheep) or corticosterone (in rats and mice) all resulted in offspring with low nephron number.^{351–353} Offspring of these models developed hypertension and impairment of renal function, including albuminuria, that was, in all cases, associated with changes in the renal renin–angiotensin system (discussed later).^{351,354,355} These outcomes occurred even following very short-term exposure (48–60 hours) that did not affect birth weight in sheep and mice.^{353,356} Prenatal administration of dexamethasone, a steroid not metabolized by 11 β -HSD2, lead to fetal growth restriction, 20% to 60% lower nephron number, glomerulomegaly, and subsequent hypertension in rats and sheep.^{103,107,185,356} Metanephric organ culture experiments have demonstrated that dexamethasone can cause a dose-dependent inhibition of branching morphogenesis in part through regulation of *GDNF* gene expression.³⁵⁷ Rats and humans with mutations in the 11 β -HSD2 gene have low levels of placental 11 β -HSD2 and give birth to low birth weight offspring in whom hypertension develops prematurely.^{358,359} Maternal low-protein diet during gestation has been shown to result in decreased placental expression of 11 β -HSD2, therefore likely increasing exposure of the fetus to maternal corticosteroids.^{34,154} Treatment of pregnant rats fed a low-protein diet with an inhibitor of steroid synthesis ameliorates the programming of hypertension and increases nephron numbers in the offspring.^{21,34,360} Although this rescue was not complete, these data strongly implicate glucocorticoids as modulators of nephrogenesis in the setting of maternal low-protein diet.³⁶⁰ Excessive fetal steroid exposure may drive inappropriate gene expression and affect growth and nephrogenesis, potentially through more rapid maturation of tissues.³⁴ Furthermore, expression of steroid-responsive receptors, including the corticosteroid responsive renal Na/K-ATPase α_1 - and β_1 -subunits, was found to be significantly increased in offspring of rats fed a low-protein diet during gestation.¹⁵⁴ In another

study, prenatal dexamethasone was associated with increased proximal tubule sodium transport, in part related to increased activity of the tubular NHE3.³⁶¹ These changes may contribute to hypertension.

In humans, there has been concern about perinatal exposure to corticosteroids; however, follow-up of subjects whose mothers participated in a randomized placebo-controlled study of antenatal betamethasone did not find any difference in blood pressure or other cardiovascular risk factors at age 30 years in exposed compared with unexposed individuals.³⁶² Although no effect on blood pressure or renal function was noted among preterm infants who had received antenatal steroids within the first 2 years of life compared with those who did not, recent evidence, however, highlights glucocorticoid therapy may have lasting effects on the brain and result in increased mental disorders at age 8 years.^{191,363,364} Longitudinal examination of the consequences of sustained glucocorticoid therapy (e.g., throughout pregnancy) has not been performed.

MATERNAL AND FETAL HYPOXIA

In addition to maternal iron deficiency (resulting in anemia) and placental insufficiency discussed earlier, a range of conditions, including high altitude, maternal smoking, and sleep apnea, can result in fetal hypoxia. These conditions have all been shown to result in low birth weight in humans, but as yet, there is little evidence to suggest long-term effects on kidney function. In mice, a short-term severe, hypoxic insult (5.5%–7.5% maternal oxygen) from E9.5–10.5 caused a CAKUT phenotype, whereas mild hypoxia in midpregnancy (12% from E12.5–14.5) resulted in low nephron endowment mediated in part through suppression of β -catenin signaling.³⁶⁵ A similar mild hypoxia in late gestation resulted in fetal growth restriction, low nephron number in male but not female offspring, and hypertension in both sexes.³⁶⁶ In a mouse model of cigarette smoke exposure, male offspring had fewer nephrons and increased albumin/creatinine ratio.³⁶⁷ Short-term cord occlusion in fetal sheep did not cause overt long-term renal damage, highlighting that the timing and duration of hypoxia are important in mediating renal outcomes.³⁶⁸

FETAL DRUG AND ALCOHOL EXPOSURE

Several medications commonly used during pregnancy or in the early postnatal period have been studied for their effects on birth weight and nephrogenesis. Among 397 pregnant women, antibiotic use was associated with a 138-g lower offspring birth weight compared with nonantibiotic use.³⁶⁹ Furthermore, analysis of methylation levels of imprinted genes showed antibiotic use to be associated with methylation at five differentially mediated regions, although methylation at only one region was associated with birth weight.³⁶⁹ The aminoglycoside antibiotic gentamicin administered to pregnant rats results in a permanent nephron deficit in offspring.³⁷⁰ Significantly lower numbers of ureteric branch points and nephrons were observed in metanephric explants cultured in the presence of gentamicin.^{371,372} In contrast, however, other investigators did not find a reduction in nephron number in rat pups administered gentamicin intraperitoneally from birth to 14 days of age.³⁷³ The β -lactams have also been

shown to result in impaired nephrogenesis.³⁷⁴ Administration of ampicillin to pregnant rats leads to an 11% average reduction in nephron endowment in offspring, as well as evidence of focal cystic tubule dilatation and interstitial inflammation. Administration of ceftriaxone *in vivo* did not result in a nephron deficit, but histologically, there was evidence of renal interstitial inflammation. The penicillins were also found to inhibit nephrogenesis in cultured metanephroi in a dose-dependent fashion, an effect that was less evident with ceftriaxone. Importantly, nephrogenesis was affected even at therapeutic doses of penicillins in the rat. The mechanism whereby the β -lactams reduce nephron endowment is likely through an increase in apoptosis in the induced mesenchyme.¹⁰⁶ Overall, antibiotic use during pregnancy may reduce nephron endowment. Further research on those antibiotics that are frequently used in human pregnancy and preterm infants is warranted.

The immunosuppressive medication cyclosporine is a known nephrotoxin in humans that crosses the placenta. Women treated with cyclosporine may have successful pregnancies, although infants tend to have birth weights in the low range, and its effect on the fetal kidney is not well described.³⁷⁵ Cyclosporine administration in varying doses and at different stages of gestation was evaluated in pregnant rabbits compared with rabbits receiving either vehicle or no drug.³⁷⁵ Cyclosporine administration in the later, but not the earlier, period of gestation resulted in smaller litters and growth-restricted pups. All pups exposed to cyclosporine *in utero* had a 25% to 33% lower nephron number than controls. The reduction in nephron number was accompanied by glomerulomegaly and was independent of birth weight. At 1 month of age, these kidneys also demonstrated foci of glomerulosclerosis. Subsequent functional evaluation of the kidneys of rabbits exposed to cyclosporine *in utero* demonstrated lower GFR at 18 and 35 weeks of age and an increase in proteinuria at 11, 18, and 35 weeks of age.¹⁷⁶ Rabbits exposed to cyclosporine *in utero* developed spontaneous hypertension by 11 weeks of age, which worsened progressively with time.¹⁷⁶ In the presence of cyclosporine, nephron formation was found to be arrested, potentially due to inhibition of conversion of metanephric mesenchyme to epithelium.¹⁰⁶

Nonsteroidal antiinflammatory drugs are sometimes used in preterm children postnatally. Administration of a cyclooxygenase 2 inhibitor, but not a cyclooxygenase 1 inhibitor, postnatally in rats and mice resulted in reduced cortical volume, impairment of nephrogenesis, and reduced glomerular diameter.³⁷⁶ Administration of indomethacin or ibuprofen postnatally did not affect nephron number in rats.³⁷³ In the preterm baboon kidney, early postnatal administration of five doses of ibuprofen (consistent with recommended dosing in preterm infants with patent ductus arteriosus) was associated with a reduction in width of the nephrogenic zone, suggesting premature termination of nephrogenesis.³⁷⁷ The impact of these medications on human nephrogenesis is not known.

Maternal alcohol consumption: Renal malformations have been reported in children with fetal alcohol syndrome.³⁷⁸ In sheep, repeated ethanol exposure during the second half of pregnancy resulted in an 11% reduction in nephron number and impaired vascular function in late pregnancy, although offspring outcomes were not examined.^{379,380} In rats, exposure to 2 days of a high dose of alcohol in mid-late

pregnancy resulted in a low nephron number and offspring with hypertension and sex-dependent impairments in GFR.³⁸¹ Using metanephric organ culture, alcohol was found to cause dose-dependent decreases in branching morphogenesis, which were prevented by retinoic acid supplementation.³⁸² An abstract has reported an impact of maternal alcohol ingestion on kidney development in children, but it is not known whether the effects were mediated by associated vitamin A deficiency or other mechanisms.⁹⁰

OBSTRUCTION OF THE DEVELOPING KIDNEY

Around one-half of all pediatric kidney transplants are required as a result of CAKUT, predominantly obstructive nephropathy, and hypoplasia/dysplasia.³⁸³ Being small for gestational age was associated with a higher risk of ESKD (OR: 2.5; 95% CI: 1.6 to 3.7) compared with normal birth weight, suggesting that nephron number acquired at birth may be an important modulator of this risk.³¹ Most cases of CAKUT result from multiple genetic, epigenetic, and fetal developmental factors such as exposure to maternal diabetes, as discussed earlier (see Table 21.8).²⁷⁴ As recently reviewed, many developmental abnormalities of the urinary tract are associated with impaired nephrogenesis, which, in turn, is compounded by obstructive injury.^{384,385} From animal studies, it has become clear that perinatal urinary obstruction may lead to reduced nephron numbers, which may exacerbate the impact of other programming factors.^{70,386} In rats, unilateral ureteral obstruction (UUO) on postnatal day 1 and relieved on day 5, or on postnatal day 14 and relieved on day 19, reduced nephron number by 50% in both groups.⁷³ Similar intervention in adult rats did not affect nephron number or tubular development. A follow-up study using this model of early-life UUO demonstrated that renin was decreased and glomerulotubular maturation was delayed.³⁸⁷ These studies demonstrate that urinary obstruction in a normal developing kidney can impact not only nephrogenesis but also tubular development, which may have long-term impacts on renal solute handling. Importantly, temporary neonatal urinary obstruction was also associated with histologic scarring and loss of function of the contralateral kidney in 1-year-old rats, suggesting consequent programming in the contralateral kidney as well.³⁸⁸ Developing and neonatal kidneys therefore appear to be highly susceptible to obstructive injury, suggesting that early relief of urinary tract obstruction may be important to preserve nephron number.

MOLECULAR PATHWAYS AFFECTED IN FETAL PROGRAMMING OF THE KIDNEY

The molecular regulation of kidney development, particularly in the mouse, has been comprehensively described elsewhere.²⁹⁸ In a number of animal models of developmental programming, changes in expression levels of genes regulating branching morphogenesis, apoptosis, and renal growth have been described (for a review, see Wang and Garrett³⁸⁹), and studies have applied microarray or deep-sequencing techniques to determine the effects of a maternal perturbation on the whole genome.^{299,313,390} Epigenetic changes are now thought to play a role in programming of disease across generations, but as yet, the evidence for epigenetic changes within the kidney is limited.³⁹¹

URETERIC BRANCHING MORPHOGENESIS

GDNF signaling through its receptor-tyrosine kinase Ret is a key ligand–receptor interaction driving ureteric budding and branching. The *c-ret* receptor is expressed on the tips of the ureteric bud branches, and knockout of this receptor in mice leads to severe renal dysplasia and reduction in nephron number.²⁹⁷ Homozygous *GDNF*-null mutant mice have complete renal agenesis and die shortly after birth.³⁹² Heterozygous *GDNF* mice, as described earlier, have reduced nephron numbers, develop glomerulomegaly, and are susceptible to hypertension.¹⁴⁹ Polymorphisms in *RET*, but not *GDNF*, are associated with newborn renal size in humans.^{121,393} As described earlier, maternal dietary vitamin A has a significant impact on nephrogenesis (Fig. 12.12). In cultured metanephroi, the expression of *c-ret* was found to be regulated by retinoic acid supplementation in a dose-dependent manner.³²⁶ *GDNF* expression was not affected by vitamin A fluctuations. Modulation of *c-ret* expression is therefore likely to be a significant pathway through which vitamin A availability regulates nephrogenesis and nephron endowment. Expression of *GDNF* has also been shown to be decreased following maternal glucocorticoid exposure³⁵⁷ and maternal alcohol consumption,³⁸¹ which may contribute to a reduction in branching morphogenesis and lower nephron endowment.

Further evidence for impairments in ureteric branching comes from a mouse model of maternal diabetes during pregnancy, where optical projection tomography was used to demonstrate a reduction in branch number and length within the developing kidney.³⁴² In another mouse model of diabetes, changes in TGF- β 1 signaling were disrupted due to increased expression of a hedgehog interacting protein.³⁹⁴ This protein was localized to differentiated metanephric mesenchyme and the ureteric epithelium during early kidney development and then was found in maturing glomerular endothelial and tubulointerstitial cells. Increased expression of this protein due to high glucose-impaired branching morphogenesis resulted in neonates with small kidneys.

APOPTOSIS

To evaluate at which stage of development a low-protein diet impacts nephrogenesis, embryonic rat metanephroi were studied at different time points.³¹³ At embryonic day 13, the metanephros has just formed, the ureteric bud has branched once, branch tips are surrounded by condensed mesenchyme that later transforms into tubule epithelium, and the ureteric stalk is surrounded by loose stromal mesenchyme.³¹³ By day 15, multiple branching cycles have occurred, and primitive nephrons begin to be formed.³¹³ At embryonic day 13, there was no difference in the number of cells in metanephroi from embryos whose mothers received low- or normal-protein diets, but by day 15, there were significantly fewer cells per metanephros in the low-protein group. In contrast, a significant increase in the number of apoptotic cells was observed in the low-protein group at day 13 but not at day 15, suggesting that increased apoptosis on day 13 likely contributed to the reduced cell numbers on day 15.³¹³ On postnatal day 1 in kidneys from offspring of mothers exposed to 50% calorie restriction, apoptosis was most evident in the nephrogenic zone, colocalizing to the mesenchyme and peritubular aggregates, suggesting a role in modulation of

nephrogenesis.³⁹⁵ In 8-week-old hypertensive low-birth-weight rat offspring of mothers subjected to a low-protein diet, the kidneys were histologically normal but also showed evidence of increased apoptosis, without an increase in proliferation, compared with normal-birth-weight controls.¹⁰⁸ The increase in apoptotic activity observed in the kidney in these studies suggests possible successive waves of apoptosis at different stages of nephrogenesis in programmed rats that may impact nephron endowment.

Several studies have suggested that altered regulation of apoptosis in the developing kidney may be due to downregulation of antiapoptotic factors (e.g., Pax-2 or Bcl-2) and/or upregulation of proapoptotic factors in response to environmental or other stimuli (e.g., Bax, p53, Fas receptor, caspase 3 and 9).^{395–399} Humans with haploinsufficiency of *PAX2* have renal coloboma syndrome, and those with certain *PAX2* polymorphisms have smaller neonatal kidney size as discussed earlier.^{306,396,400} *PAX2* is an antiapoptotic transcriptional regulator that is highly expressed in the branching ureteric tree as well as in foci of induced nephrogenic mesenchyme during kidney development.³⁹⁶ Heterozygous mice with *Pax2* mutations were very small at birth and had significant reductions in nephron number. In addition, there was a significant increase in apoptotic cell death in the developing kidneys. Subsequently, the same group⁴⁰⁰ demonstrated that loss of *Pax2* antiapoptotic activity reduced ureteric branching and increased ureteric apoptosis. Similarly, loss of the antiapoptotic factor *Bcl-2* or gain of function of the proapoptotic factor p53 are both associated with a significant reduction in nephron number, associated with increased apoptosis in metanephric blastemas, in *Bcl-2* knockout mice and *p53* transgenic mice.^{397,401}

Mutant mouse models, although providing evidence that an increase in apoptosis results in reduced nephron numbers, do not, however, address the impact of environmental factors on renal development. Pham and associates⁴⁰² examined gene expression in the kidneys of offspring of rats subjected to uterine artery ligation during gestation. These authors found a 25% reduction in glomerular number was associated with increased evidence of apoptosis and increased proapoptotic caspase-3 activity in the kidney at birth. Furthermore, they found evidence of increased mRNA expression of the proapoptotic genes *Bax* and *p53* and decreased expression of the antiapoptotic gene *Bcl-2*. These authors also found evidence of hypomethylation of the *p53* gene, which, in addition to a decrease in *Bcl-2* expression, would lead to an increase in *p53* activity, suggesting epigenetic programming of a proapoptotic milieu as a potential modulator of nephron endowment. Increased apoptosis was also observed in a similar rat model during late gestation (E20) and 1 week after birth (PN7), suggesting inappropriate apoptosis even after birth.⁴⁰³ Of great interest in that study, the growth-restricted offspring, but not control offspring, had evidence of ongoing nephrogenesis at PN7, suggesting a delay in kidney development in the growth-restricted group. Gene microarray studies in rodent models of maternal iron deficiency³⁹⁰ and diabetes⁴⁰⁴ have also identified apoptotic signaling pathways in fetal kidneys.

IMPACT OF SEX

In some experimental models and human studies, although not all, programming effects on blood pressure and kidney

function appear to be different between males and females, especially at young ages. In female rats with similar programmed reductions in nephron numbers, blood pressures are often not as high or increase much later than in males.^{108,405} Often a secondary challenge such as pregnancy, however, will “unmask” a disease phenotype in females. For example, males but not females born small as a result of bilateral uterine vessel ligation develop hypertension and insulin resistance.^{111,406,407} However, growth-restricted females develop glucose intolerance during pregnancy.⁴⁰⁸ Sex hormones may, in part, explain these differences. Growth-restricted males (induced by uterine artery ligation) have elevated testosterone levels compared with controls, and hypertension can be abrogated by castration.⁴⁰⁵ Such changes are not observed in male offspring of protein-restricted mothers, however, pointing to the intricacies of the programming models. In female rats growth restricted by placental insufficiency, hypertension only develops late, but onset can be accelerated by ovariectomy.⁴⁰⁵ These data suggest that in the uterine ischemia programming model, testosterone exacerbates and estrogen protects against hypertension. Sex differences in relative expression of components of the RAAS appear to participate in programming of hypertension, potentially differentially altering the balance between vasoconstriction and vasodilation and sodium handling.^{159,405} Furthermore, growth-restricted male rats exhibit increased markers of renal oxidative stress compared with controls, which are absent in similarly programmed females, and antioxidant treatment normalized blood pressure in the male rats.^{170,409} Multiple other suggested mechanisms underlying these gender differences have been reviewed by Ojeda et al., as outlined in Fig. 21.14.^{170,405}

Sex differences may also originate in utero due to differences in growth of male and female fetuses and sex-specific placental responses to maternal perturbations (reviewed in Kalisch-Smith et al.⁴¹⁰). In particular, the placenta has an important role in preventing excess maternal glucocorticoids from reaching the fetus via the activity of placental 11 β -HSD2. This enzyme has the ability to convert the active glucocorticoid into an inactive form, as discussed earlier. The placenta of female fetuses is known to have higher levels of this enzyme compared with males,⁴¹¹ and as such, the male fetus is likely to be exposed to more glucocorticoids. Furthermore, in women treated with synthetic glucocorticoids when threatening to deliver preterm, placental 11 β -HSD2 was elevated more in placentas of females than males.⁴¹² The sex of the fetus has not always been taken into account when performing studies on renal development; however, a recent metaanalysis demonstrated the adult human kidney showed sex-specific expression of more than 200 genes.⁴¹³

POTENTIAL FOR RESCUE OF NEPHRON NUMBER

Given the evidence for developmental programming of hypertension and kidney disease and the associations with birth weight, preterm birth, other intrauterine exposures, and the impact of nutrition in early childhood, it is possible that interventions could be designed to modulate developmentally programmed changes in the kidney and reduce long-term disease risk. Optimization of maternal health and nutrition prior to and during pregnancy to attenuate any risk factors for low birth weight and preterm birth is the most obvious

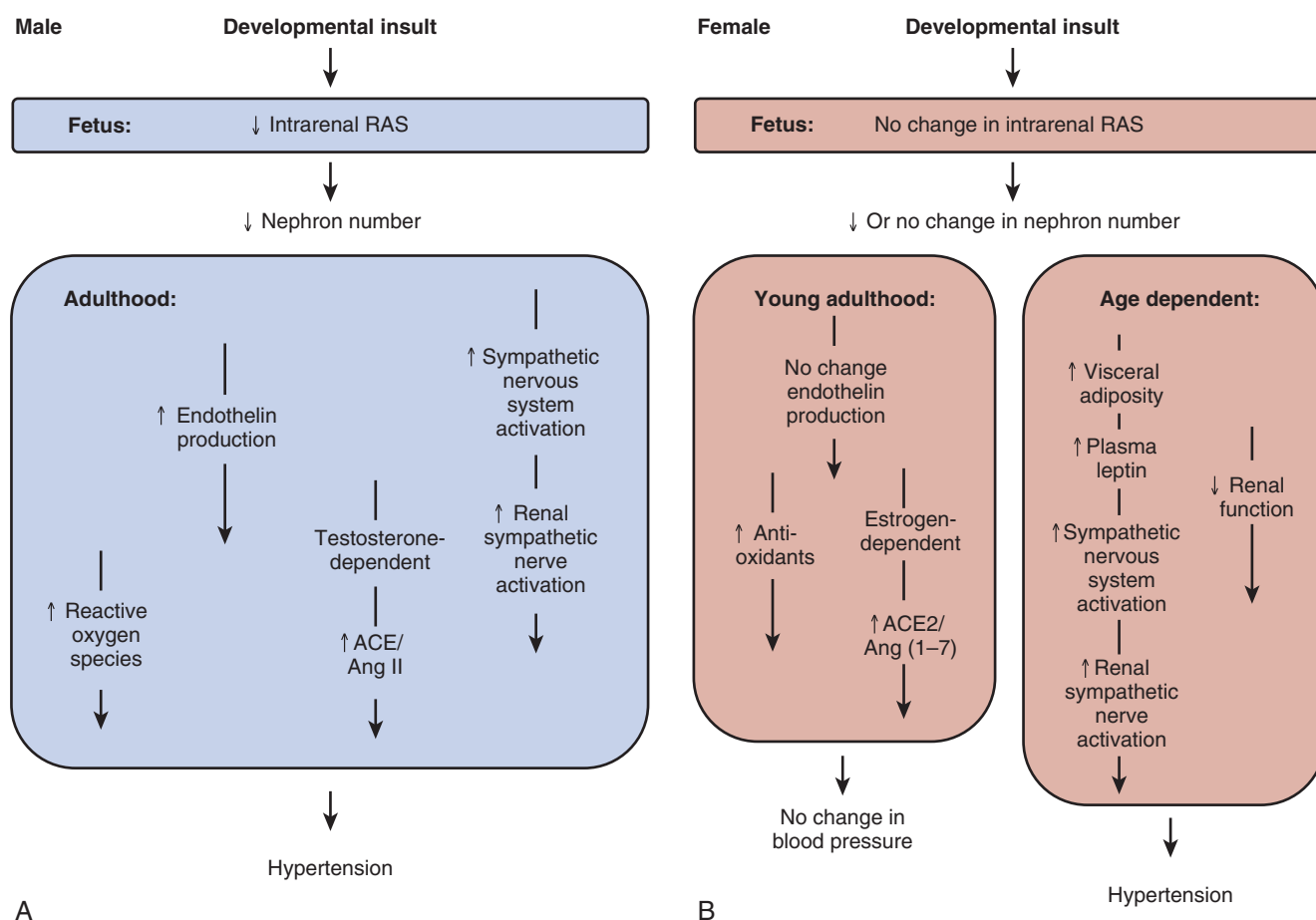


Fig. 21.14 Potential differing mechanisms impacting developmental programming of blood pressure in (A) males, and (B) females. These mechanisms may be due to the influence of the hormonal milieu on the renin-angiotensin system (RAS) due to innate sex differences in production of reactive oxygen species or endothelin, or impacted by increased susceptibility that occurs with age and the development of age-dependent increases in adiposity leading to activation of the sympathetic renal nerves (females). The fetus also exhibits innate sex differences in expression of the intrarenal RAS, which may or may not (as in females) reduce nephron number. ACE, Angiotensin-converting enzyme; Ang, angiotensin. (From Ojeda NB, Intapad S, Alexander BT. Sex differences in the developmental programming of hypertension. *Acta Physiol (Oxf)*. 2014;210:307–316.)

intervention, as it has been estimated that intrauterine factors determine around 60% of the variation in birth weight.⁴¹⁴ In addition, minimization of nephrotoxin exposure and attention to neonatal nutrition in preterm infants are important to permit optimal nephrogenesis after birth. Specific interventions that may augment nephron number per se have been investigated, some of which are clinically feasible, while others are still in research stages (see Table 21.4). Interventions to modulate other aspects of developmental programming in the kidney have not yet been reported, but the assumption would be that the impact may be similar to those affecting nephrogenesis.

Prevention is likely more realistic than rescue of low nephron number. Ouabain is a highly specific ligand for Na/K-ATPase, the activity of which is known to be reduced in erythrocytes of low-birth-weight young men.^{415,416} Na/K-ATPase is a ubiquitously expressed plasma membrane protein that regulates the release of calcium waves and thereby is an important regulator of early development.⁴¹⁵ In vitro addition of ouabain to the medium of metanephroi in culture was found to abrogate the effect of serum starvation on ureteric branching.⁴¹⁵ Similarly, in vivo, ouabain administration throughout pregnancy prevented the reduction in nephron

number in rats subjected to maternal low-protein diet.⁴¹⁵ Whether ouabain can rescue nephron number if given late in pregnancy was not addressed in this study. Supplementation of glycine, urea, or alanine to a maternal low-protein diet prevented development of low nephron numbers in rat offspring, but only glycine supplementation prevented subsequent hypertension.³⁴ Intriguingly, water restriction of rat mothers during gestation resulted in augmentation of normal nephron number but also induced hypertension in the offspring. The authors implicate vasopressin as a mediator of this programming effect.⁴¹⁷

Vitamin A deficiency is common among women in poorer nations and in animals is associated with reduced nephron number as discussed earlier.^{326,418} In rats exposed to maternal low-protein diet, nephron number was restored to normal by one dose of retinoic acid given to the pregnant dams during early nephrogenesis.⁴¹⁹ In preterm baboons, however, administration of retinoic acid postnatally did not rescue nephron number, suggesting that vitamin A is likely necessary earlier in gestation.³²⁷ These baboons also received postnatal antibiotics, which may have confounded the effect of the vitamin A.

Postnatal nutrition is an important modulator of kidney development, especially in preterm infants. Restoration of normal protein intake by cross-fostering growth-restricted pups onto normal mothers after birth reduced nephron number and prevented hypertension compared with those fed by protein-deficient mothers.¹¹²

A maternal single kidney may impact fetal kidney development. In offspring of rats that had undergone nephrectomy prior to pregnancy, nephron numbers were increased at birth, although not at 6 weeks.^{420,421} A circulating renotrophic factor in the mother may therefore accelerate nephrogenesis but does not appear to affect final nephron endowment. How these observations would apply in humans is difficult to extrapolate, as outcomes may be different depending on maternal age, whether she has a congenital or acquired single kidney, or a kidney transplant with the attendant required medications, which, in turn, may affect nephrogenesis.

Modulation of regression of nephron number, although still hypothetical, has been suggested as a potential pathway to augment final nephron number.⁴²² Glomerular number was evaluated from postnatal days 7 to 28 in normal mice. Maximal nephron number was seen at day 7, with a subsequent regression and plateau at day 18. Such a time course would need to be studied in growth-restricted animals before any potential intervention to inhibit this regression could be tested.

CATCH-UP GROWTH

Postnatal nutrition is important for infant growth, especially in the setting of preterm birth or growth restriction, and can impact nephron number and long-term renal function, as discussed earlier.^{187,224} Cross-fostering of growth-restricted newborn rats (induced by placental insufficiency) onto normal mothers permitted restoration of normal nephron number and prevented subsequent hypertension, demonstrating the potential “rescue” effect of adequate postnatal nutrition.¹¹² Postnatal overfeeding of low-birth-weight rats, induced by reduction of litter size to three pups, however, did not augment low nephron numbers, and with aging, rats became obese and hypertensive and developed renal injury.⁴²³ In this model, despite the mother being switched to a normal-protein diet at time of delivery, the pups may remain somewhat protein deficient despite consuming larger quantities of milk, as opposed to cross-fostering, which provides normal milk immediately, which may explain why nephron number remained low and underscores the importance of diet composition. In contrast, overfeeding of normal-birth-weight rats led to higher than normal nephron numbers, despite which high blood pressure and renal injury developed over time.¹³⁷ These animal data suggest that restoration of normal dietary components after growth restriction may permit some reversal of programmed changes, but overfeeding appears harmful.

In diverse populations worldwide, rapid “catch-up” growth (defined as upward crossing of weight centiles), or an increase in body mass index, even in children of normal birth weight, is associated with higher blood pressures and increased cardiovascular risk.^{424–426} On the other hand, catch-up growth has been advocated in poorer countries to improve child survival from infectious diseases and reduce stunting and malnutrition.⁴²⁷ Among 7-year-old children who had been preterm, GFRs were reduced in those who had experienced

either intrauterine or extrauterine growth restriction (i.e., inadequate growth early postbirth), highlighting the importance of adequate early postnatal nutrition for kidney development.²²⁴ Nutrition in preterm infants is challenging, and balancing benefits and risks of rapid growth must be considered, including optimizing neurodevelopment and metabolic outcomes.⁴²⁸

The timing of catch-up growth in early infancy and childhood, which tends to occur rapidly in low-birth-weight children when adequate nutrition is available, appears a crucial factor in determining this long-term risk.^{426,427,429,430} The importance of birth weight and catch-up growth was examined in a cohort of British 22-year-olds, in whom systolic blood pressure was observed to increase by 1.3 mm Hg (95% CI: 0.3 to 2.3 mm Hg) for each SD decrease in birth weight and to increase by 1.6 mm Hg (95% CI: 0.6 to 2.7 mm Hg) for each SD increase in weight gain between the ages of 1 and 10 years.⁴³¹ Such observations have been reproduced in several populations, together with evidence of increased arterial stiffness and greater prevalence of cardiovascular risk factors in early childhood after rapid growth, with children who had been of low birth weight but became overweight being at highest risk.^{208,424,429} The consequences of catch-up growth may, however, differ if growth is linear (i.e., growth in height) or in weight, and may differ between developed and developing countries where health priorities differ.⁴³² Thus far, it appears that among those born with low birth weight or preterm, the development of overweight or obesity is the predominant risk factor.^{208,426,429,430,433} Low birth weight independently predicted both proteinuria and obesity in a rural Canadian cohort, demonstrating likely simultaneous programming of multiple risk factors for kidney disease.⁴³⁴ Risk factors for childhood overweight and obesity also include high birth weight and exposure to gestational diabetes.⁴³⁵ Obesity, in turn, is a risk factor for renal disease.^{436,437} Finding the balance where postnatal nutrition is optimized to improve short-term survival and reduce long-term risk of chronic disease requires further study. In general, avoidance of overweight through diet and exercise are safe principles.^{438,439}

IMPACT OF EARLY GROWTH ON KIDNEY FUNCTION

Grijalva-Eternod⁴⁴⁰ and colleagues developed a model to test whether “mismatch” between a small kidney and a (relatively) larger body, as would occur with catch-up growth and overweight after being born small, is associated with hypertension. Birth weight was presumed to reflect the homeostatic metabolic capacity of the kidney and childhood body composition to reflect the metabolic load. When applied to a birth cohort of children, the model found that a high metabolic load, relative to innate metabolic capacity, was associated with higher blood pressures (Fig. 21.15). Consistent with this hypothesis, proteinuric renal disease progressed faster in children who had been born prematurely and became obese compared with those who were not obese.⁴⁴¹ In this study, all obese children, whether born at term or preterm, were found to have glomerulomegaly, although kidney size remained small in all those born preterm, even among the obese. Similarly, excessive weight gain was a predictor of worse renal function at age 7.5 years in a cohort of very low-birth-weight preterm infants who had experienced AKI.⁴⁴²

Accelerated senescence has been proposed as a potential mechanism whereby catch-up growth may increase the risk of

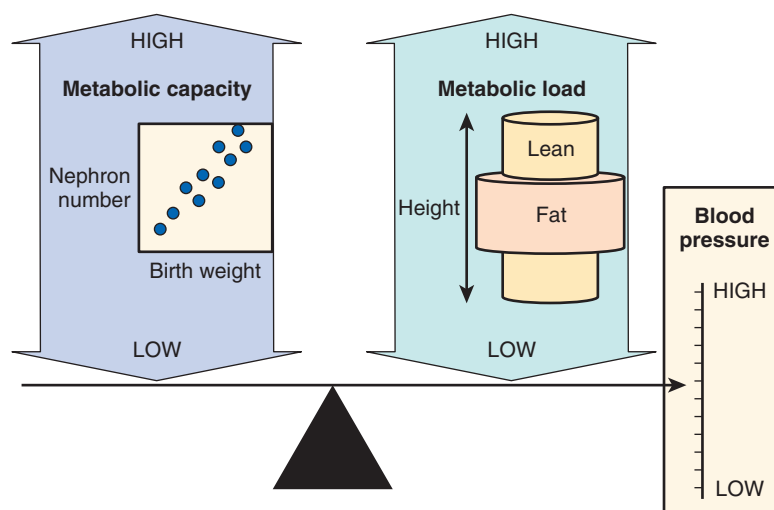


Fig. 21.15 Diagram illustrating the concepts of metabolic capacity and metabolic load in relation to blood pressure. Metabolic capacity, meaning the capacity to maintain normal metabolic homeostasis, is dependent on nephron number, which scales positively with birth weight. Metabolic load, meaning the demand placed on homeostatic systems, is dependent on body size and adiposity, and hence on height, lean mass, and fat mass. Here they are portrayed using a cylinder model in which cylinder lengths are proportional to height, and the volumes are proportional to the masses of lean and fat mass. A high load and low capacity are each predicted to increase blood pressure. (From Grijalva-Eternod CS, Lawlor DA, Wells JC. Testing a capacity-load model for hypertension: disentangling early and late growth effects on childhood blood pressure in a prospective birth cohort. *PLoS One*. 2013;8:e56078.)

cardiovascular and renal disease.⁴⁴³ Senescence is a state of cellular growth arrest that naturally occurs with age but may be accelerated in the face of stress, mediated by upregulation of cell cycle inhibitors (p53, p21, and p16INK4a) and progressive shortening of telomeres.⁴⁴⁴ Increased expression of senescence markers has been observed in diseased human kidneys.⁴⁴⁵ In animals, rapid weight gain in growth-restricted animals was associated with evidence of accelerated senescence in the kidney and cardiovascular system and premature death, consistent with accelerated aging.^{443,446,447} Senescence markers are not more highly expressed at birth in growth-restricted versus normal birth weight animals but increase more rapidly as the animals age.⁴⁴⁶ Hyperfiltration in a small kidney, exacerbated by high metabolic demand from increasing body size, likely contributes to ongoing injury and progressive senescence.^{445,446} Senescence markers have not been studied in low-birth-weight human kidneys, but leukocyte telomere length was found to be significantly shorter among 5-year-old Bangladeshi children who had been of low birth weight compared with those of normal birth weight, lending support to this hypothesis.^{448,449} Oxidative stress is a driver of senescence, and in small-for-gestational-age children, markers of oxidative stress were higher in those who experienced catch-up growth compared with controls, suggesting oxidative stress as a possible initiator of accelerated senescence.⁴⁵⁰ A programmed link between low nephron number, catch-up growth, and accelerated senescence as a potential mediator of hypertension and renal disease in humans has yet to be proven.

Given the normal age-related loss of nephrons and potential for accelerated senescence in the kidney, it could be expected that the impact of prenatal programming would be compounded by age.⁴¹ This has not yet been extensively studied; however, a single Japanese study found that among elderly subjects with ESKD, diabetic nephropathy was more common among those who had had a low birth weight.⁴⁵¹ This observa-

tion may suggest a programming effect of diabetes, kidney disease, or both and requires further study.²⁷⁵

MATERNAL HEALTH AND INTERGENERATIONAL EFFECTS OF PROGRAMMING

Maternal health and nutrition are crucial determinants of a healthy pregnancy, offspring birth weight, and fetal kidney development.²⁷⁹ Socioeconomic and structural factors are important determinants of maternal health and thereby the health of future generations.¹ Maternal prepregnancy underweight is associated with a higher odds of having a small-for-gestational-age (OR: 1.81; 95% CI: 1.76 to 1.87) or low-birth-weight infant (OR: 1.47; 95% CI: 1.27 to 1.71), as well as a higher risk of preterm birth.^{452,453} In contrast, being overweight before pregnancy increases the risk of high birth weight compared with nonobese women (OR: 1.67; 95% CI: 1.42 to 1.97).⁴⁵³ The risks of gestational diabetes and preeclampsia double with prepregnancy maternal overweight.⁴⁵² Preconception care and weight loss may positively impact these adverse outcomes, although around two-thirds of preterm births remain unexplained.^{452,454–456}

Maternal risk factors for low birth weight or preterm birth are highlighted in Table 21.15, and global prevalences of some of these factors are outlined in Table 21.16.¹ Short maternal stature, maternal underweight, or iron deficiency are risk factors for offspring low birth weight.^{284,457–459} Maternal vitamin A levels are associated with newborn kidney size and nephron number.⁴⁶⁰ Supplementation of various nutrients in pregnant women has been shown to reduce the risks of low birth weight or preterm birth, but the long-term impact on childhood or later blood pressure and kidney function are not yet clear (see Table 21.14).^{1,457,461} Maternal smoking, alcohol, and caffeine consumption during pregnancy may also increase

Table 21.15 Maternal Factors Associated With Birth Weight and Preterm Delivery

Maternal Factor	
Developmental	Maternal birth weight <2.5 kg or >4.0 kg Short stature, stunting (height <145 cm)
Behavioral	Cigarette smoking Alcohol consumption Substance and/or drug abuse
Demographic	Age <18 years or >40 years Ethnicity
Health-related	Undernutrition, low maternal BMI Iron deficiency Malaria Diabetes or gestational diabetes mellitus Hypertension Preeclampsia, eclampsia Chronic kidney disease, transplant, dialysis Birth before term Multiple gestations Multiparous (≥3) Assisted reproduction Infections Obesity
Social	Highly active antiretroviral therapy for HIV Prenatal care Unplanned pregnancy, birth spacing Teenage pregnancy Marriage during childhood Conflict, war, stress Environmental conditions Education level Poverty
Environmental	Seasonal variations in nutrient availability Toxin or pollutant exposure

BMI, Body mass index.

From Luyckx VA, Brenner BM. Birth weight, malnutrition and kidney-associated outcomes—a global concern. *Nat Rev Nephrol.* 2015;11:135–149, Box 2.

the risk of low birth weight and preterm birth and have been associated with changes in childhood blood pressures and kidney size and function.^{462–467} Infections during pregnancy or chronic maternal illness all increase the risks of adverse pregnancy outcomes (Table 21.15).^{303,468,469} Acute infections such as malaria are highly prevalent in lower-income regions and may contribute to a large number of low-birth-weight or preterm births annually.⁴⁷⁰ In high-income countries, delays in pregnancy, use of assisted reproduction, or chronic maternal illness are increasingly common, ranging from 3%–20% in different studies.^{471,472} Maternal chronic disease increases the risk of having a preterm birth or small-for-gestational-age infant; therefore, planning of pregnancies in such women is important to minimize these risks.^{456,473} Specifically, pregnant women with CKD have a higher risk of low birth weight, small for gestational age, and preterm birth.^{474,475} CKD is also a major risk factor for preeclampsia, in addition to maternal preexisting hypertension, diabetes, anemia, heart disease, and urinary tract infections/pyelonephritis.⁴⁷⁶ Teenage pregnancies

are at higher risk of preeclampsia, low birth weight, and preterm birth.^{454,477,478} These factors therefore may all impact fetal programming.

In a population-based cohort of women, the odds of pregnancy-associated gestational diabetes or gestational hypertension (including preeclampsia or eclampsia) among those who had been born preterm were significantly increased and were higher with lower gestational age, suggesting a “dose–response” relationship with degree of maternal prematurity (OR: 1.95; 95% CI: 1.54 to 2.47 if mother born at <32 weeks; OR: 1.14; 95% CI: 1.03 to 1.25 if mother born at 32–36 weeks).¹⁹⁰ Gestational hypertension and preeclampsia, in turn, are risk factors for low birth weight and preterm birth, and gestational diabetes is a risk factor for high birth weight; therefore, the offspring of these pregnancies are likely at risk for programmed hypertension and renal dysfunction, including gestational hypertension.⁴⁷⁶ The programming cycle could therefore continue across generations. Maternal low birth weight or preterm birth are also risk factors for having low-birth-weight or preterm infants.^{1,479,480} A direct intergenerational programming effect of maternal preterm birth on subsequent offspring preterm birth is supported by the observed increasing risk of preterm birth with lower maternal gestational age and the absence of risk if the father was born preterm.⁴⁸¹ Maternal obesity and gestational diabetes are increasing worldwide and are established risk factors for both low and high birth weight.^{482–484} Obesity in pregnant women who had themselves been born with low birth weight was associated with a 3.65-fold increased risk of preterm delivery compared with those who were not obese.⁴⁸⁰ Maternal low or high birth weight is also associated with a higher risk of developing gestational diabetes.⁴⁸⁵ Maternal birth circumstances and subsequent obesity therefore affect risk of adverse pregnancy outcomes and may program the health of their offspring. The mechanisms, however, remain unclear.^{391,408,486,487}

In rats, maternal protein restriction results in transgenerational programming persisting into the F2 generation.⁴⁸⁸ These effects have been proposed to be mediated largely by changes in DNA methylation, determined by amino acid availability, resulting in epigenetic changes in gene expression.²¹ Whether these epigenetic changes can be transmitted through the germline and persist in the offspring, or whether a mother who had been subjected to adverse intrauterine events may experience changes in renal function and blood pressure during pregnancy that may, in turn, de novo impact the development of her offspring is still a point of debate.^{391,408,486} The latter hypothesis seems more likely.

IMPACT OF NEPHRON ENDOWMENT IN TRANSPLANTATION

Nephron endowment in transplantation is relevant for both the donor and the recipient. Removal of a kidney in a healthy donor implies loss of 50% of original nephron endowment, and gain of a kidney in a recipient may provide more or less than half of the recipient's original nephron endowment, depending on relative sizes of donor and recipient, periprocedural nephron loss, and other factors. Consideration of nephron endowment prior to transplantation could impact long-term renal health in both donors and recipients.

Table 21.16 Global Incidence/Prevalence of Programming Risk Factors

Maternal Risk Factors	Global Incidence/Prevalence (%)	Reference
Undernutrition women 20–49 years (BMI <18)	12	280, 457
Short stature women of reproductive age (<155 cm)	19.7–68.5	457, 459
Malaria during pregnancy	41.2 (LMIC)	470
Hepatitis B infection (sAg) in women of childbearing age	1.5–8.5 (2005)	563
Anemia in pregnant women (Hb <110 g/L)	38.2 (95 CI: 34.3–42.0)	331, 457
Iron deficiency in pregnant women	19.2 (95 CI: 17.1–21.5)	
Vitamin A deficiency in pregnant women (night blindness)	7.8 (95 CI: 6.5–9.1)	
Zinc deficiency (general population)	17.3 (95 CI: 15.9–18.8)	
Preeclampsia	4.6 (95 CI: 2.7–8.2)	564
Eclampsia	1.4 (95 CI: 1.0–2.0)	
Hyperglycemia in pregnancy women 20–49 years (diabetes/gestational diabetes)	16.9	565
Maternal chronic illness	15.8 (Denmark, 2013)	471
Maternal overweight (BMI 25–29.9 kg/m ²)	30	457, 566, 567
Maternal obesity (BMI ≥30 kg/m ²)	11	457, 482
Maternal smoking	9–16 HIC 2.6 (LMIC)	568, 569
Assisted reproduction technology	2–3 annual births HIC	570, 571
Teenage pregnancy (<19 years)	15 million per year	478, 572, 573
Infant/Child Risk Factors		
Low birth weight	15	44
Preterm birth	11.1	44, 574
Small for gestational age (<10th percentile)	12.1	44
High birth weight (>4 kg)	0.5–14.9 (LMIC)	40
Not weighed at birth	48	43
Child <5 years of age	6.0 (95 CI: 5.1, 7.1)	575
Overweight/obesity, 2016		
Childhood stunting (2016)	22.9 (95 CI: 21.1, 24.7)	

BMI, Body mass index; HIC, high-income country; LMIC, low- and middle-income country; sAg, surface antigen–positive hepatitis B.

IMPLICATIONS OF NEPHRON ENDOWMENT FOR THE DONOR

Living donation, after appropriate donor screening, has generally been presumed to be safe, although recent studies have described some risk of ESKD after kidney donation, highlighting the urgent need to better understand predictors of renal function postdonation in living donors.^{489–492}

In predominantly Caucasian cohorts, hypertension and proteinuria do increase over time in living donors, but renal function remains generally well preserved over the first decades.^{64,493,494}

In a cross-sectional retrospective study of donors followed for up to 40 years, however, donor GFRs were found to decline after 15–17 years postdonation; therefore, duration of follow-up is important to fully understand potential associations with risk.⁴⁹⁵ Obesity significantly increases this risk over time.⁴⁹⁶ There are donor groups, including the young and obese, the older donor, and some ethnic groups, that may be at increased risk of long-term renal dysfunction but have not been well represented in the current literature.^{489,497}

A recent analysis of the U.S. donor population estimated the risk of ESKD among average donors to be 34 cases per 10,000 donors; however, this risk increased to 256 cases per 10,000 in higher-risk donors and was associated with black race, elevated body mass index, donation to a first-degree relative, and with age among nonblack donors.⁴⁹⁷ Black race

was associated with the greatest risk of ESKD (HR: 2.96; 95% CI: 2.25 to 3.89).⁴⁹⁷ Whether developmental programming of nephron number in a prospective donor may impact these risks has not been studied. Among African Americans, we now know that many donors may carry *APOL1* risk genotypes; therefore, this risk may be multifactorial. In support of a programmed risk of accelerated loss of renal function postdonation, Aboriginal Australian donors were found to have a significantly increased risk of hypertension, renal dysfunction, and ESKD at a median of 16 years postdonation compared with Caucasians (Table 21.17).⁴⁹⁸ Similarly, hypertension and proteinuria were much more prevalent among Aboriginal compared with Caucasian Canadian donors at 20 years of follow-up, and African Americans as well as Hispanics had more hypertension and CKD after donation than white Americans (Table 21.17).^{499–501} These data are troubling because the donors were screened prior to donation; therefore, donor nephrectomy can be implicated in the disease process. Importantly, African Americans and Aboriginal Australian populations have lower birth weights than their Caucasian counterparts, and Canadian Aboriginals have higher birth weights; therefore, developmental programming of the kidney may be a risk factor contributing to poorer outcomes postdonation.⁵⁰² Given the increasing need for donors worldwide, better understanding of renal risk in these populations is urgently needed.

Table 21.17 Hypertension and Renal Function in Living Kidney Donors at Risk of Adverse Renal Programming

Population	United States		Australia		Canada		Germany	
	Black Donor	White Donor	Indigenous Donors	Nonindigenous Donors	Aboriginal Donors	White Donors	BW ≤2.5 kg	BW >2.5 kg
Donor number	12,387	71,769	22	28	38	76	18	73
Programmed risk	LBW prematurity	ref	LBW	ref	HBW (offspring of DM pregnancies)	ref	LBW	ref
HT	—	—	50%	6%	42%	19%	39%	15%
Proteinuria	—	—	81%	6%	21%	4%	81%	35%
↓ GFR	—	—	81%	38%	Not different		Not different	
ESKD	74.4 per 10,000 vs. 23.9 per 10,000 in nondonors	22.7 per 10,000 vs. 0.0 per 10,000 in nondonors	19%	0%	1	0	0	0
Follow-up (years)	7.6 (IQR 3.9–11.5)		16.1 (1.27–20.2)	6.37 (2.54–21.2)	14.6 ± 9.3	13.4 ± 9.5	≥5	≥1–3
Reference	490		498		501		503	

BW, Body weight; DM, diabetes mellitus; ESKD, end-stage kidney disease; GFR, glomerular filtration rate; HBW, high birth rate; IQR, interquartile range; LBW, low birth weight.

Reproduced from Low Birth Weight and Nephron Number Working Group. The impact of kidney development on the life course: a consensus document for action. *Nephron*. 2017;136:3–49.

A study from Germany found that GFRs were lower and proteinuria and hypertension were more frequent at 5 years postdonation among donors with birth weights <2.5 kg (see Table 21.17).⁵⁰³ A group of potential donors that has not been studied is women who have had preeclampsia, who are known to be at an increased, albeit small, risk of ESKD.⁵⁰⁴ As discussed earlier, the risk of preeclampsia is increased in women who had been of low birth weight or preterm; therefore, a proportion of women with preeclampsia were likely at risk of renal programming.¹⁹⁰ The risk of gestational hypertension and preeclampsia is also increased among female donors, suggesting that a reduced nephron mass may be a risk factor.⁵⁰⁵ Caution should therefore be advised in considering women with a history of preeclampsia, low birth weight, or having been born preterm as kidney donors, as each of these factors may be a marker for programmed renal risk.¹

IMPLICATIONS OF NEPHRON ENDOWMENT FOR THE RECIPIENT

Prescription of donor kidneys is largely based on immunologic matching. In animal experiments of renal transplantation, however, the impact of transplanted nephron mass, independent of immunologic factors, on the subsequent development of chronic allograft nephropathy has been demonstrated.^{506–510} Despite such evidence, prescription of kidneys on the basis of the physiologic capacity of the donor organ to meet the metabolic needs of the recipient has not generally been considered.⁵¹¹ More and more data are accumulating, however, suggesting a significant impact of transplanted renal mass on long-term posttransplantation outcomes.

Demographic and anthropomorphic factors associated with late renal allograft loss include donor age, sex, and race, as well as recipient BSA.^{512–514} In general, kidneys from older,

female, and African American donors fare worse.^{54,77,95,515} Indirectly, these observations suggest that the intrinsic nephron endowment of the transplanted kidney is likely to play a role in the development of chronic allograft nephropathy. As nephron numbers are not yet measurable in vivo, several investigators have compared recipient and donor BSA as surrogates for metabolic demand and kidney size; others have used kidney weights or renal volumetric measurements by ultrasound as surrogates for nephron mass.^{516–520} Importantly, although kidney mass and kidney volume have been thought to be proportional to nephron number, and measures of body size tend to be proportional to kidney size, these relationships do vary in strength of association⁵⁷; therefore, these data should be interpreted cautiously. In general, however, the preponderance of evidence does support the hypothesis that small kidneys perform less well when transplanted into larger recipients.^{516–520}

A retrospective analysis of 32,083 patients who received a first cadaver kidney found that large recipients of kidneys from small donors had a 43% increased risk of late allograft failure compared with medium-sized recipients receiving kidneys from medium-sized donors.⁵¹⁹ Outcomes were best in small recipients receiving kidneys from large donors. Subsequently, among 69,737 deceased donor kidney transplants, a severe recipient/donor size mismatch (BSA ratio >1.38) was associated with higher 10-year graft loss compared with closer matches, and the risk was doubled in the case of extended criteria donors with severe mismatches (22% vs. 10%, respectively).⁵²¹ Similar findings were seen among recipients of older (>60 years) versus younger kidneys.⁵²² These data suggest that donor and recipient size should be considered in organ allocation decisions, especially if the donor kidney is known to be “suboptimal.” Smaller studies have not consistently found similar results, however, potentially reflecting lack of statistical power.^{518,519}

Kidney size, however, may not always be directly proportional to BSA; therefore, ratios of donor to recipient BSA may not be an ideal method of estimating nephron mass to recipient mismatch. Kidney weight may be a better surrogate for nephron mass.^{54,523} Using this parameter, Kim and associates⁵²⁴ analyzed the ratio of donor kidney weight to recipient body weight (DKW/RBW) in 259 live-donor transplants. These authors found that a higher DKW/RBW of greater than 4.5 g/kg was significantly associated with improved allograft function at 3 years compared with a ratio of less than 3.0 g/kg. A similar study including 964 recipients of cadaveric kidneys, in whom proteinuria and Cockcroft-Gault creatinine clearances were also calculated, found that 10% of the subjects were “strongly” mismatched, having a DKW/RBW ratio of less than 2 g/kg.⁵¹⁵ The DKW/RBW ratio was lowest when male recipients received kidneys from female donors. The risk of having proteinuria higher than 0.5 g/24 h was significantly greater, and developed earlier, in those with DKW/RBW below 2 g/kg as compared with those with higher ratios. Proteinuria was present in 50% of those with DKW/RBW less than 2 g/kg, 33% of those with DKW/RBW of 2 to 4 g/kg, and 23% in those with DKW/RBW of 4 g/kg or greater. At 5 years follow-up, however, there was no difference in graft survival among the three DKW/RBW groups, but the authors conceded that longer follow-up was needed.⁵¹⁵ Subsequent analysis of the same cohort 5 years later showed that GFR declined more rapidly after 7 years in the low versus the high DKW/RBW group, suggesting that the smaller kidneys were likely hyperfiltering early on, which initiated the cycle of progressive nephron loss (see Fig. 21.16).⁵²⁵

In an attempt to more accurately reflect transplanted nephron mass, other investigators have used renal ultrasonography to measure cadaveric transplant kidney (Tx) cross-sectional area in relation to recipient body weight (W) to calculate a “nephron dose index,” Tx/W.⁵²⁶ These authors found that during the first 5 years after transplantation, serum creatinine was significantly lower in patients with a high Tx/W

compared with those with lower values, with a trend toward better graft survival. A similar analysis using renal volume determined by CT angiographic volumetry (done pretransplant in living donors) to recipient BSA found that post-transplant GFRs during the first year correlated with recipient GFR, more significantly among those with a donor/recipient BSA ratio of 1 or less.⁵²⁷ A small kidney transplanted into a large recipient may not have an adequate capacity to meet the metabolic needs of the recipient without imposing glomerular hyperfiltration, which ultimately leads to nephron loss and eventual allograft failure.^{528,529}

Transplanted nephron mass may be a function of congenital endowment and attrition of nephrons with age but is also impacted by peritransplant renal injury (i.e., donor hypotension, prolonged cold and warm ischemia, nephrotoxic immunosuppressive drugs). All of these factors must be closely considered, in addition to immunologic matching, in selection of appropriate recipients in whom the allograft is likely to function for the longest time and therefore provide best possible improvement in quality of life.

GLOBAL HEALTH RELEVANCE OF RENAL PROGRAMMING

The known risk factors—low birth weight, small for gestational age, preterm birth, high birth weight—are prevalent worldwide, in both low- and high-income settings, although underlying causes may differ (see Table 21.16). Millions of infants born each year are at risk, therefore, of future hypertension and kidney disease as well as other programmed conditions. How much developmental programming contributes to the growing burden of chronic noncommunicable diseases worldwide is unknown but may be significant, especially in regions where access to preventive health care is limited. In the current era of sustainable development goals, it may be possible to reduce the impact of developmental programming on future generations through a multisectoral approach encompassing not only the health system but education, infrastructure, and reducing social inequities. Strategies to reduce the impact of developmental programming in individual infants should be simple to implement and, as reviewed in depth elsewhere,¹ include documentation of birth weight and gestational age in each child to identify those at risk; heightened awareness of the risk of neonatal AKI in preterm infants and implementation of strategies to reduce these risks; and education of parents about healthy lifestyle choices to prevent obesity and stunting, low birth weight, small for gestational age, and preterm. Children born to mothers with preeclampsia, diabetes, or obesity should be screened for hypertension and proteinuria, especially if they become overweight; as adults these individuals require long-term screening, early intervention with renoprotective strategies, and strong emphasis on healthy lifestyles.

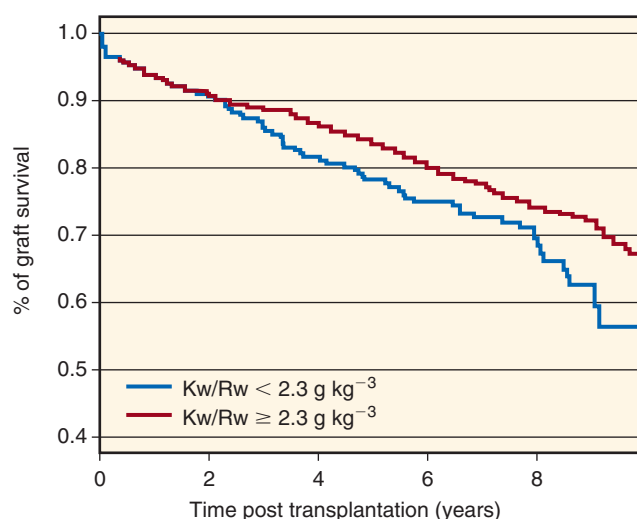


Fig. 21.16 Correlation between kidney weight (Kw)-to-recipient weight (Rw) ratio and long-term graft survival in renal transplantation. Graft survival declines faster with a Kw/Rw <2.3 g/kg ($P = .016$). (From Giral M, Foucher Y, Karam G, et al. Kidney and recipient weight incompatibility reduces long-term graft survival. *J Am Soc Nephrol*. 2010;21:1022–1029.)

CONCLUSION

The association between an adverse fetal environment and subsequent hypertension as well as kidney disease in later life is now quite compelling and appears to be mediated, at least in part, by impaired nephrogenesis and suboptimal

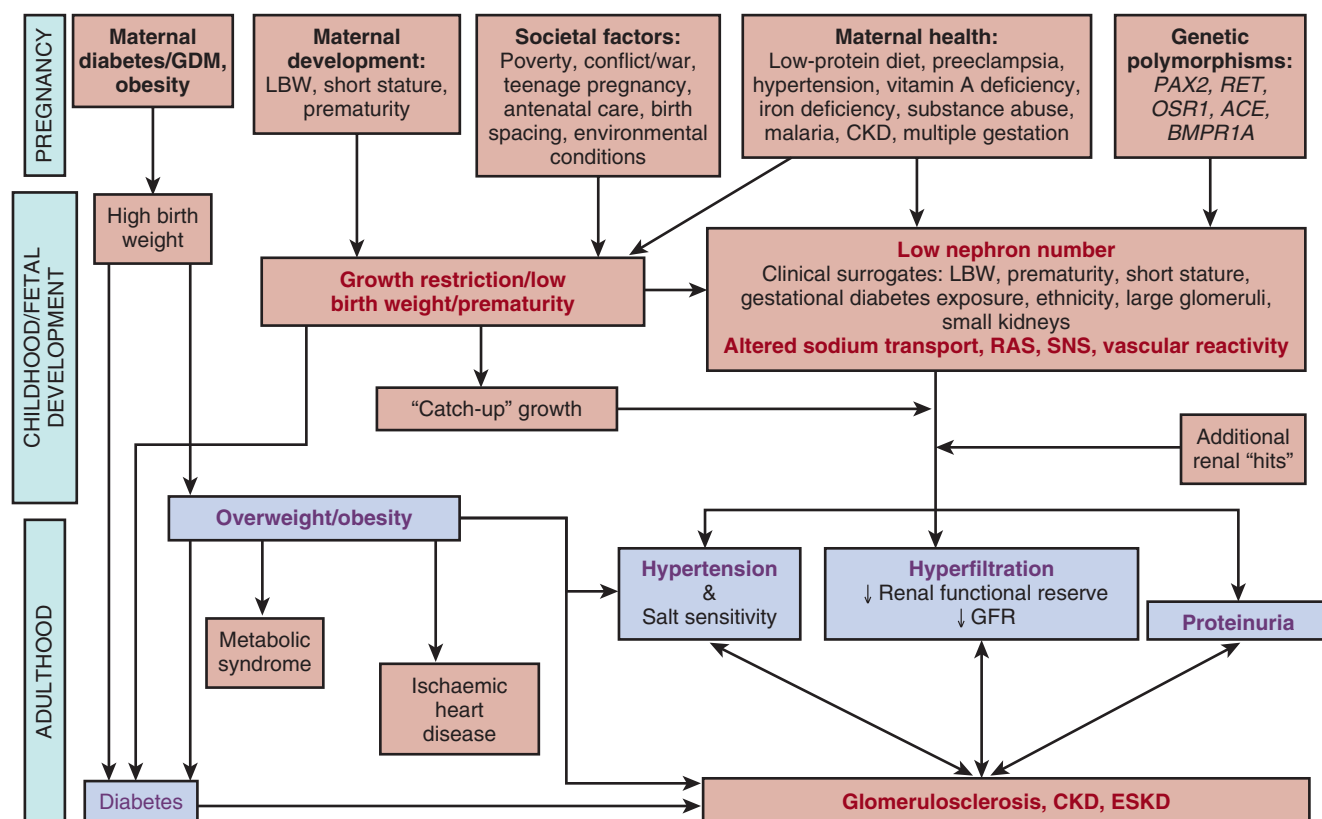


Fig. 21.17 Proposed mechanisms of fetal programming of hypertension and renal disease. CKD, Chronic kidney disease; ESKD, end-stage kidney disease; GDM, gestational diabetes mellitus; GFR, glomerular filtration rate; LBW, low birth weight; RAS, renin-angiotensin system; SNS, sympathetic nervous system. (Adapted from Luyckx VA, Bertram JF, Brenner BM, et al. Effect of fetal and child health on kidney development and long-term risk of hypertension and kidney disease. *Lancet* 2013;382:273–283.)

nephron endowment (Fig. 21.17). Concomitant glomerular hypertrophy and altered expression of sodium transporters in the programmed kidney also contribute to the vicious circle of glomerular hypertension, glomerular injury, and sclerosis, leading to worsening hypertension and ongoing renal injury. In addition, numerous other factors, such as increased oxidative stress, renal inflammation, accelerated senescence, and catch-up body growth, are all likely contributors to ongoing nephron loss and eventual renal disease.^{284,530–533} The number of nephrons in humans without kidney disease varies widely, suggesting that a significant proportion of the general population, especially in areas where high or low birth weights are prevalent, may be at increased risk of developing later-life hypertension and renal dysfunction. Measurement of nephron number in vivo remains an obstacle, with the best surrogate markers thus far being a low birth weight, high birth weight, preterm birth, being small for gestational age, and, in the absence of other known renal diseases, a reduced kidney volume on ultrasound, especially in children, and glomerular enlargement on kidney biopsy. A kidney with a reduced complement of nephrons would have less renal reserve to adapt to dietary excesses or to compensate for renal injury. The molecular mechanisms through which fetal programming exerts its effects on nephrogenesis are varied and likely complementary and intertwined. Although in some animal studies, nephron number and blood pressure can be “rescued” by good postnatal nutrition or vitamin A

supplementation, applicability of these findings to humans requires further study. The fact that even seemingly minor influences, such as composition of maternal diet during fetal life, can have major consequences on renal development in the offspring underscores the critical importance of optimization of perinatal care and early nutrition, which could have a major impact on population health in the future.

Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

KEY REFERENCES

1. Low Birth W, Nephron Number Working G. The impact of kidney development on the life course: a consensus document for action. *Nephron*. 2017;136(1):3–49.
15. Barker DJ. Developmental origins of adult health and disease. *J Epidemiol Community Health*. 2004;58(2):114–115.
26. Barker DJ, Hales CN, Fall CH, et al. Type 2 (non-insulin-dependent) diabetes mellitus, hypertension and hyperlipidaemia (syndrome X): relation to reduced fetal growth. *Diabetologia*. 1993;36(1):62–67.
32. White SL, Perkovic V, Cass A, et al. Is low birth weight an antecedent of CKD in later life? A systematic review of observational studies. *Am J Kidney Dis*. 2009;54(2):248–261.
34. Langley-Evans S, Langley-Evans A, Marchand M. Nutritional programming of blood pressure and renal morphology. *Arch Physiol Biochem*. 2003;111(1):8–16.
36. de Jong F, Monuteaux MC, van Elburg RM, et al. Systematic review and meta-analysis of preterm birth and later systolic blood pressure. *Hypertension*. 2012;59(2):226–234.
37. Mu M, Wang SF, Sheng J, et al. Birth weight and subsequent blood pressure: a meta-analysis. *Arch Cardiovasc Dis*. 2012;105(2):99–113.

38. Nelson RG, Morgenstern H, Bennett PH. Intrauterine diabetes exposure and the risk of renal disease in diabetic pima Indians. *Diabetes*. 1998;47(9):1489–1493.
48. Brenner BM, Garcia DL, Anderson S. Glomeruli and blood pressure. Less of one, more the other? *Am J Hypertens*. 1988;1(4 Pt 1):335–347.
58. Puelles VG, Hoy WE, Hughson MD, et al. Glomerular number and size variability and risk for kidney disease. *Curr Opin Nephrol Hypertens*. 2011;20(1):7–15.
75. Kett MM, Denton KM. Renal programming: cause for concern? *Am J Physiol Regul Integr Comp Physiol*. 2011;300(4):R791–R803.
87. Hughson M, Farris AB, Douglas-Denton R, et al. Glomerular number and size in autopsy kidneys: the relationship to birth weight. *Kidney Int*. 2003;63(6):2113–2122.
99. Hoy WE, Hughson MD, Diouf B, et al. Distribution of volumes of individual glomeruli in kidneys at autopsy: association with physical and clinical characteristics and with ethnic group. *Am J Nephrol*. 2011;33(suppl 1):15–20.
108. Vehaskari VM, Aviles DH, Manning J. Prenatal programming of adult hypertension in the rat. *Kidney Int*. 2001;59(1):238–245.
112. Wlodek ME, Mibus A, Tan A, et al. Normal lactational environment restores nephron endowment and prevents hypertension after placental restriction in the rat. *J Am Soc Nephrol*. 2007;18(6):1688–1696.
114. Baum M. Role of the kidney in the prenatal and early postnatal programming of hypertension. *Am J Physiol Renal Physiol*. 2010;298(2):F235–F247.
121. Zhang Z, Quinlan J, Hoy W, et al. A common RET variant is associated with reduced newborn kidney size and function. *J Am Soc Nephrol*. 2008;19(10):2027–2034.
122. Rodriguez MM, Gomez AH, Abitbol CL, et al. Histomorphometric analysis of postnatal glomerulogenesis in extremely preterm infants. *Pediatr Dev Pathol*. 2004;7(1):17–25.
159. Moritz KM, Cuffe JS, Wilson LB, et al. Review: sex specific programming: a critical role for the renal renin-angiotensin system. *Placenta*. 2010;31(suppl):S40–S46.
170. Ojeda NB, Intapad S, Alexander BT. Sex differences in the developmental programming of hypertension. *Acta physiologica*. 2014;210(2):307–316.
181. Jones SE, Bilous RW, Flyvbjerg A, et al. Intra-uterine environment influences glomerular number and the acute renal adaptation to experimental diabetes. *Diabetologia*. 2001;44(6):721–728.
187. Abitbol CL, Rodriguez MM. The long-term renal and cardiovascular consequences of prematurity. *Nat Rev Nephrol*. 2012;8(5):265–274.
190. Boivin A, Luo ZC, Audibert F, et al. Pregnancy complications among women born preterm. *CMAJ*. 2012;184(16):1777–1784.
209. Khalsa DD, Beydoun HA, Carmody JB. Prevalence of chronic kidney disease risk factors among low birth weight adolescents. *Pediatr Nephrol*. 2016;31(9):1509–1516.
217. Bergvall N, Iliadou A, Johansson S, et al. Genetic and shared environmental factors do not confound the association between birth weight and hypertension: a study among Swedish twins. *Circulation*. 2007;115(23):2931–2938.
223. Schreuder MF, Wilhelm AJ, Bokenkamp A, et al. Impact of gestational age and birth weight on amikacin clearance on day 1 of life. *Clin J Am Soc Nephrol*. 2009;4(11):1774–1778.
224. Bacchetta J, Harambat J, Dubourg L, et al. Both extrauterine and intrauterine growth restriction impair renal function in children born very preterm. *Kidney Int*. 2009;76(4):445–452.
230. Keijzer-Veen MG, Schrevel M, Finken MJ, et al. Microalbuminuria and lower glomerular filtration rate at young adult age in subjects born very premature and after intrauterine growth retardation. *J Am Soc Nephrol*. 2005;16(9):2762–2768.
235. Hoy WE, White AV, Tipiloura B, et al. The multideterminant model of renal disease in a remote Australian aboriginal population in the context of early life risk factors: lower birth weight, childhood post-streptococcal glomerulonephritis, and current body mass index influence levels of albuminuria in young aboriginal adults. *Clin Nephrol*. 2015;83(7 suppl 1):75–81.
244. Selewski DT, Charlton JR, Jetton JG, et al. Neonatal acute kidney injury. *Pediatrics*. 2015;136(2):e463–e473.
246. Sutherland M, Ryan D, Black MJ, et al. Long-term renal consequences of preterm birth. *Clin Perinatol*. 2014;41(3):561–573.
274. Hsu CW, Yamamoto KT, Henry RK, et al. Prenatal risk factors for childhood CKD. *J Am Soc Nephrol*. 2014;25(9):2105–2111.
276. Hanson M, Gluckman P. Developmental origins of noncommunicable disease: population and public health implications. *Am J Clin Nutr*. 2011;94(6 suppl):1754S–1758S.
284. Luyckx VA, Brenner BM. Birth weight, malnutrition and kidney-associated outcomes—a global concern. *Nat Rev Nephrol*. 2015;11(3):135–149.
299. Little MH, McMahon AP. Mammalian kidney development: principles, progress, and projections. *Cold Spring Harb Perspect Biol*. 2012;4(5).
300. Moritz KM, Wintour EM, Black MJ, et al. Factors influencing mammalian kidney development: implications for health in adult life. *Adv Anat Embryol Cell Biol*. 2008;196:1–78.
303. Luyckx VA. Preterm birth and its impact on renal health. *Semin Nephrol*. 2017;37(4):311–319.
315. Langley-Evans SC. Nutritional programming of disease: unravelling the mechanism. *J Anat*. 2009;215(1):36–51.
341. Amri K, Freund N, Vilar J, et al. Adverse effects of hyperglycemia on kidney development in rats: in vivo and in vitro studies. *Diabetes*. 1999;48(11):2240–2245.
348. Jackson CM, Alexander BT, Roach L, et al. Exposure to maternal overnutrition and a high-fat diet during early postnatal development increases susceptibility to renal and metabolic injury later in life. *Am J Physiol Renal Physiol*. 2012;302(6):F774–F783.
385. Chevalier RL. Congenital urinary tract obstruction: the long view. *Adv Chronic Kidney Dis*. 2015;22(4):312–319.
391. O'Sullivan L, Combes AN, Little MH, et al. Epigenetics and developmental programming of adult onset diseases. *Pediatr Nephrol*. 2012;27(12):2175–2182.
405. Ojeda NB. Prenatal programming of hypertension: role of sympathetic response to physical stress. *Hypertension*. 2013;61(1):16–17.
427. Jain V, Singhal A. Catch up growth in low birth weight infants: striking a healthy balance. *Rev Endocr Metab Disord*. 2012;13(2):141–147.
440. Grijalva-Eternod CS, Lawlor DA, Wells JC. Testing a capacity-load model for hypertension: disentangling early and late growth effects on childhood blood pressure in a prospective birth cohort. *PLoS ONE*. 2013;8(2):e56078.
441. Abitbol CL, Chandar J, Rodriguez MM, et al. Obesity and preterm birth: additive risks in the progression of kidney disease in children. *Pediatr Nephrol*. 2009;24(7):1363–1370.
442. Abitbol CL, Bauer CR, Montane B, et al. Long-term follow-up of extremely low birth weight infants with neonatal renal failure. *Pediatr Nephrol*. 2003;18(9):887–893.
479. Boivin A, Luo ZC, Audibert F, et al. Risk for preterm and very preterm delivery in women who were born preterm. *Obstet Gynecol*. 2015;125(5):1177–1184.
486. Aiken CE, Ozanne SE. Transgenerational developmental programming. *Hum Reprod Update*. 2014;20(1):63–75.
489. Mueller TF, Luyckx VA. The natural history of residual renal function in transplant donors. *J Am Soc Nephrol*. 2012;23(9):1462–1466.
525. Giral M, Foucher Y, Karam G, et al. Kidney and recipient weight incompatibility reduces long-term graft survival. *J Am Soc Nephrol*. 2010;21(6):1022–1029.

REFERENCES

- Low Birth W, Nephron Number Working G. The impact of kidney development on the life course: a consensus document for action. *Nephron*. 2017;136(1):3–49.
- Kemper MJ, Muller-Wiefel DE. Renal function in congenital anomalies of the kidney and urinary tract. *Curr Opin Urol*. 2001;11(6):571–575.
- Schreuder MF, Langemeijer ME, Bokenkamp A, et al. Hypertension and microalbuminuria in children with congenital solitary kidneys. *J Paediatr Child Health*. 2008;44(6):363–368.
- Genovese G, Friedman DJ, Ross MD, et al. Association of trypanolytic apol1 variants with kidney disease in African Americans. *Science*. 2010;329(5993):841–845.
- Bianchi G. Genetic variations of tubular sodium reabsorption leading to “primary” hypertension: from gene polymorphism to clinical symptoms. *Am J Physiol Regul Integr Comp Physiol*. 2005;289(6):R1536–R1549.
- Hubner N, Yagil C, Yagil Y. Novel integrative approaches to the identification of candidate genes in hypertension. *Hypertension*. 2006;47(1):1–5.
- Kaplan NM, Opie LH. Controversies in hypertension. *Lancet*. 2006;367(9505):168–176.
- Lifton RP, Wilson FH, Choate KA, et al. Salt and blood pressure: new insight from human genetic studies. *Cold Spring Harb Symp Quant Biol*. 2002;67:445–450.
- Tzur S, Rosset S, Shemer R, et al. Missense mutations in the APOL1 gene are highly associated with end stage kidney disease risk previously attributed to the MYH9 gene. *Hum Genet*. 2010;128(3):345–350.
- Hoy WE, Rees M, Kile E, et al. Low birthweight and renal disease in Australian aborigines. *Lancet*. 1998;352(9143):1826–1827.
- Lackland DT. Mechanisms and fetal origins of kidney disease. *J Am Soc Nephrol*. 2005;16(9):2531–2532.
- Nelson RG, Morgenstern H, Bennett PH. Birth weight and renal disease in Pima Indians with type 2 diabetes mellitus. *Am J Epidemiol*. 1998;148(7):650–656.
- Norton JM, Moxey-Mims MM, Eggers PW, et al. Social determinants of racial disparities in CKD. *J Am Soc Nephrol*. 2016;27(9):2576–2595.
- Luyckx VA, Bertram JF, Brenner BM, et al. Effect of fetal and child health on kidney development and long-term risk of hypertension and kidney disease. *Lancet*. 2013;382(9888):273–283.
- Barker DJ. Developmental origins of adult health and disease. *J Epidemiol Community Health*. 2004;58(2):114–115.
- Whincup PH, Kaye SJ, Owen CG, et al. Birth weight and risk of type 2 diabetes: a systematic review. *JAMA*. 2008;300(24):2886–2897.
- Hales CN, Barker DJ. Type 2 (non-insulin-dependent) diabetes mellitus: the thrifty phenotype hypothesis. *Diabetologia*. 1992;35(7):595–601.
- Demicheva E, Crispi F. Long-term follow-up of intrauterine growth restriction: cardiovascular disorders. *Fetal Diagn Ther*. 2014;36(2):143–153.
- Parkinson JR, Hyde MJ, Gale C, et al. Preterm birth and the metabolic syndrome in adult life: a systematic review and meta-analysis. *Pediatrics*. 2013;131(4):e1240–e1263.
- Rothermund L, Lorenz M, Schnieper A, et al. Impact of nephron number dosing on cardiorenal damage and effects of ACE inhibition. *Am J Hypertens*. 2011;24(4):474–481.
- McMillen IC, Robinson JS. Developmental origins of the metabolic syndrome: prediction, plasticity, and programming. *Physiol Rev*. 2005;85(2):571–633.
- Drake AJ, Walker BR. The intergenerational effects of fetal programming: non-genomic mechanisms for the inheritance of low birth weight and cardiovascular risk. *J Endocrinol*. 2004;180(1):1–16.
- Nathanielsz PW, Poston L, Taylor PD. In utero exposure to maternal obesity and diabetes: animal models that identify and characterize implications for future health. *Clin Perinatol*. 2007;34(4):515–526, v.
- Forsdahl A. Are poor living conditions in childhood and adolescence an important risk factor for arteriosclerotic heart disease? *Br J Prev Soc Med*. 1977;31(2):91–95.
- Kermack WO, McKendrick AG, McKinlay PL. Death-rates in Great Britain and Sweden. Some general regularities and their significance. *Lancet*. 1934;i:698–703.
- Barker DJ, Hales CN, Fall CH, et al. Type 2 (non-insulin-dependent) diabetes mellitus, hypertension and hyperlipidaemia (syndrome X): relation to reduced fetal growth. *Diabetologia*. 1993;36(1):62–67.
- Bellinger L, Langley-Evans SC. Fetal programming of appetite by exposure to a maternal low protein diet in the rat. *Clin Sci*. 2005;109(4):413–420.
- Gardner DS, Tingey K, Van Bon BW, et al. Programming of glucose-insulin metabolism in adult sheep after maternal undernutrition. *Am J Physiol Regul Integr Comp Physiol*. 2005;289(4):R947–R954.
- Wust S, Entringer S, Federenko IS, et al. Birth weight is associated with salivary cortisol responses to psychosocial stress in adult life. *Psychoneuroendocrinology*. 2005;30(6):591–598.
- Li S, Chen SC, Shlipak M, et al. Low birth weight is associated with chronic kidney disease only in men. *Kidney Int*. 2008;73(5):637–642.
- Vikse BE, Irgens LM, Leivestad T, et al. Low birth weight increases risk for end-stage renal disease. *J Am Soc Nephrol*. 2008;19(1):151–157.
- White SL, Perkovic V, Cass A, et al. Is low birth weight an antecedent of CKD in later life? A systematic review of observational studies. *Am J Kidney Dis*. 2009;54(2):248–261.
- Zetterstrom K, Lindeberg S, Haglund B, et al. Being born small for gestational age increases the risk of severe pre-eclampsia. *BJOG*. 2007;114(3):319–324.
- Langley-Evans S, Langley-Evans A, Marchand M. Nutritional programming of blood pressure and renal morphology. *Arch Physiol Biochem*. 2003;111(1):8–16.
- Vehaskari VM, Woods LL. Prenatal programming of hypertension: lessons from experimental models. *J Am Soc Nephrol*. 2005;16(9):2545–2556.
- de Jong F, Monuteaux MC, van Elburg RM, et al. Systematic review and meta-analysis of preterm birth and later systolic blood pressure. *Hypertension*. 2012;59(2):226–234.
- Mu M, Wang SF, Sheng J, et al. Birth weight and subsequent blood pressure: a meta-analysis. *Arch Cardiovasc Dis*. 2012;105(2):99–113.
- Nelson RG, Morgenstern H, Bennett PH. Intrauterine diabetes exposure and the risk of renal disease in diabetic Pima Indians. *Diabetes*. 1998;47(9):1489–1493.
- Zhang Y, Li H, Liu SJ, et al. The associations of high birth weight with blood pressure and hypertension in later life: a systematic review and meta-analysis. *Hypertens Res*. 2013;36(8):725–735.
- Koyanagi A, Zhang J, Dagvadorj A, et al. Macrosomia in 23 developing countries: an analysis of a multicountry, facility-based, cross-sectional survey. *Lancet*. 2013;381(9865):476–483.
- Ruggajo P, Skrunes R, Svarstad E, et al. Familial factors, low birth weight, and development of ESRD: a nationwide registry study. *Am J Kidney Dis*. 2016;67(4):601–608.
- Blencowe H, Cousens S, Chou D, et al. Born too soon: the global epidemiology of 15 million preterm births. *Reprod Health*. 2013;10(suppl 1):S2.
- World Health Organisation. *Global nutrition targets 2025: low birth weight policy brief (WHO/NMH/NHD/14.5)*. 2014. http://www.who.int/nutrition/publications/globaltargets2025_policybrief_lbww/en/.
- Lee AC, Katz J, Blencowe H, et al. National and regional estimates of term and preterm babies born small for gestational age in 138 low-income and middle-income countries in 2010. *Lancet Glob Health*. 2013;1(1):e26–e36.
- Guyton AC, Coleman TG, Young DB, et al. Salt balance and long-term blood pressure control. *Annu Rev Med*. 1980;31:15–27.
- Guyton AC, Young DB, DeClue JW, et al. Fluid balance, renal function, and blood pressure. *Clin Nephrol*. 1975;4(4):122–126.
- Guidi E, Menghetti D, Milani S, et al. Hypertension may be transplanted with the kidney in humans: a long-term historical prospective follow-up of recipients grafted with kidneys coming from donors with or without hypertension in their families. *J Am Soc Nephrol*. 1996;7(8):1131–1138.
- Brenner BM, Garcia DL, Anderson S. Glomeruli and blood pressure. Less of one, more the other? *Am J Hypertens*. 1988;1(4 Pt 1):335–347.
- Hsu CY, Lin F, Vittinghoff E, et al. Racial differences in the progression from chronic renal insufficiency to end-stage renal disease in the United States. *J Am Soc Nephrol*. 2003;14(11):2902–2907.
- Lackland DT, Bendall HE, Osmond C, et al. Low birth weights contribute to high rates of early-onset chronic renal failure in the southeastern United States. *Arch Intern Med*. 2000;160(10):1472–1476.
- Lackland DT, Egan BM, Ferguson PL. Low birth weight as a risk factor for hypertension. *J Clin Hypertens (Greenwich)*. 2003;5(2):133–136.
- Lackland DT, Egan BM, Syddall HE, et al. Associations between birth weight and antihypertensive medication in black and white medicated recipients. *Hypertension*. 2002;39(1):179–183.
- Bertram JF. Counting in the kidney. *Kidney Int*. 2001;59(2):792–796.

54. Nyengaard JR, Bendtsen TF. Glomerular number and size in relation to age, kidney weight, and body surface in normal man. *Anat Rec*. 1992;232(2):194–201.
55. Kett MM, Bertram JF. Nephron endowment and blood pressure: what do we really know? *Curr Hypertens Rep*. 2004;6(2):133–139.
56. Sterio DC. The unbiased estimation of number and sizes of arbitrary particles using the disector. *J Microsc*. 1984;134(Pt 2):127–136.
57. Bueters RR, van de Kar NC, Schreuder MF. Adult renal size is not a suitable marker for nephron numbers: an individual patient data meta-analysis. *Kidney Blood Press Res*. 2013;37(6):540–546.
58. Puelles VG, Hoy WE, Hughson MD, et al. Glomerular number and size variability and risk for kidney disease. *Curr Opin Nephrol Hypertens*. 2011;20(1):7–15.
59. Keller G, Zimmer G, Mall G, et al. Nephron number in patients with primary hypertension. *N Engl J Med*. 2003;348(2):101–108.
60. Hoy WE, Douglas-Denton RN, Hughson MD, et al. A stereological study of glomerular number and volume: preliminary findings in a multiracial study of kidneys at autopsy. *Kidney Int Suppl*. 2003;83:S31–S37.
61. Denic A, Lieske JC, Chakkera HA, et al. The substantial loss of nephrons in healthy human kidneys with aging. *J Am Soc Nephrol*. 2017;28(1):313–320.
62. Bhatena DB, Julian BA, McMorow RG, et al. Focal sclerosis of hypertrophied glomeruli in solitary functioning kidneys of humans. *Am J Kidney Dis*. 1985;5(5):226–232.
63. Kasike BL, Ma JZ, Louis TA, et al. Long-term effects of reduced renal mass in humans. *Kidney Int*. 1995;48(3):814–819.
64. Ibrahim HN, Foley R, Tan L, et al. Long-term consequences of kidney donation. *N Engl J Med*. 2009;360(5):459–469.
65. Moritz KM, Wintour EM, Dodic M. Fetal uninephrectomy leads to postnatal hypertension and compromised renal function. *Hypertension*. 2002;39(6):1071–1076.
66. Woods LL, Weeks DA, Rasch R. Hypertension after neonatal uninephrectomy in rats precedes glomerular damage. *Hypertension*. 2001;38(3):337–342.
67. Singh RR, Denton KM, Bertram JF, et al. Development of cardiovascular disease due to renal insufficiency in male sheep following fetal unilateral nephrectomy. *J Hypertens*. 2009;27(2):386–396.
68. Nyengaard JR. Number and dimensions of rat glomerular capillaries in normal development and after nephrectomy. *Kidney Int*. 1993;43(5):1049–1057.
69. Wang X, Johnson AC, Sasser JM, et al. Spontaneous one-kidney rats are more susceptible to develop hypertension by DOCA-NaCl and subsequent kidney injury compared with uninephrectomized rats. *Am J Physiol Renal Physiol*. 2016;310(10):F1054–F1064.
70. Sergio M, Galarreta CI, Thornhill BA, et al. The fate of nephrons in congenital obstructive nephropathy: adult recovery is limited by nephron number despite early release of obstruction. *J Urol*. 2015;194(5):1463–1472.
71. Abou Jaoude P, Dubourg L, Bacchetta J, et al. Congenital versus acquired solitary kidney: is the difference relevant? *Nephrol Dial Transplant*. 2011;26(7):2188–2194.
72. Abi Khalil C, Travert F, Fetita S, et al. Fetal exposure to maternal type 1 diabetes is associated with renal dysfunction at adult age. *Diabetes*. 2010;59(10):2631–2636.
73. Chevalier RL, Thornhill BA, Chang AY, et al. Recovery from release of ureteral obstruction in the rat: relationship to nephrogenesis. *Kidney Int*. 2002;61(6):2033–2043.
74. Douglas-Denton R, Moritz KM, Bertram JF, et al. Compensatory renal growth after unilateral nephrectomy in the ovine fetus. *J Am Soc Nephrol*. 2002;13(2):406–410.
75. Kett MM, Denton KM. Renal programming: cause for concern? *Am J Physiol Regul Integr Comp Physiol*. 2011;300(4):R791–R803.
76. Lenihan CR, Busque S, Derby G, et al. Longitudinal study of living kidney donor glomerular dynamics after nephrectomy. *J Clin Invest*. 2015;125(3):1311–1318.
77. Fulladosa X, Moreso F, Narvaez JA, et al. Estimation of total glomerular number in stable renal transplants. *J Am Soc Nephrol*. 2003;14(10):2662–2668.
78. Tan JC, Busque S, Workeneh B, et al. Effects of aging on glomerular function and number in living kidney donors. *Kidney Int*. 2010;78(7):686–692.
79. Lenihan CR, Busque S, Derby G, et al. The association of predonation hypertension with glomerular function and number in older living kidney donors. *J Am Soc Nephrol*. 2015;26(6):1261–1267.
80. Beeman SC, Cullen-McEwen LA, Puelles VG, et al. MRI-based glomerular morphology and pathology in whole human kidneys. *Am J Physiol Renal Physiol*. 2014;306(11):F1381–F1390.
81. Beeman SC, Mandarino LJ, Georges JF, et al. Cationized ferritin as a magnetic resonance imaging probe to detect microstructural changes in a rat model of non-alcoholic steatohepatitis. *Magn Reson Med*. 2013;70(6):1728–1738.
82. Beeman SC, Zhang M, Gubhaju L, et al. Measuring glomerular number and size in perfused kidneys using MRI. *Am J Physiol Renal Physiol*. 2011;300(6):F1454–F1457.
83. Heilmann M, Neudecker S, Wolf I, et al. Quantification of glomerular number and size distribution in normal rat kidneys using magnetic resonance imaging. *Nephrol Dial Transplant*. 2012;27(1):100–107.
84. Baldeomar EJ, Charlton JR, Beeman SC, et al. Phenotyping by magnetic resonance imaging nondestructively measures glomerular number and volume distribution in mice with and without nephron reduction. *Kidney Int*. 2016;89(2):498–505.
85. Klingberg A, Hasenberg A, Ludwig-Portugall I, et al. Fully automated evaluation of total glomerular number and capillary tuft size in nephritic kidneys using lightsheet microscopy. *J Am Soc Nephrol*. 2017;28(2):452–459.
86. Cortes P, Zhao X, Dumler F, et al. Age-related changes in glomerular volume and hydroxyproline content in rat and human. *J Am Soc Nephrol*. 1992;2(12):1716–1725.
87. Hughson M, Farris AB, Douglas-Denton R, et al. Glomerular number and size in autopsy kidneys: the relationship to birth weight. *Kidney Int*. 2003;63(6):2113–2122.
88. Puelles VG, Douglas-Denton RN, Cullen-McEwen LA, et al. Podocyte number in children and adults: associations with glomerular size and numbers of other glomerular resident cells. *J Am Soc Nephrol*. 2015;26(9):2277–2288.
89. Hoy WE, Hughson MD, Zimanyi M, et al. Distribution of volumes of individual glomeruli in kidneys at autopsy: association with age, nephron number, birth weight and body mass index. *Clin Nephrol*. 2010;74(S1):105–112.
90. Hoy WE, Hughson MD, Bertram JF, et al. Nephron number, hypertension, renal disease, and renal failure. *J Am Soc Nephrol*. 2005;16(9):2557–2564.
91. Manalich R, Reyes L, Herrera M, et al. Relationship between weight at birth and the number and size of renal glomeruli in humans: a histomorphometric study. *Kidney Int*. 2000;58(2):770–773.
92. Zimanyi MA, Hoy WE, Douglas-Denton RN, et al. Nephron number and individual glomerular volumes in male Caucasian and African American subjects. *Nephrol Dial Transplant*. 2009;24(8):2428–2433.
93. Abdi R, Dong VM, Rubel JR, et al. Correlation between glomerular size and long-term renal function in patients with substantial loss of renal mass. *J Urol*. 2003;170(1):42–44.
94. Abdi R, Slakey D, Kittur D, et al. Baseline glomerular size as a predictor of function in human renal transplantation. *Transplantation*. 1998;66(3):329–333.
95. Abdi R, Slakey D, Kittur D, et al. Heterogeneity of glomerular size in normal donor kidneys: impact of race. *Am J Kidney Dis*. 1998;32(1):43–46.
96. Hoy WE, Wang Z, VanBuynder P, et al. The natural history of renal disease in Australian Aborigines. Part 1. Changes in albuminuria and glomerular filtration rate over time. *Kidney Int*. 2001;60(1):243–248.
97. Hoy WE, Wang Z, VanBuynder P, et al. The natural history of renal disease in Australian aborigines. Part 2. Albuminuria predicts natural death and renal failure. *Kidney Int*. 2001;60(1):249–256.
98. Lemley KV. A basis for accelerated progression of diabetic nephropathy in Pima Indians. *Kidney Int Suppl*. 2003;83:S38–S42.
99. Hoy WE, Hughson MD, Diouf B, et al. Distribution of volumes of individual glomeruli in kidneys at autopsy: association with physical and clinical characteristics and with ethnic group. *Am J Nephrol*. 2011;33(suppl 1):15–20.
100. Lu MC, Halfon N. Racial and ethnic disparities in birth outcomes: a life-course perspective. *Matern Child Health J*. 2003;7(1):13–30.
101. Schmidt K, Pesce C, Liu Q, et al. Large glomerular size in Pima Indians: lack of change with diabetic nephropathy. *J Am Soc Nephrol*. 1992;3(2):229–235.
102. Young RJ, Hoy WE, Kincaid-Smith P, et al. Glomerular size and glomerulosclerosis in Australian aborigines. *Am J Kidney Dis*. 2000;36(3):481–489.
103. Celsi G, Kistner A, Aizman R, et al. Prenatal dexamethasone causes oligonephronia, sodium retention, and higher blood pressure in the offspring. *Pediatr Res*. 1998;44(3):317–322.

104. Gilbert T, Lelievre-Pegorier M, Merlet-Benichou C. Long-term effects of mild oligonephronia induced in utero by gentamicin in the rat. *Pediatr Res*. 1991;30(5):450–456.
105. Langley-Evans SC. Intrauterine programming of hypertension in the rat: nutrient interactions. *Comp Biochem Physiol A Physiol*. 1996;114(4):327–333.
106. Merlet-Benichou C. Influence of fetal environment on kidney development. *Int J Dev Biol*. 1999;43(5 Spec No):453–456.
107. Ortiz LA, Quan A, Weinberg A, et al. Effect of prenatal dexamethasone on rat renal development. *Kidney Int*. 2001;59(5):1663–1669.
108. Vehaskari VM, Aviles DH, Manning J. Prenatal programming of adult hypertension in the rat. *Kidney Int*. 2001;59(1):238–245.
109. Woods LL, Ingelfinger JR, Nyengaard JR, et al. Maternal protein restriction suppresses the newborn renin-angiotensin system and programs adult hypertension in rats. *Pediatr Res*. 2001;49(4):460–467.
110. Alexander BT. Placental insufficiency leads to development of hypertension in growth-restricted offspring. *Hypertension*. 2003;41(3):457–462.
111. Moritz KM, Mazzuca MQ, Siebel AL, et al. Uteroplacental insufficiency causes a nephron deficit, modest renal insufficiency but no hypertension with ageing in female rats. *J Physiol*. 2009;587(Pt 11):2635–2646.
112. Wlodek ME, Mibus A, Tan A, et al. Normal lactational environment restores nephron endowment and prevents hypertension after placental restriction in the rat. *J Am Soc Nephrol*. 2007;18(6):1688–1696.
113. Stelloh C, Allen KP, Mattson DL, et al. Prematurity in mice leads to reduction in nephron number, hypertension, and proteinuria. *Transl Res*. 2012;159(2):80–89.
114. Baum M. Role of the kidney in the prenatal and early postnatal programming of hypertension. *Am J Physiol Renal Physiol*. 2010;298(2):F235–F247.
115. Sampogna RV, Schneider L, Al-Awqati Q. Developmental programming of branching morphogenesis in the kidney. *J Am Soc Nephrol*. 2015;26(10):2414–2422.
116. Boubred F, Daniel L, Buffat C, et al. The magnitude of nephron number reduction mediates intrauterine growth-restriction-induced long term chronic renal disease in the rat. A comparative study in two experimental models. *J Transl Med*. 2016;14(1):331.
117. Bertram JF, Douglas-Denton RN, Diouf B, et al. Human nephron number: implications for health and disease. *Pediatr Nephrol*. 2011;26(9):1529–1533.
118. Faa G, Gerosa C, Fanni D, et al. Marked interindividual variability in renal maturation of preterm infants: lessons from autopsy. *J Matern Fetal Neonatal Med*. 2010;23(suppl 3):129–133.
119. Sutherland MR, Gubhaju L, Moore L, et al. Accelerated maturation and abnormal morphology in the preterm neonatal kidney. *J Am Soc Nephrol*. 2011;22(7):1365–1374.
120. Hughson MD, Douglas-Denton R, Bertram JF, et al. Hypertension, glomerular number, and birth weight in African Americans and white subjects in the southeastern United States. *Kidney Int*. 2006;69(4):671–678.
121. Zhang Z, Quinlan J, Hoy W, et al. A common RET variant is associated with reduced newborn kidney size and function. *J Am Soc Nephrol*. 2008;19(10):2027–2034.
122. Rodriguez MM, Gomez AH, Abitbol CL, et al. Histomorphometric analysis of postnatal glomerulogenesis in extremely preterm infants. *Pediatr Dev Pathol*. 2004;7(1):17–25.
123. Hinchliffe SA, Howard CV, Lynch MR, et al. Renal developmental arrest in sudden infant death syndrome. *Pediatr Pathol*. 1993;13(3):333–343.
124. Hinchliffe SA, Lynch MR, Sargent PH, et al. The effect of intrauterine growth retardation on the development of renal nephrons. *Br J Obstet Gynaecol*. 1992;99(4):296–301.
125. Lackland DT, Barker DJ. Birth weight: a predictive medicine consideration for the disparities in CKD. *Am J Kidney Dis*. 2009;54(2):191–193.
126. Nyengaard JR, Bendtsen TF, Mogensen CE. Low birth weight—is it associated with few and small glomeruli in normal subjects and NIDDM patients? *Diabetologia*. 1996;39(12):1634–1637.
127. Deutinger J, Bartl W, Pfersmann C, et al. Fetal kidney volume and urine production in cases of fetal growth retardation. *J Perinat Med*. 1987;15(3):307–315.
128. Kurjak A, Kirkinen P, Latin V, et al. Ultrasonic assessment of fetal kidney function in normal and complicated pregnancies. *Am J Obstet Gynecol*. 1981;141(3):266–270.
129. Silver LE, Decamps PJ, Korst LM, et al. Intrauterine growth restriction is accompanied by decreased renal volume in the human fetus. *Am J Obstet Gynecol*. 2003;188(5):1320–1325.
130. Kandasamy Y, Smith R, Wright IM, et al. Extra-uterine renal growth in preterm infants: oligonephropathy and prematurity. *Pediatr Nephrol*. 2013;28(9):1791–1796.
131. Schmidt IM, Damgaard IN, Boisen KA, et al. Increased kidney growth in formula-fed versus breast-fed healthy infants. *Pediatr Nephrol*. 2004;19(10):1137–1144.
132. Iyengar A, Nesargi S, George A, et al. Are low birth weight neonates at risk for suboptimal renal growth and function during infancy? *BMC Nephrol*. 2016;17(1):100.
133. Bakker H, Gaillard R, Franco OH, et al. Fetal and infant growth patterns and kidney function at school age. *J Am Soc Nephrol*. 2014;25(11):2607–2615.
134. Brennan S, Kandasamy Y. Renal parenchymal thickness as a measure of renal growth in low-birth-weight infants versus normal-birth-weight infants. *Ultrasound Med Biol*. 2013;39(12):2315–2320.
135. Spencer J, Wang Z, Hoy W. Low birth weight and reduced renal volume in aboriginal children. *Am J Kidney Dis*. 2001;37(5):915–920.
136. Rakow A, Johansson S, Legnevall L, et al. Renal volume and function in school-age children born preterm or small for gestational age. *Pediatr Nephrol*. 2008;23(8):1309–1315.
137. Boubred F, Buffat C, Feuerstein JM, et al. Effects of early postnatal hypernutrition on nephron number and long-term renal function and structure in rats. *Am J Physiol Renal Physiol*. 2007;293(6):F1944–F1949.
138. Morton JS, Cooke CL, Davidge ST. In utero origins of hypertension: mechanisms and targets for therapy. *Physiol Rev*. 2016;96(2):549–603.
139. Manning J, Vehaskari VM. Postnatal modulation of prenatally programmed hypertension by dietary Na and ACE inhibition. *Am J Physiol Regul Integr Comp Physiol*. 2005;288(1):R80–R84.
140. Sanders MW, Fazzi GE, Janssen GM, et al. High sodium intake increases blood pressure and alters renal function in intrauterine growth-retarded rats. *Hypertension*. 2005;46(1):71–75.
141. Woods LL, Weeks DA, Rasch R. Programming of adult blood pressure by maternal protein restriction: role of nephrogenesis. *Kidney Int*. 2004;65(4):1339–1348.
142. de Boer MP, Ijzerman RG, de Jongh RT, et al. Birth weight relates to salt sensitivity of blood pressure in healthy adults. *Hypertension*. 2008;51(4):928–932.
143. Simonetti GD, Raio L, Surbek D, et al. Salt sensitivity of children with low birth weight. *Hypertension*. 2008;52(4):625–630.
144. Zimanyi MA, Bertram JF, Black MJ. Does a nephron deficit in rats predispose to salt-sensitive hypertension? *Kidney Blood Press Res*. 2004;27(4):239–247.
145. Gilbert JS. Sex, salt, and senescence: sorting out mechanisms of the developmental origins of hypertension. *Hypertension*. 2008;51(4):997–999.
146. Salazar F, Reverte V, Saez F, et al. Age- and sodium-sensitive hypertension and sex-dependent renal changes in rats with a reduced nephron number. *Hypertension*. 2008;51(4):1184–1189.
147. Magalhaes JC, da Silveira AB, Mota DL, et al. Renal function in juvenile rats subjected to prenatal malnutrition and chronic salt overload. *Exp Physiol*. 2006;91(3):611–619.
148. Nehiri T, Duong Van Huyen JP, Viltard M, et al. Exposure to maternal diabetes induces salt-sensitive hypertension and impairs renal function in adult rat offspring. *Diabetes*. 2008;57(8):2167–2175.
149. Ruta LA, Dickinson H, Thomas MC, et al. High-salt diet reveals the hypertensive and renal effects of reduced nephron endowment. *Am J Physiol Renal Physiol*. 2010;298(6):F1384–F1392.
150. Walker KA, Cai X, Caruana G, et al. High nephron endowment protects against salt-induced hypertension. *Am J Physiol Renal Physiol*. 2012;303(2):F253–F258.
151. Stewart T, Ascani J, Craver RD, et al. Role of postnatal dietary sodium in prenatally programmed hypertension. *Pediatr Nephrol*. 2009;24(9):1727–1733.
152. Manning J, Beutler K, Knepper MA, et al. Upregulation of renal BSC1 and TSC in prenatally programmed hypertension. *Am J Physiol Renal Physiol*. 2002;283(1):F202–F206.
153. Dagan A, Habib S, Gattineni J, et al. Prenatal programming of rat thick ascending limb chloride transport by low-protein diet and dexamethasone. *Am J Physiol Regul Integr Comp Physiol*. 2009;297(1):R93–R99.
154. Bertram C, Trowern AR, Copin N, et al. The maternal diet during pregnancy programs altered expression of the glucocorticoid

- receptor and type 2 11 β -hydroxysteroid dehydrogenase: potential molecular mechanisms underlying the programming of hypertension in utero. *Endocrinology*. 2001;142(7):2841–2853.
155. Luzardo R, Silva PA, Einicker-Lamas M, et al. Metabolic programming during lactation stimulates renal Na⁺ transport in the adult offspring due to an early impact on local angiotensin II pathways. *PLoS ONE*. 2011;6(7):e21232.
 156. Dagan A, Kwon HM, Dwarakanath V, et al. Effect of renal denervation on prenatal programming of hypertension and renal tubular transporter abundance. *Am J Physiol Renal Physiol*. 2008;295(1):F29–F34.
 157. Cheng CJ, Lozano G, Baum M. Prenatal programming of rat cortical collecting tubule sodium transport. *Am J Physiol Renal Physiol*. 2012;302(6):F674–F678.
 158. Guron G, Friberg P. An intact renin-angiotensin system is a prerequisite for normal renal development. *J Hypertens*. 2000;18(2):123–137.
 159. Moritz KM, Cuffe JS, Wilson LB, et al. Review: sex specific programming: a critical role for the renal renin-angiotensin system. *Placenta*. 2010;31(suppl):S40–S46.
 160. Grigore D, Ojeda NB, Robertson EB, et al. Placental insufficiency results in temporal alterations in the renin angiotensin system in male hypertensive growth restricted offspring. *Am J Physiol Regul Integr Comp Physiol*. 2007;293(2):R804–R811.
 161. Tran S, Chen YW, Chenier I, et al. Maternal diabetes modulates renal morphogenesis in offspring. *J Am Soc Nephrol*. 2008;19(5):943–952.
 162. Woods LL, Rasch R. Perinatal ANG II programs adult blood pressure, glomerular number, and renal function in rats. *Am J Physiol*. 1998;275(5 Pt 2):R1593–R1599.
 163. McCausland JE, Bertram JF, Ryan GB, et al. Glomerular number and size following chronic angiotensin II blockade in the postnatal rat. *Exp Nephrol*. 1997;5(3):201–209.
 164. Zhang SL, Moini B, Ingelfinger JR. Angiotensin II increases Pax-2 expression in fetal kidney cells via the AT2 receptor. *J Am Soc Nephrol*. 2004;15(6):1452–1465.
 165. Langley-Evans SC, Jackson AA. Captopril normalises systolic blood pressure in rats with hypertension induced by fetal exposure to maternal low protein diets. *Comp Biochem Physiol A Physiol*. 1995;110(3):223–228.
 166. McMullen S, Gardner DS, Langley-Evans SC. Prenatal programming of angiotensin II type 2 receptor expression in the rat. *Br J Nutr*. 2004;91(1):133–140.
 167. Sahajpal V, Ashton N. Increased glomerular angiotensin II binding in rats exposed to a maternal low protein diet in utero. *J Physiol*. 2005;563(Pt 1):193–201.
 168. Vehaskari VM, Stewart T, Lafont D, et al. Kidney angiotensin and angiotensin receptor expression in prenatally programmed hypertension. *Am J Physiol Renal Physiol*. 2004;287(2):F262–F267.
 169. Dasinger JH, Intapad S, Backstrom MA, et al. Intrauterine growth restriction programs an accelerated age-related increase in cardiovascular risk in male offspring. *Am J Physiol Renal Physiol*. 2016;311(2):F312–F319.
 170. Ojeda NB, Intapad S, Alexander BT. Sex differences in the developmental programming of hypertension. *Acta physiologica*. 2014;210(2):307–316.
 171. Ajala AR, Almeida SS, Rangel M, et al. Association of ACE gene insertion/deletion polymorphism with birth weight, blood pressure levels, and ACE activity in healthy children. *Am J Hypertens*. 2012;25(7):827–832.
 172. Alexander BT, Hendon AE, Ferril G, et al. Renal denervation abolishes hypertension in low-birth-weight offspring from pregnant rats with reduced uterine perfusion. *Hypertension*. 2005;45(4):754–758.
 173. Intapad S, Tull FL, Brown AD, et al. Renal denervation abolishes the age-dependent increase in blood pressure in female intrauterine growth-restricted rats at 12 months of age. *Hypertension*. 2013;61(4):828–834.
 174. Franco Mdo C, Arruda RM, Fortes ZB, et al. Severe nutritional restriction in pregnant rats aggravates hypertension, altered vascular reactivity, and renal development in spontaneously hypertensive rats offspring. *J Cardiovasc Pharmacol*. 2002;39(3):369–377.
 175. Paixao AD, Maciel CR, Teles MB, et al. Regional Brazilian diet-induced low birth weight is correlated with changes in renal hemodynamics and glomerular morphometry in adult age. *Biol Neonate*. 2001;80(3):239–246.
 176. Tendron-Franzin A, Gouyon JB, Guignard JP, et al. Long-term effects of in utero exposure to cyclosporin A on renal function in the rabbit. *J Am Soc Nephrol*. 2004;15(10):2687–2693.
 177. Sanders MW, Fazzi GE, Janssen GM, et al. Reduced uteroplacental blood flow alters renal arterial reactivity and glomerular properties in the rat offspring. *Hypertension*. 2004;43(6):1283–1289.
 178. van Onna M, Houben AJ, Kroon AA, et al. Asymmetry of renal blood flow in patients with moderate to severe hypertension. *Hypertension*. 2003;41(1):108–113.
 179. Chen W, Srinivasan SR, Hallman DM, et al. The relationship between birthweight and longitudinal changes of blood pressure is modulated by beta-adrenergic receptor genes: the Bogalusa Heart Study. *J Biomed Biotechnol*. 2010;2010:543514.
 180. Gilbert JS, Lang AL, Grant AR, et al. Maternal nutrient restriction in sheep: hypertension and decreased nephron number in offspring at 9 months of age. *J Physiol*. 2005;565(Pt 1):137–147.
 181. Jones SE, Bilous RW, Flyvbjerg A, et al. Intra-uterine environment influences glomerular number and the acute renal adaptation to experimental diabetes. *Diabetologia*. 2001;44(6):721–728.
 182. Jones SE, White KE, Flyvbjerg A, et al. The effect of intrauterine environment and low glomerular number on the histological changes in diabetic glomerulosclerosis. *Diabetologia*. 2006;49(1):191–199.
 183. Macconi D, Bonomelli M, Benigni A, et al. Pathophysiologic implications of reduced podocyte number in a rat model of progressive glomerular injury. *Am J Pathol*. 2006;168(1):42–54.
 184. Wiggins RC. The spectrum of podocytopathies: a unifying view of glomerular diseases. *Kidney Int*. 2007;71(12):1205–1214.
 185. Martins JP, Monteiro JC, Paixao AD. Renal function in adult rats subjected to prenatal dexamethasone. *Clin Exp Pharmacol Physiol*. 2003;30(1–2):32–37.
 186. Plank C, Ostreicher I, Hartner A, et al. Intrauterine growth retardation aggravates the course of acute mesangioproliferative glomerulonephritis in the rat. *Kidney Int*. 2006;70(11):1974–1982.
 187. Abitbol CL, Rodriguez MM. The long-term renal and cardiovascular consequences of prematurity. *Nat Rev Nephrol*. 2012;8(5):265–274.
 188. Spence D, Stewart MC, Alderdice FA, et al. Intra-uterine growth restriction and increased risk of hypertension in adult life: a follow-up study of 50-year-olds. *Public Health*. 2012;126(7):561–565.
 189. Duncan AF, Heyne RJ, Morgan JS, et al. Elevated systolic blood pressure in preterm very-low-birth-weight infants ≤ 3 years of life. *Pediatr Nephrol*. 2011;26(7):1115–1121.
 190. Boivin A, Luo ZC, Audibert F, et al. Pregnancy complications among women born preterm. *CMAJ*. 2012;184(16):1777–1784.
 191. Crowther CA, Doyle LW, Haslam RR, et al. Outcomes at 2 years of age after repeat doses of antenatal corticosteroids. *N Engl J Med*. 2007;357(12):1179–1189.
 192. Keijzer-Veen MG, Kleinveld HA, Lequin MH, et al. Renal function and size at young adult age after intrauterine growth restriction and very premature birth. *Am J Kidney Dis*. 2007;50(4):542–551.
 193. Rossi P, Tauzin L, Marchand E, et al. Respective roles of preterm birth and fetal growth restriction in blood pressure and arterial stiffness in adolescence. *J Adolesc Health*. 2011;48(5):520–522.
 194. Smal JC, Uiterwaal CS, Bruinse HW, et al. Inverse relationship between birth weight and blood pressure in growth-retarded but not in appropriate for gestational age infants during the first week of life. *Neonatology*. 2009;96(2):86–92.
 195. Dalziel SR, Parag V, Rodgers A, et al. Cardiovascular risk factors at age 30 following pre-term birth. *Int J Epidemiol*. 2007;36(4):907–915.
 196. Law CM, de Swiet M, Osmond C, et al. Initiation of hypertension in utero and its amplification throughout life. *Br Med J*. 1993;306(6869):24–27.
 197. Richardson LJ, Hussey JM, Strutz KL. Origins of disparities in cardiovascular disease: birth weight, body mass index, and young adult systolic blood pressure in the national longitudinal study of adolescent health. *Ann Epidemiol*. 2011;21(8):598–607.
 198. Hemachandra AH, Klebanoff MA, Furth SL. Racial disparities in the association between birth weight in the term infant and blood pressure at age 7 years: results from the collaborative perinatal project. *J Am Soc Nephrol*. 2006;17(9):2576–2581.
 199. Rostand SG, Cliver SP, Goldenberg RL. Racial disparities in the association of foetal growth retardation to childhood blood pressure. *Nephrol Dial Transplant*. 2005;20(8):1592–1597.
 200. Vancheri F, Alletto M, Burgio A, et al. Inverse relationship between fetal growth and arterial pressure in children and adults. *G Ital Cardiol*. 1995;25(7):833–841.
 201. Zhao M, Shu XO, Jin F, et al. Birthweight, childhood growth and hypertension in adulthood. *Int J Epidemiol*. 2002;31(5):1043–1051.
 202. Cruickshank JK, Mzayek F, Liu L, et al. Origins of the “black/white” difference in blood pressure: roles of birth weight, postnatal growth,

- early blood pressure, and adolescent body size: the Bogalusa Heart Study. *Circulation*. 2005;111(15):1932–1937.
203. Stewart CP, Christian P, Schulze KJ, et al. Size at birth is associated with blood pressure but not insulin resistance in 6-8 year old children in rural Nepal. *J Dev Origins Health Dis*. 2010;1(2):114–122.
 204. Ben-Shlomo Y, McCarthy A, Hughes R, et al. Immediate postnatal growth is associated with blood pressure in young adulthood: the Barry Caerphilly Growth Study. *Hypertension*. 2008;52(4):638–644.
 205. Hemachandra AH, Howards PP, Furth SL, et al. Birth weight, postnatal growth, and risk for high blood pressure at 7 years of age: results from the Collaborative Perinatal Project. *Pediatrics*. 2007;119(6):e1264–e1270.
 206. Bowers K, Liu G, Wang P, et al. Birth weight, postnatal weight change, and risk for high blood pressure among Chinese children. *Pediatrics*. 2011;127(5):e1272–e1279.
 207. Rottevel J, van Weissenbruch MM, Twisk JW, et al. Infant and childhood growth patterns, insulin sensitivity, and blood pressure in prematurely born young adults. *Pediatrics*. 2008;122(2):313–321.
 208. Vohr BR, Allan W, Katz KH, et al. Early predictors of hypertension in prematurely born adolescents. *Acta Paediatr*. 2010;99(12):1812–1818.
 209. Khalsa DD, Beydoun HA, Carmody JB. Prevalence of chronic kidney disease risk factors among low birth weight adolescents. *Pediatr Nephrol*. 2016;31(9):1509–1516.
 210. Zhao Y, Wang SF, Mu M, et al. Birth weight and overweight/obesity in adults: a meta-analysis. *Eur J Pediatr*. 2012;171(12):1737–1746.
 211. Aceti A, Santhakumaran S, Logan KM, et al. The diabetic pregnancy and offspring blood pressure in childhood: a systematic review and meta-analysis. *Diabetologia*. 2012;55(11):3114–3127.
 212. Fraser A, Nelson SM, Macdonald-Wallis C, et al. Hypertensive disorders of pregnancy and cardiometabolic health in adolescent offspring. *Hypertension*. 2013;62(3):614–620.
 213. Miettola S, Hartikainen AL, Vaarasmaki M, et al. Offspring's blood pressure and metabolic phenotype after exposure to gestational hypertension in utero. *Eur J Epidemiol*. 2013;28(1):87–98.
 214. Davis EF, Lazdam M, Lewandowski AJ, et al. Cardiovascular risk factors in children and young adults born to preeclamptic pregnancies: a systematic review. *Pediatrics*. 2012;129(6):e1552–e1561.
 215. Lewandowski AJ, Davis EF, Yu G, et al. Elevated blood pressure in preterm-born offspring associates with a distinct antiangiogenic state and microvascular abnormalities in adult life. *Hypertension*. 2015;65:607–614.
 216. Gamborg M, Byberg L, Rasmussen F, et al. Birth weight and systolic blood pressure in adolescence and adulthood: meta-regression analysis of sex- and age-specific results from 20 nordic studies. *Am J Epidemiol*. 2007;166(6):634–645.
 217. Bergvall N, Iliadou A, Johansson S, et al. Genetic and shared environmental factors do not confound the association between birth weight and hypertension: a study among Swedish twins. *Circulation*. 2007;115(23):2931–2938.
 218. Levine RS, Hennekens CH, Jesse MJ. Blood pressure in prospective population based cohort of newborn and infant twins. *Br Med J*. 1994;308(6924):298–302.
 219. Poulter NR, Chang CL, MacGregor AJ, et al. Association between birth weight and adult blood pressure in twins: historical cohort study. *BMJ*. 1999;319(7221):1330–1333.
 220. Hoy WE, Hughson MD, Singh GR, et al. Reduced nephron number and glomerulomegaly in Australian Aborigines: a group at high risk for renal disease and hypertension. *Kidney Int*. 2006;70(1):104–110.
 221. Kanzaki G, Puelles VG, Cullen-McEwen LA, et al. New insights on glomerular hyperfiltration: a Japanese autopsy study. *JCI Insight*. 2017;2(19).
 222. Perala MM, Moltchanova E, Kaartinen NE, et al. The association between salt intake and adult systolic blood pressure is modified by birth weight. *Am J Clin Nutr*. 2011;93(2):422–426.
 223. Schreuder MF, Wilhelm AJ, Bokenkamp A, et al. Impact of gestational age and birth weight on amikacin clearance on day 1 of life. *Clin J Am Soc Nephrol*. 2009;4(11):1774–1778.
 224. Bacchetta J, Harambat J, Dubourg L, et al. Both extrauterine and intrauterine growth restriction impair renal function in children born very preterm. *Kidney Int*. 2009;76(4):445–452.
 225. Ingelfinger JR. Weight for gestational age as a baseline predictor of kidney function in adulthood. *Am J Kidney Dis*. 2008;51(1):1–4.
 226. Kandasamy Y, Smith R, Wright IM, et al. Relationships between glomerular filtration rate and kidney volume in low-birth-weight neonates. *J Nephrol*. 2013;26(5):894–898.
 227. Bruel A, Roze JC, Flamant C, et al. Critical serum creatinine values in very preterm newborns. *PLoS ONE*. 2013;8(12):e84892.
 228. Rodriguez-Soriano J, Aguirre M, Oliveros R, et al. Long-term renal follow-up of extremely low birth weight infants. *Pediatr Nephrol*. 2005;20(5):579–584.
 229. Franco MC, Nishida SK, Sesso R. GFR estimated from cystatin C versus creatinine in children born small for gestational age. *Am J Kidney Dis*. 2008;51(6):925–932.
 230. Keijzer-Veen MG, Schrevel M, Finken MJ, et al. Microalbuminuria and lower glomerular filtration rate at young adult age in subjects born very premature and after intrauterine growth retardation. *J Am Soc Nephrol*. 2005;16(9):2762–2768.
 231. Johnson MJ, Wootton SA, Leaf AA, et al. Preterm birth and body composition at term equivalent age: a systematic review and meta-analysis. *Pediatrics*. 2012;130(3):e640–e649.
 232. Silverwood RJ, Pierce M, Hardy R, et al. Low birth weight, later renal function, and the roles of adulthood blood pressure, diabetes, and obesity in a British birth cohort. *Kidney Int*. 2013;84(6):1262–1270.
 233. Gielen M, Pinto-Sietsma SJ, Zeegers MP, et al. Birth weight and creatinine clearance in young adult twins: influence of genetic, prenatal, and maternal factors. *J Am Soc Nephrol*. 2005;16(8):2471–2476.
 234. Hoy WE, Rees M, Kile E, et al. A new dimension to the Barker hypothesis: low birthweight and susceptibility to renal disease. *Kidney Int*. 1999;56(3):1072–1077.
 235. Hoy WE, White AV, Tipiloura B, et al. The multideterminant model of renal disease in a remote Australian Aboriginal population in the context of early life risk factors: lower birth weight, childhood post-streptococcal glomerulonephritis, and current body mass index influence levels of albuminuria in young aboriginal adults. *Clin Nephrol*. 2015;83(7 suppl 1):75–81.
 236. Hoy WE, White AV, Tipiloura B, et al. The influence of birthweight, past poststreptococcal glomerulonephritis and current body mass index on levels of albuminuria in young adults: the multideterminant model of renal disease in a remote Australian aboriginal population with high rates of renal disease and renal failure. *Nephrol Dial Transplant*. 2016;31(6):971–977.
 237. Vieux R, Hascoet JM, Franck P, et al. Increased albuminuria in 4-year-old preterm-born children with normal height. *J Pediatr*. 2012;160(6):923–928.e1.
 238. Painter RC, Roseboom TJ, van Montfrans GA, et al. Microalbuminuria in adults after prenatal exposure to the Dutch famine. *J Am Soc Nephrol*. 2005;16(1):189–194.
 239. Dyck RF, Bingham WT, Lim H, et al. Decreased urine albumin:creatinine ratios in infants of diabetic mothers: does exposure to diabetic pregnancies alter fetal renal development? *J Dev Origins Health Dis*. 2011;2(5):265–271.
 240. Carmody JB, Charlton JR. Short-term gestation, long-term risk: prematurity and chronic kidney disease. *Pediatrics*. 2013;131(6):1168–1179.
 241. Jetton JG, Askenazi DJ. Acute kidney injury in the neonate. *Clin Perinatol*. 2014;41(3):487–502.
 242. Carmody JB, Swanson JR, Rhone ET, et al. Recognition and reporting of AKI in very low birth weight infants. *Clin J Am Soc Nephrol*. 2014;9(12):2036–2043.
 243. Carmody JB, Seckeler MD, Ballengee CR, et al. Pre-operative renal volume predicts peak creatinine after congenital heart surgery in neonates. *Cardiol Young*. 2014;24(5):831–839.
 244. Selewski DT, Charlton JR, Jetton JG, et al. Neonatal acute kidney injury. *Pediatrics*. 2015;136(2):e463–e473.
 245. Rhone ET, Carmody JB, Swanson JR, et al. Nephrotoxic medication exposure in very low birth weight infants. *J Matern Fetal Neonatal Med*. 2014;27(14):1485–1490.
 246. Sutherland M, Ryan D, Black MJ, et al. Long-term renal consequences of preterm birth. *Clin Perinatol*. 2014;41(3):561–573.
 247. Hanna MH, Askenazi DJ, Selewski DT. Drug-induced acute kidney injury in neonates. *Curr Opin Pediatr*. 2016;28(2):180–187.
 248. Girardi A, Raschi E, Galletti S, et al. Drug-induced renal damage in preterm neonates: state of the art and methods for early detection. *Drug Saf*. 2015;38(6):535–551.
 249. Cuzzolin L, Fanos V, Pinna B, et al. Postnatal renal function in preterm newborns: a role of diseases, drugs and therapeutic interventions. *Pediatr Nephrol*. 2006;21(7):931–938.
 250. Vieux R, Fresson J, Guillemin F, et al. Perinatal drug exposure and renal function in very preterm infants. *Arch Dis Child Fetal Neonatal Ed*. 2011;96(4):F290–F295.

251. Magnani C, Moretti S, Ammenti A. Neonatal chronic renal failure associated with maternal ingestion of nimesulide as analgesic. *Eur J Obstet Gynecol Reprod Biol.* 2004;116(2):244–245.
252. Giapros V, Papadimitriou P, Challa A, et al. The effect of intrauterine growth retardation on renal function in the first two months of life. *Nephrol Dial Transplant.* 2007;22(1):96–103.
253. Askenazi DJ, Koralkar R, Patil N, et al. Acute kidney injury urine biomarkers in very low-birth-weight infants. *Clin J Am Soc Nephrol.* 2016;11(9):1527–1535.
254. Bruel A, Roze JC, Quere MP, et al. Renal outcome in children born preterm with neonatal acute renal failure: IRENEO—a prospective controlled study. *Pediatr Nephrol.* 2016;31(12):2365–2373.
255. Kwint P, Klimek M, Drozdz D, et al. Assessment of long-term renal complications in extremely low birth weight children. *Pediatr Nephrol.* 2011;26(7):1095–1103.
256. Mammen C, Al Abbas A, Skippen P, et al. Long-term risk of CKD in children surviving episodes of acute kidney injury in the intensive care unit: a prospective cohort study. *Am J Kidney Dis.* 2012;59(4):523–530.
257. Nishizaki N, Hirano D, Nishizaki Y, et al. Increased urinary angiotensinogen is an effective marker of chronic renal impairment in very low birth weight children. *Clin Exp Nephrol.* 2014;18(4):642–648.
258. Chaturvedi S, Ng KH, Mammen C. The path to chronic kidney disease following acute kidney injury: a neonatal perspective. *Pediatr Nephrol.* 2017;32(2):227–241.
259. Hodgins JB, Rasoulpour M, Markowitz GS, et al. Very low birth weight is a risk factor for secondary focal segmental glomerulosclerosis. *Clin J Am Soc Nephrol.* 2009;4(1):71–76.
260. Hibino S, Abe Y, Watanabe S, et al. Proteinuria caused by glomerular hypertension during adolescence associated with extremely premature birth: a report of two cases. *Pediatr Nephrol.* 2015;30(10):1889–1892.
261. Hayashi A, Santo Y, Satomura K. Proteinuria and glomerular hypertrophy in extremely low-birthweight children. *Pediatr Int.* 2014;56(6):860–864.
262. Duncan RC, Bass PS, Garrett PJ, et al. Weight at birth and other factors influencing progression of idiopathic membranous nephropathy. *Nephrol Dial Transplant.* 1994;9(1):875.
263. Garrett P, Sandeman D, Reza M, et al. Weight at birth and renal disease in adulthood. *Nephrol Dial Transplant.* 1993;8:920.
264. Na YW, Yang HJ, Choi JH, et al. Effect of intrauterine growth retardation on the progression of nephrotic syndrome. *Am J Nephrol.* 2002;22(5–6):463–467.
265. Sandeman D, Reza M, Phillips DI, et al. Why do some type 1 diabetic patients develop nephropathy? A possible role of birth weight. *Diabet Med.* 1992;9(suppl 1):36A.
266. Zidar N, Cavic MA, Kenda RB, et al. Effect of intrauterine growth retardation on the clinical course and prognosis of IgA glomerulonephritis in children. *Nephron.* 1998;79(1):28–32.
267. Zidar N, Cor A, Premru Srsen T, et al. Is there an association between glomerular density and birth weight in healthy humans? *Nephron.* 1998;80(1):97–98.
268. Teeninga N, Schreuder MF, Bokenkamp A, et al. Influence of low birth weight on minimal change nephrotic syndrome in children, including a meta-analysis. *Nephrol Dial Transplant.* 2008;23(5):1615–1620.
269. Orskov B, Christensen KB, Feldt-Rasmussen B, et al. Low birth weight is associated with earlier onset of end-stage renal disease in Danish patients with autosomal dominant polycystic kidney disease. *Kidney Int.* 2012;81(9):919–924.
270. Rajan T, Barbour SJ, White CT, et al. Low birth weight and nephron mass and their role in the progression of chronic kidney disease: a case report on identical twins with alport disease. *Nephrol Dial Transplant.* 2011;26(12):4136–4139.
271. Rossing P, Tarnow L, Nielsen FS, et al. Short stature and diabetic nephropathy. *BMJ.* 1995;310(6975):296–297.
272. Jacobsen P, Rossing P, Tarnow L, et al. Birth weight—a risk factor for progression in diabetic nephropathy? *J Intern Med.* 2003;253(3):343–350.
273. Hirano D, Ishikura K, Uemura O, et al. Association between low birth weight and childhood-onset chronic kidney disease in Japan: a combined analysis of a nationwide survey for paediatric chronic kidney disease and the National Vital Statistics Report. *Nephrol Dial Transplant.* 2016;31(11):1895–1900.
274. Hsu CW, Yamamoto KT, Henry RK, et al. Prenatal risk factors for childhood CKD. *J Am Soc Nephrol.* 2014;25(9):2105–2111.
275. Pavkov ME, Hanson RL, Knowler WC, et al. Effect of intrauterine diabetes exposure on the incidence of end-stage renal disease in young adults with type 2 diabetes. *Diabetes Care.* 2010;33(11):2396–2398.
276. Hanson M, Gluckman P. Developmental origins of noncommunicable disease: population and public health implications. *Am J Clin Nutr.* 2011;94(6 suppl):1754S–1758S.
277. Ruggajo P, Svarstad E, Leh S, et al. Low birth weight and risk of progression to end stage renal disease in IgA nephropathy—a retrospective registry-based cohort study. *PLoS ONE.* 2016;11(4):e0153819.
278. Couser WG, Remuzzi G, Mendis S, et al. The contribution of chronic kidney disease to the global burden of major noncommunicable diseases. *Kidney Int.* 2011;80(12):1258–1270.
279. World Health Organisation Regional Office for Europe. *Good Maternal Nutrition. The Best Start in Life.* 2016. http://www.euro.who.int/_data/assets/pdf_file/0008/313667/Good-maternal-nutrition-The-best-start-in-life.pdf?ua=1. Accessed November 3, 2016.
280. Ahmed T, Hossain M, Sanin KI. Global burden of maternal and child undernutrition and micronutrient deficiencies. *Ann Nutr Metab.* 2012;61(suppl 1):8–17.
281. Chang HH, Larson J, Blencowe H, et al. Preventing preterm births: analysis of trends and potential reductions with interventions in 39 countries with very high human development index. *Lancet.* 2013;381(9862):223–234.
282. Sharma D, Shastri S, Sharma P. Intrauterine growth restriction: antenatal and postnatal aspects. *Clin Med Insights Pediatr.* 2016;10:67–83.
283. Christian P. Micronutrients, birth weight, and survival. *Annu Rev Nutr.* 2010;30:83–104.
284. Luyckx VA, Brenner BM. Birth weight, malnutrition and kidney-associated outcomes—a global concern. *Nat Rev Nephrol.* 2015;11(3):135–149.
285. Hoy WE, Bertram JF, Denton RD, et al. Nephron number, glomerular volume, renal disease and hypertension. *Curr Opin Nephrol Hypertens.* 2008;17(3):258–265.
286. Uauy R, Kain J, Corvalan C. How can the Developmental Origins of Health and Disease (DOHaD) hypothesis contribute to improving life in developing countries? *Am J Clin Nutr.* 2011;94(6 suppl):1759S–1764S.
287. Roderick PJ, Jeffrey RF, Yuen HM, et al. Smaller kidney size at birth in South Asians: findings from the Born in Bradford Birth Cohort study. *Nephrol Dial Transplant.* 2016;31(3):455–465.
288. Kandasamy Y, Smith R, Wright IM, et al. Reduced nephron endowment in the neonates of indigenous Australian peoples. *J Dev Orig Health Dis.* 2014;5(1):31–35.
289. Cassidy-Bushrow AE, Wegienka G, Barone CJ 2nd, et al. Race-specific relationship of birth weight and renal function among healthy young children. *Pediatr Nephrol.* 2012;27(8):1317–1323.
290. Liyanage T, Ninomiya T, Jha V, et al. Worldwide access to treatment for end-stage kidney disease: a systematic review. *Lancet.* 2015;385(9981):1975–1982.
291. McNamara BJ, Diouf B, Douglas-Denton RN, et al. A comparison of nephron number, glomerular volume and kidney weight in Senegalese Africans and African Americans. *Nephrol Dial Transplant.* 2010;25(5):1514–1520.
292. Douglas-Denton RN, McNamara BJ, Hoy WE, et al. Does nephron number matter in the development of kidney disease? *Ethn Dis.* 2006;16(2 suppl 2):S2-40-5.
293. Hult M, Tornhammar P, Ueda P, et al. Hypertension, diabetes and overweight: looming legacies of the Biafran famine. *PLoS ONE.* 2010;5(10):e13582.
294. Painter RC, Roseboom TJ, Bleker OP. Prenatal exposure to the Dutch famine and disease in later life: an overview. *Reprod Toxicol.* 2005;20(3):345–352.
295. Stein AD, Zybert PA, van der Pal-de Bruin K, et al. Exposure to famine during gestation, size at birth, and blood pressure at age 59 y: evidence from the Dutch famine. *Eur J Epidemiol.* 2006;21(10):759–765.
296. Hoy WE, Hughson MD, Kopp JB, et al. APOL1 risk alleles are associated with exaggerated age-related changes in glomerular number and volume in African-American adults: an autopsy study. *J Am Soc Nephrol.* 2015;26(12):3179–3189.
297. Clark AT, Bertram JF. Molecular regulation of nephron endowment. *Am J Physiol.* 1999;276(4 Pt 2):F485–F497.
298. Costantini F, Kopan R. Patterning a complex organ: branching morphogenesis and nephron segmentation in kidney development. *Dev Cell.* 2010;18(5):698–712.

299. Little MH, McMahon AP. Mammalian kidney development: principles, progress, and projections. *Cold Spring Harb Perspect Biol.* 2012;4(5).
300. Moritz KM, Wintour EM, Black MJ, et al. Factors influencing mammalian kidney development: implications for health in adult life. *Adv Anat Embryol Cell Biol.* 2008;196:1–78.
301. Walker KA, Bertram JF. Kidney development: core curriculum 2011. *Am J Kidney Dis.* 2011;57(6):948–958.
302. Schreuder MF, Nyengaard JR, Fodor M, et al. Glomerular number and function are influenced by spontaneous and induced low birth weight in rats. *J Am Soc Nephrol.* 2005;16(10):2913–2919.
303. Luyckx VA. Preterm birth and its impact on renal health. *Semin Nephrol.* 2017;37(4):311–319.
304. Simeoni U, Zetterstrom R. Long-term circulatory and renal consequences of intrauterine growth restriction. *Acta Paediatr.* 2005;94(7):819–824.
305. Sakurai H, Nigam SK. In vitro branching tubulogenesis: implications for developmental and cystic disorders, nephron number, renal repair, and nephron engineering. *Kidney Int.* 1998;54(1):14–26.
306. Quinlan J, Lemire M, Hudson T, et al. A common variant of the PAX2 gene is associated with reduced newborn kidney size. *J Am Soc Nephrol.* 2007;18(6):1915–1921.
307. Kaczmarczyk M, Goracy I, Loniewska B, et al. Association of BMPRIA polymorphism, but not BMP4, with kidney size in full-term newborns. *Pediatr Nephrol.* 2012.
308. El Kares R, Manolescu DC, Lakhal-Chaieb L, et al. A human ALDH1A2 gene variant is associated with increased newborn kidney size and serum retinoic acid. *Kidney Int.* 2010;78(1):96–102.
309. Mugford JW, Sipila P, McMahon JA, et al. Osr1 expression demarcates a multi-potent population of intermediate mesoderm that undergoes progressive restriction to an Osr1-dependent nephron progenitor compartment within the mammalian kidney. *Dev Biol.* 2008;324(1):88–98.
310. Zhang Z, Iglesias D, Eliopoulos N, et al. A variant OSR1 allele which disturbs OSR1 mRNA expression in renal progenitor cells is associated with reduction of newborn kidney size and function. *Hum Mol Genet.* 2011;20(21):4167–4174.
311. Barker DJ, Gluckman PD, Godfrey KM, et al. Fetal nutrition and cardiovascular disease in adult life. *Lancet.* 1993;341(8850):938–941.
312. Godfrey KM, Forrester T, Barker DJ, et al. Maternal nutritional status in pregnancy and blood pressure in childhood. *Br J Obstet Gynaecol.* 1994;101(5):398–403.
313. Welham SJ, Riley PR, Wade A, et al. Maternal diet programs embryonic kidney gene expression. *Physiol Genomics.* 2005;22(1):48–56.
314. Zeman FJ. Effects of maternal protein restriction on the kidney of the newborn young of rats. *J Nutr.* 1968;94(2):111–116.
315. Langley-Evans SC. Nutritional programming of disease: unravelling the mechanism. *J Anat.* 2009;215(1):36–51.
316. Woods LL, Ingelfinger JR, Rasch R. Modest maternal protein restriction fails to program adult hypertension in female rats. *Am J Physiol Regul Integr Comp Physiol.* 2005;289(4):R1131–R1136.
317. Burton GJ, Fowden AL, Thornburg KL. Placental origins of chronic disease. *Physiol Rev.* 2016;96(4):1509–1565.
318. Langley-Evans SC, Phillips GJ, Benediktsson R, et al. Protein intake in pregnancy, placental glucocorticoid metabolism and the programming of hypertension in the rat. *Placenta.* 1996;17(2–3):169–172.
319. Khorram O, Khorram N, Momeni M, et al. Maternal undernutrition inhibits angiogenesis in the offspring: a potential mechanism of programmed hypertension. *Am J Physiol Regul Integr Comp Physiol.* 2007;293(2):R745–R753.
320. Denisenko O, Lin B, Louey S, et al. Maternal malnutrition and placental insufficiency induce global downregulation of gene expression in fetal kidneys. *J Dev Origins Health Dis.* 2011;2(2):124–133.
321. Altobelli G, Bogdarina IG, Stupka E, et al. Genome-wide methylation and gene expression changes in newborn rats following maternal protein restriction and reversal by folic acid. *PLoS ONE.* 2013;8(12):e82989.
322. Lisle SJ, Lewis RM, Petry CJ, et al. Effect of maternal iron restriction during pregnancy on renal morphology in the adult rat offspring. *Br J Nutr.* 2003;90(1):33–39.
323. Drake KA, Sauerbry MJ, Blohowiak SE, et al. Iron deficiency and renal development in the newborn rat. *Pediatr Res.* 2009;66(6):619–624.
324. Tomat AL, Inerra F, Veiras L, et al. Moderate zinc restriction during fetal and postnatal growth of rats: effects on adult arterial blood pressure and kidney. *Am J Physiol Regul Integr Comp Physiol.* 2008;295(2):R543–R549.
325. Mishra V, Thapa S, Retherford RD, et al. Effect of iron supplementation during pregnancy on birthweight: evidence from Zimbabwe. *Food Nutr Bull.* 2005;26(4):338–347.
326. Merlet-Benichou C, Vilar J, Lelievre-Pegorier M, et al. Role of retinoids in renal development: pathophysiological implication. *Curr Opin Nephrol Hypertens.* 1999;8(1):39–43.
327. Sutherland MR, Gubhaju L, Yoder BA, et al. The effects of postnatal retinoic acid administration on nephron endowment in the preterm baboon kidney. *Pediatr Res.* 2009;65(4):397–402.
328. Boyce AC, Palmer-Aronsten BJ, Kim MY, et al. Maternal vitamin D deficiency programmes adult renal renin gene expression and renal function. *J Dev Orig Health Dis.* 2013;4(5):368–376.
329. Reichetzer C, Chen H, Foller M, et al. Maternal vitamin D deficiency and fetal programming—lessons learned from humans and mice. *Kidney Blood Press Res.* 2014;39(4):315–329.
330. Nascimento FA, Ceciliano TC, Aguila MB, et al. Maternal vitamin D deficiency delays glomerular maturity in F1 and F2 offspring. *PLoS ONE.* 2012;7(8):e41740.
331. Bhutta ZA, Das JK, Rizvi A, et al. Evidence-based interventions for improvement of maternal and child nutrition: what can be done and at what cost? *Lancet.* 2013;382(9890):452–477.
332. Vaidya A, Saville N, Shrestha BP, et al. Effects of antenatal multiple micronutrient supplementation on children's weight and size at 2 years of age in Nepal: follow-up of a double-blind randomised controlled trial. *Lancet.* 2008;371(9611):492–499.
333. Sommer A, Tarwojjo I, Djunaedi E, et al. Impact of vitamin A supplementation on childhood mortality. A randomised controlled community trial. *Lancet.* 1986;1(8491):1169–1173.
334. da Silva Lopes K, Ota E, Shakya P, et al. Effects of nutrition interventions during pregnancy on low birth weight: an overview of systematic reviews. *BMJ Glob Health.* 2017;2(3):e000389.
335. Stewart CP, Christian P, Schulze KJ, et al. Antenatal micronutrient supplementation reduces metabolic syndrome in 6- to 8-year-old children in rural Nepal. *J Nutr.* 2009;139(8):1575–1581.
336. Lee LM, Leung CY, Tang WW, et al. A paradoxical teratogenic mechanism for retinoic acid. *Proc Natl Acad Sci U S A.* 2012;109(34):13668–13673.
337. Hawkesworth S, Wagatsuma Y, Kahn AI, et al. Combined food and micronutrient supplements during pregnancy have limited impact on child blood pressure and kidney function in rural Bangladesh. *J Nutr.* 2013;143(5):728–734.
338. Amri K, Freund N, Van Huyen JP, et al. Altered nephrogenesis due to maternal diabetes is associated with increased expression of IGF-II/ mannose-6-phosphate receptor in the fetal kidney. *Diabetes.* 2001;50(5):1069–1075.
339. Doublier S, Amri K, Seurin D, et al. Overexpression of human insulin-like growth factor binding protein-1 in the mouse leads to nephron deficit. *Pediatr Res.* 2001;49(5):660–666.
340. Kistner A, Sigurdsson J, Niklasson A, et al. Neonatal IGF-1/IGFBP-1 axis and retinopathy of prematurity are associated with increased blood pressure in preterm children. *Acta Paediatr.* 2014;103(2):149–156.
341. Amri K, Freund N, Vilar J, et al. Adverse effects of hyperglycemia on kidney development in rats: in vivo and in vitro studies. *Diabetes.* 1999;48(11):2240–2245.
342. Hokke SN, Armitage JA, Puelles VG, et al. Altered ureteric branching morphogenesis and nephron endowment in offspring of diabetic and insulin-treated pregnancy. *PLoS ONE.* 2013;8(3):e58243.
343. Tain YL, Lee WC, Hsu CN, et al. Asymmetric dimethylarginine is associated with developmental programming of adult kidney disease and hypertension in offspring of streptozotocin-treated mothers. *PLoS ONE.* 2013;8(2):e55420.
344. Rocha SO, Gomes GN, Forti AL, et al. Long-term effects of maternal diabetes on vascular reactivity and renal function in rat male offspring. *Pediatr Res.* 2005;58(6):1274–1279.
345. Magaton A, Gil FZ, Casarini DE, et al. Maternal diabetes mellitus—early consequences for the offspring. *Pediatr Nephrol.* 2007;22(1):37–43.
346. Blackmore HL, Ozanne SE. Maternal diet-induced obesity and offspring cardiovascular health. *J Dev Orig Health Dis.* 2013;4(5):338–347.
347. Armitage JA, Lakasing L, Taylor PD, et al. Developmental programming of aortic and renal structure in offspring of rats fed fat-rich diets in pregnancy. *J Physiol.* 2005;565(Pt 1):171–184.
348. Jackson CM, Alexander BT, Roach L, et al. Exposure to maternal overnutrition and a high-fat diet during early postnatal development increases susceptibility to renal and metabolic injury later in life. *Am J Physiol Renal Physiol.* 2012;302(6):F774–F783.

349. Samuelsson AM, Morris A, Igosheva N, et al. Evidence for sympathetic origins of hypertension in juvenile offspring of obese rats. *Hypertension*. 2010;55(1):76–82.
350. Hokke S, Puelles VG, Armitage JA, et al. Maternal fat feeding augments offspring nephron endowment in mice. *PLoS ONE*. 2016;11(8):e0161578.
351. Moritz KM, De Matteo R, Dodic M, et al. Prenatal glucocorticoid exposure in the sheep alters renal development in utero: implications for adult renal function and blood pressure control. *Am J Physiol Regul Integr Comp Physiol*. 2011;301(2):R500–R509.
352. Singh RR, Cuffe JS, Moritz KM. Short- and long-term effects of exposure to natural and synthetic glucocorticoids during development. *Clin Exp Pharmacol Physiol*. 2012;39(11):979–989.
353. O'Sullivan L, Cuffe JS, Koning A, et al. Excess prenatal corticosterone exposure results in albuminuria, sex-specific hypotension, and altered heart rate responses to restraint stress in aged adult mice. *Am J Physiol Renal Physiol*. 2015;308(10):F1065–F1073.
354. O'Sullivan L, Cuffe JS, Paravicini TM, et al. Prenatal exposure to dexamethasone in the mouse alters cardiac growth patterns and increases pulse pressure in aged male offspring. *PLoS ONE*. 2013;8(7):e69149.
355. Singh RR, Cullen-McEwen LA, Kett MM, et al. Prenatal corticosterone exposure results in altered AT1/AT2, nephron deficit and hypertension in the rat offspring. *J Physiol*. 2007;579(Pt 2):503–513.
356. Wintour EM, Moritz KM, Johnson K, et al. Reduced nephron number in adult sheep, hypertensive as a result of prenatal glucocorticoid treatment. *J Physiol*. 2003;549(Pt 3):929–935.
357. Singh RR, Moritz KM, Bertram JF, et al. Effects of dexamethasone exposure on rat metanephric development: in vitro and in vivo studies. *Am J Physiol Renal Physiol*. 2007;293(2):F548–F554.
358. Dave-Sharma S, Wilson RC, Harbison MD, et al. Examination of genotype and phenotype relationships in 14 patients with apparent mineralocorticoid excess. *J Clin Endocrinol Metab*. 1998;83(7):2244–2254.
359. Seckl JR, Meaney MJ. Glucocorticoid programming. *Ann N Y Acad Sci*. 2004;1032:63–84.
360. Habib S, Gattineni J, Twombly K, et al. Evidence that prenatal programming of hypertension by dietary protein deprivation is mediated by fetal glucocorticoid exposure. *Am J Hypertens*. 2010.
361. Dagan A, Gattineni J, Cook V, et al. Prenatal programming of rat proximal tubule Na⁺/H⁺ exchanger by dexamethasone. *Am J Physiol Regul Integr Comp Physiol*. 2007;292(3):R1230–R1235.
362. Dalziel SR, Walker NK, Parag V, et al. Cardiovascular risk factors after antenatal exposure to betamethasone: 30-year follow-up of a randomised controlled trial. *Lancet*. 2005;365(9474):1856–1862.
363. Carballo-Magdaleno D, Guizaz-Mendoza JM, Amador-Licona N, et al. Renal function, renal volume, and blood pressure in infants with antecedent of antenatal steroids. *Pediatr Nephrol*. 2011;26(10):1851–1856.
364. Khalife N, Glover V, Taanila A, et al. Prenatal glucocorticoid treatment and later mental health in children and adolescents. *PLoS ONE*. 2013;8(11):e81394.
365. Wilkinson LJ, Neal CS, Singh RR, et al. Renal developmental defects resulting from in utero hypoxia are associated with suppression of ureteric beta-catenin signaling. *Kidney Int*. 2015;87(5):975–983.
366. Walton SL, Bielefeldt-Ohmann H, Singh RR, et al. Prenatal hypoxia leads to hypertension, renal renin-angiotensin system activation and exacerbates salt-induced pathology in a sex-specific manner. *Sci Rep*. 2017;7(1):8241.
367. Al-Odat I, Chen H, Chan YL, et al. The impact of maternal cigarette smoke exposure in a rodent model on renal development in the offspring. *PLoS ONE*. 2014;9(7):e103443.
368. O'Connell AE, Boyce AC, Lumbers ER, et al. The effects of asphyxia on renal function in fetal sheep at midgestation. *J Physiol*. 2003;552(Pt 3):933–943.
369. Vidal AC, Murphy SK, Murtha AP, et al. Associations between antibiotic exposure during pregnancy, birth weight and aberrant methylation at imprinted genes among offspring. *Int J Obes (Lond)*. 2013;37(7):907–913.
370. Gilbert T, Lelievre-Pegorier M, Merlet-Benichou C. Immediate and long-term renal effects of fetal exposure to gentamicin. *Pediatr Nephrol*. 1990;4(4):445–450.
371. Gilbert T, Gaonach S, Moreau E, et al. Defect of nephrogenesis induced by gentamicin in rat metanephric organ culture. *Lab Invest*. 1994;70(5):656–666.
372. Gilbert T, Cibert C, Moreau E, et al. Early defect in branching morphogenesis of the ureteric bud in induced nephron deficit. *Kidney Int*. 1996;50(3):783–795.
373. Kent AL, Douglas-Denton R, Shadbolt B, et al. Indomethacin, ibuprofen and gentamicin administered during late stages of glomerulogenesis do not reduce glomerular number at 14 days of age in the neonatal rat. *Pediatr Nephrol*. 2009;24(6):1143–1149.
374. Nathanson S, Moreau E, Merlet-Benichou C, et al. In utero and in vitro exposure to beta-lactams impair kidney development in the rat. *J Am Soc Nephrol*. 2000;11(5):874–884.
375. McKay DB, Josephson MA. Pregnancy in recipients of solid organs—effects on mother and child. *N Engl J Med*. 2006;354(12):1281–1293.
376. Komhoff M, Wang JL, Cheng HF, et al. Cyclooxygenase-2-selective inhibitors impair glomerulogenesis and renal cortical development. *Kidney Int*. 2000;57(2):414–422.
377. Sutherland MR, Yoder BA, McCurnin D, et al. Effects of ibuprofen treatment on the developing preterm baboon kidney. *Am J Physiol Renal Physiol*. 2012;302(10):F1286–F1292.
378. Qazi Q, Masakawa A, Milman D, et al. Renal anomalies in fetal alcohol syndrome. *Pediatrics*. 1979;63(6):886–889.
379. Gray SP, Kenna K, Bertram JF, et al. Repeated ethanol exposure during late gestation decreases nephron endowment in fetal sheep. *Am J Physiol Regul Integr Comp Physiol*. 2008;295(2):R568–R574.
380. Parkington HC, Kenna KR, Sozo F, et al. Maternal alcohol consumption in pregnancy enhances arterial stiffness and alters vasodilator function that varies between vascular beds in fetal sheep. *J Physiol*. 2014;592(12):2591–2603.
381. Gray SP, Denton KM, Cullen-McEwen L, et al. Prenatal exposure to alcohol reduces nephron number and raises blood pressure in progeny. *J Am Soc Nephrol*. 2010;21(11):1891–1902.
382. Gray SP, Cullen-McEwen LA, Bertram JF, et al. Mechanism of alcohol-induced impairment in renal development: could it be reduced by retinoic acid? *Clin Exp Pharmacol Physiol*. 2012;39(9):807–813.
383. North American Pediatric Renal Trials and Collaborative Studies (NAPRTCS). *Annual Transplant Report*. 2014. <https://web.emmes.com/study/ped/annlrept/annualrept2014.pdf>.
384. Chevalier RL. Prognostic factors and biomarkers of congenital obstructive nephropathy. *Pediatr Nephrol*. 2016;31(9):1411–1420.
385. Chevalier RL. Congenital urinary tract obstruction: the long view. *Adv Chronic Kidney Dis*. 2015;22(4):312–319.
386. Puddu M, Fanos V, Podda F, et al. The kidney from prenatal to adult life: perinatal programming and reduction of number of nephrons during development. *Am J Nephrol*. 2009;30(2):162–170.
387. Forbes MS, Thornhill BA, Galarreta CI, et al. Chronic unilateral ureteral obstruction in the neonatal mouse delays maturation of both kidneys and leads to late formation of atubular glomeruli. *Am J Physiol Renal Physiol*. 2013;305(12):F1736–F1746.
388. Chevalier RL, Thornhill BA, Chang AY. Unilateral ureteral obstruction in neonatal rats leads to renal insufficiency in adulthood. *Kidney Int*. 2000;58(5):1987–1995.
389. Wang X, Garrett MR. Nephron number, hypertension, and CKD: physiological and genetic insight from humans and animal models. *Physiol Genomics*. 2017;49(3):180–192.
390. Swali A, McMullen S, Hayes H, et al. Processes underlying the nutritional programming of embryonic development by iron deficiency in the rat. *PLoS ONE*. 2012;7(10):e48133.
391. O'Sullivan L, Combes AN, Little MH, et al. Epigenetics and developmental programming of adult onset diseases. *Pediatr Nephrol*. 2012;27(12):2175–2182.
392. Cullen-McEwen LA, Drago J, Bertram JF. Nephron endowment in glial cell line-derived neurotrophic factor (GDNF) heterozygous mice. *Kidney Int*. 2001;60(1):31–36.
393. Zhang Z, Quinlan J, Grote D, et al. Common variants of the glial cell-derived neurotrophic factor gene do not influence kidney size of the healthy newborn. *Pediatr Nephrol*. 2009;24(6):1151–1157.
394. Zhao XP, Liao MC, Chang SY, et al. Maternal diabetes modulates kidney formation in murine progeny: the role of hedgehog interacting protein (HHIP). *Diabetologia*. 2014;57(9):1986–1996.
395. Taft SA, Nast CC, Desai M, et al. Maternal undernutrition upregulates apoptosis in offspring nephrogenesis. *J Dev Origins Health Dis*. 2012;2(4):226–235.
396. Porteous S, Torban E, Cho NP, et al. Primary renal hypoplasia in humans and mice with PAX2 mutations: evidence of increased apoptosis in fetal kidneys of Pax2(1Neu) +/- mutant mice. *Hum Mol Genet*. 2000;9(1):1–11.

397. Sorenson CM, Rogers SA, Korsmeyer SJ, et al. Fulminant metanephric apoptosis and abnormal kidney development in bcl-2-deficient mice. *Am J Physiol*. 1995;268(1 Pt 2):F73–F81.
398. Torban E, Eccles MR, Favor J, et al. PAX2 suppresses apoptosis in renal collecting duct cells. *Am J Pathol*. 2000;157(3):833–842.
399. Welham SJ, Wade A, Woolf AS. Protein restriction in pregnancy is associated with increased apoptosis of mesenchymal cells at the start of rat metanephrogenesis. *Kidney Int*. 2002;61(4):1231–1242.
400. Dziarmaga A, Clark P, Stayner C, et al. Ureteric bud apoptosis and renal hypoplasia in transgenic PAX2-Bax fetal mice mimics the renal-coloboma syndrome. *J Am Soc Nephrol*. 2003;14(11):2767–2774.
401. Godley LA, Kopp JB, Eckhaus M, et al. Wild-type p53 transgenic mice exhibit altered differentiation of the ureteric bud and possess small kidneys. *Genes Dev*. 1996;10(7):836–850.
402. Pham TD, MacLennan NK, Chiu CT, et al. Uteroplacental insufficiency increases apoptosis and alters p53 gene methylation in the full-term IUGR rat kidney. *Am J Physiol Regul Integr Comp Physiol*. 2003;285(5):R962–R970.
403. Cuffe JS, Briffa JF, Rosser S, et al. Uteroplacental insufficiency in rats induces renal apoptosis and delays nephrogenesis completion. *Acta Physiol (Oxf)*. 2018;222(3). doi:10.1111/apha.12982.
404. Lin CY, Lin TY, Lee MC, et al. Hyperglycemia: GDNF-EGFR pathway target renal epithelial cell migration and apoptosis in diabetic renal embryopathy. *PLoS ONE*. 2013;8(2):e56731.
405. Ojeda NB. Prenatal programming of hypertension: role of sympathetic response to physical stress. *Hypertension*. 2013;61(1):16–17.
406. Wlodk ME, Westcott K, Siebel AL, et al. Growth restriction before or after birth reduces nephron number and increases blood pressure in male rats. *Kidney Int*. 2008;74(2):187–195.
407. Siebel AL, Mibus A, De Blasio MJ, et al. Improved lactational nutrition and postnatal growth ameliorates impairment of glucose tolerance by uteroplacental insufficiency in male rat offspring. *Endocrinology*. 2008;149(6):3067–3076.
408. Gallo LA, Tran M, Moritz KM, et al. Developmental programming: variations in early growth and adult disease. *Clin Exp Pharmacol Physiol*. 2013;40(11):795–802.
409. Ojeda NB, Hennington BS, Williamson DT, et al. Oxidative stress contributes to sex differences in blood pressure in adult growth-restricted offspring. *Hypertension*. 2012;60(1):114–122.
410. Kalisch-Smith JJ, Simmons DG, Dickinson H, et al. Review: sexual dimorphism in the formation, function and adaptation of the placenta. *Placenta*. 2017;54:10–16.
411. Clifton VL, Murphy VE. Maternal asthma as a model for examining fetal sex-specific effects on maternal physiology and placental mechanisms that regulate human fetal growth. *Placenta*. 2004;25(supplA):S45–S52.
412. Stark MJ, Wright IM, Clifton VL. Sex-specific alterations in placental 11 β -hydroxysteroid dehydrogenase 2 activity and early postnatal clinical course following antenatal betamethasone. *Am J Physiol Regul Integr Comp Physiol*. 2009;297(2):R510–R514.
413. Mayne BT, Bianco-Miotto T, Buckberry S, et al. Large scale gene expression Meta-analysis reveals tissue-specific, sex-biased gene expression in humans. *Front Genet*. 2016;7:183.
414. Guilloteau P, Zabielski R, Hammon HM, et al. Adverse effects of nutritional programming during prenatal and early postnatal life, some aspects of regulation and potential prevention and treatments. *J Physiol Pharmacol*. 2009;60(suppl 3):17–35.
415. Li J, Khodus GR, Kruusmaki M, et al. Ouabain protects against adverse developmental programming of the kidney. *Nat Commun*. 2010;1:42.
416. Vasarhelyi B, Dobos M, Reusz GS, et al. Normal kidney function and elevated natriuresis in young men born with low birth weight. *Pediatr Nephrol*. 2000;15(1–2):96–100.
417. Mansano R, Desai M, Garg A, et al. Enhanced nephrogenesis in offspring of water-restricted rat dams. *Am J Obstet Gynecol*. 2007;196(5):480.e1–480.e6.
418. World Health Organization. *Global Prevalence of Vitamin A Deficiency in Populations at Risk 1995-2005: WHO Global Database of Vitamin A Deficiency*. Geneva: World Health Organization; 2009.
419. Makrakis J, Zimanyi MA, Black MJ. Retinoic acid enhances nephron endowment in rats exposed to maternal protein restriction. *Pediatr Nephrol*. 2007;22(11):1861–1867.
420. Averbukh Z, Bogin E, Cohn M, et al. The renotropic factor, a persistent stimulus that crosses the placenta in mice. *J Physiol*. 1988;404:31–38.
421. Okada T, Yamagishi T, Morikawa Y. Morphometry of the kidney in rat pups from uninephrectomized mothers. *Anat Rec*. 1994;240(1):120–124.
422. Zhong J, Perrien DS, Yang HC, et al. Maturational regression of glomeruli determines the nephron population in normal mice. *Pediatr Res*. 2012;72(3):241–248.
423. Boubred F, Daniel L, Buffat C, et al. Early postnatal overfeeding induces early chronic renal dysfunction in adult male rats. *Am J Physiol Renal Physiol*. 2009;297(4):F943–F951.
424. Andersen LG, Angquist L, Eriksson JG, et al. Birth weight, childhood body mass index and risk of coronary heart disease in adults: combined historical cohort studies. *PLoS ONE*. 2010;5(11):e14126.
425. Barker DJ, Osmond C, Forsen TJ, et al. Trajectories of growth among children who have coronary events as adults. *N Engl J Med*. 2005;353(17):1802–1809.
426. Fall CH, Sachdev HS, Osmond C, et al. Adult metabolic syndrome and impaired glucose tolerance are associated with different patterns of BMI gain during infancy: data from the New Delhi birth cohort. *Diabetes Care*. 2008;31(12):2349–2356.
427. Jain V, Singhal A. Catch up growth in low birth weight infants: striking a healthy balance. *Rev Endocr Metab Disord*. 2012;13(2):141–147.
428. Harding JE, Cormack BE, Alexander T, et al. Advances in nutrition of the newborn infant. *Lancet*. 2017;389(10079):1660–1668.
429. Bansal N, Ayoola OO, Gemmell I, et al. Effects of early growth on blood pressure of infants of British European and South Asian origin at one year of age: the Manchester children's growth and vascular health study. *J Hypertens*. 2008;26(3):412–418.
430. Bhargava SK, Sachdev HS, Fall CH, et al. Relation of serial changes in childhood body-mass index to impaired glucose tolerance in young adulthood. *N Engl J Med*. 2004;350(9):865–875.
431. Law CM, Shiell AW, Newsome CA, et al. Fetal, infant, and childhood growth and adult blood pressure: a longitudinal study from birth to 22 years of age. *Circulation*. 2002;105(9):1088–1092.
432. Yang Z, Huffman SL. Nutrition in pregnancy and early childhood and associations with obesity in developing countries. *Matern Child Nutr*. 2013;9(suppl 1):105–119.
433. Adair LS, Fall CH, Osmond C, et al. Associations of linear growth and relative weight gain during early life with adult health and human capital in countries of low and middle income: findings from five birth cohort studies. *Lancet*. 2013;382(9891):525–534.
434. Oster RT, Luyckx VA, Toth EL. Birth weight predicts both proteinuria and overweight/obesity in a rural population of Aboriginal and non-Aboriginal Canadians. *J Dev Origins Health Dis*. 2013;4(2):139–145.
435. Skelton JA, Irby MB, Grzywacz JG, et al. Etiologies of obesity in children: nature and nurture. *Pediatr Clin North Am*. 2011;58(6):1333–1354, ix.
436. Wang Y, Chen X, Klag MJ, et al. Epidemic of childhood obesity: implications for kidney disease. *Adv Chronic Kidney Dis*. 2006;13(4):336–351.
437. Vivante A, Golan E, Tzur D, et al. Body mass index in 1.2 million adolescents and risk for end-stage renal disease. *Arch Intern Med*. 2012;172(21):1644–1650.
438. Laaksonen DE, Lakka HM, Lynch J, et al. Cardiorespiratory fitness and vigorous leisure-time physical activity modify the association of small size at birth with the metabolic syndrome. *Diabetes Care*. 2003;26(7):2156–2164.
439. Siebel AL, Carey AL, Kingwell BA. Can exercise training rescue the adverse cardiometabolic effects of low birth weight and prematurity? *Clin Exp Pharmacol Physiol*. 2012;39(11):944–957.
440. Grijalva-Eternod CS, Lawlor DA, Wells JC. Testing a capacity-load model for hypertension: disentangling early and late growth effects on childhood blood pressure in a prospective birth cohort. *PLoS ONE*. 2013;8(2):e56078.
441. Abitbol CL, Chandar J, Rodriguez MM, et al. Obesity and preterm birth: additive risks in the progression of kidney disease in children. *Pediatr Nephrol*. 2009;24(7):1363–1370.
442. Abitbol CL, Bauer CR, Montane B, et al. Long-term follow-up of extremely low birth weight infants with neonatal renal failure. *Pediatr Nephrol*. 2003;18(9):887–893.
443. Ozanne SE, Hales CN. Lifespan: catch-up growth and obesity in male mice. *Nature*. 2004;427(6973):411–412.
444. Tarry-Adkins JL, Ozanne SE. Mechanisms of early life programming: current knowledge and future directions. *Am J Clin Nutr*. 2011;94(suppl):1765S–1771S.
445. Melk A. Senescence of renal cells: molecular basis and clinical implications. *Nephrol Dial Transplant*. 2003;18(12):2474–2478.

446. Luyckx VA, Compston CA, Simmen T, et al. Accelerated senescence in kidneys of low-birth-weight rats after catch-up growth. *Am J Physiol Renal Physiol*. 2009;297(6):F1697–F1705.
447. Tarry-Adkins JL, Martin-Gronert MS, Chen JH, et al. Maternal diet influences DNA damage, aortic telomere length, oxidative stress, and antioxidant defense capacity in rats. *FASEB J*. 2008;22(6):2037–2044.
448. Akkad A, Hastings R, Konje JC, et al. Telomere length in small-for-gestational-age babies. *BJOG*. 2006;113(3):318–323.
449. Raqib R, Alam DS, Sarker P, et al. Low birth weight is associated with altered immune function in rural Bangladeshi children: a birth cohort study. *Am J Clin Nutr*. 2007;85(3):845–852.
450. Mohn A, Chiavaroli V, Cerruto M, et al. Increased oxidative stress in prepubertal children born small for gestational age. *J Clin Endocrinol Metab*. 2007;92(4):1372–1378.
451. Nagai K, Saito C, Yamagata K. Birth weight and end-stage diabetic nephropathy in later life: a Japanese multicenter study. *Ther Apher Dial*. 2014;18(1):111–112.
452. Dean SV, Lassi ZS, Imam AM, et al. Preconception care: closing the gap in the continuum of care to accelerate improvements in maternal, newborn and child health. *Reprod Health*. 2014;11(suppl 3):S1.
453. Yu Z, Han S, Zhu J, et al. Pre-pregnancy body mass index in relation to infant birth weight and offspring overweight/obesity: a systematic review and meta-analysis. *PLoS ONE*. 2013;8(4):e61627.
454. Dean SV, Lassi ZS, Imam AM, et al. Preconception care: promoting reproductive planning. *Reprod Health*. 2014;11(suppl 3):S2.
455. Ferrero DM, Larson J, Jacobsson B, et al. Cross-country individual participant analysis of 4.1 million singleton births in 5 countries with very high human development index confirms known associations but provides no biologic explanation for 2/3 of all preterm births. *PLoS ONE*. 2016;11(9):e0162506.
456. Lassi ZS, Imam AM, Dean SV, et al. Preconception care: screening and management of chronic disease and promoting psychological health. *Reprod Health*. 2014;11(suppl 3):S5.
457. Black RE, Victora CG, Walker SP, et al. Maternal and child undernutrition and overweight in low-income and middle-income countries. *Lancet*. 2013;382(9890):427–451.
458. Christian P, Lee SE, Donahue Angel M, et al. Risk of childhood undernutrition related to small-for-gestational age and preterm birth in low- and middle-income countries. *Int J Epidemiol*. 2013;42(5):1340–1355.
459. Kozuki N, Katz J, Lee AC, et al. Short maternal stature increases risk of small-for-gestational-age and preterm births in low- and middle-income countries: individual participant data meta-analysis and population attributable fraction. *J Nutr*. 2015;145(11):2542–2550.
460. Goodyer P, Kurpad A, Rekha S, et al. Effects of maternal vitamin A status on kidney development: a pilot study. *Pediatr Nephrol*. 2007;22(2):209–214.
461. Bhutta ZA, Das JK, Bahl R, et al. Can available interventions end preventable deaths in mothers, newborn babies, and stillbirths, and at what cost? *Lancet*. 2014;384(9940):347–370.
462. Caputo C, Wood E, Jabbour L. Impact of fetal alcohol exposure on body systems: a systematic review. *Birth Defects Res C Embryo Today*. 2016;108(2):174–180.
463. Chen LW, Wu Y, Neelakantan N, et al. Maternal caffeine intake during pregnancy is associated with risk of low birth weight: a systematic review and dose-response meta-analysis. *BMC Med*. 2014;12:174.
464. Cnattingius S, Granath F, Petersson G, et al. The influence of gestational age and smoking habits on the risk of subsequent preterm deliveries. *N Engl J Med*. 1999;341(13):943–948.
465. Hogberg L, Cnattingius S, Lundholm C, et al. Effects of maternal smoking during pregnancy on offspring blood pressure in late adolescence. *J Hypertens*. 2012;30(4):693–699.
466. Kooijman MN, Bakker H, Franco OH, et al. Fetal smoke exposure and kidney outcomes in school-aged children. *Am J Kidney Dis*. 2015;66(3):412–420.
467. Nykjaer C, Alwan NA, Greenwood DC, et al. Maternal alcohol intake prior to and during pregnancy and risk of adverse birth outcomes: evidence from a British cohort. *J Epidemiol Community Health*. 2014;68(6):542–549.
468. Sharma D, Shastri S, Farahbakhsh N, et al. Intrauterine growth restriction—part 1. *J Matern Fetal Neonatal Med*. 2016;29(24):3977–3987.
469. Liu J, Zhang S, Liu M, et al. Maternal pre-pregnancy infection with hepatitis B virus and the risk of preterm birth: a population-based cohort study. *Lancet Glob Health*. 2017;5(6):e624–e632.
470. Walker PG, Ter Kuile FO, Garske T, et al. Estimated risk of placental infection and low birthweight attributable to *plasmodium falciparum* malaria in Africa in 2010: a modelling study. *Lancet Glob Health*. 2014;2(8):e460–e467.
471. Jolving LR, Nielsen J, Kesmodel US, et al. Prevalence of maternal chronic diseases during pregnancy—a nationwide population based study from 1989 to 2013. *Acta Obstet Gynecol Scand*. 2016;95(11):1295–1304.
472. Kersten I, Lange AE, Haas JP, et al. Chronic diseases in pregnant women: prevalence and birth outcomes based on the SNIP-study. *BMC Pregnancy Childbirth*. 2014;14:75.
473. Ota E, Ganchimeg T, Morisaki N, et al. Risk factors and adverse perinatal outcomes among term and preterm infants born small-for-gestational-age: secondary analyses of the WHO Multi-Country Survey on Maternal and Newborn Health. *PLoS ONE*. 2014;9(8):e105155.
474. Hladunewich MA, Melamad N, Bramham K. Pregnancy across the spectrum of chronic kidney disease. *Kidney Int*. 2016;89(5):995–1007.
475. Piccoli GB, Minelli F, Versino E, et al. Pregnancy in dialysis patients in the new millennium: a systematic review and meta-regression analysis correlating dialysis schedules and pregnancy outcomes. *Nephrol Dial Transplant*. 2016;31(11):1915–1934.
476. Bilano VL, Ota E, Ganchimeg T, et al. Risk factors of pre-eclampsia/eclampsia and its adverse outcomes in low- and middle-income countries: a WHO secondary analysis. *PLoS ONE*. 2014;9(3):e91198.
477. Fall CH, Sachdev HS, Osmond C, et al. Association between maternal age at childbirth and child and adult outcomes in the offspring: a prospective study in five low-income and middle-income countries (COHORTS collaboration). *Lancet Glob Health*. 2015;3(7):e366–e377.
478. Ganchimeg T, Ota E, Morisaki N, et al. Pregnancy and childbirth outcomes among adolescent mothers: a World Health Organization multicountry study. *BJOG*. 2014;121(suppl 1):40–48.
479. Boivin A, Luo ZC, Audibert F, et al. Risk for preterm and very preterm delivery in women who were born preterm. *Obstet Gynecol*. 2015;125(5):1177–1184.
480. De B, Lin S, Lohsoonthorn V, et al. Risk of preterm delivery in relation to maternal low birth weight. *Acta Obstet Gynecol*. 2007;86:565–571.
481. Wilcox AJ, Skjaerven R, Terje Lie R. Familial patterns of preterm delivery: maternal and fetal contributions. *Am J Epidemiol*. 2007;167(4):474–479.
482. Poston L, Caleyachetty R, Cnattingius S, et al. Preconceptional and maternal obesity: epidemiology and health consequences. *Lancet Diabetes Endocrinol*. 2016;4(12):1025–1036.
483. Goldenberg RL, McClure EM, Harrison MS, et al. Diabetes during pregnancy in low- and middle-income countries. *Am J Perinatol*. 2016;33(13):1227–1235.
484. Luyckx VA, Perico N, Somaschini M, et al. A developmental approach to the prevention of hypertension and kidney disease: a report from the Low Birth Weight and Nephron Number Working Group. *Lancet*. 2017;390(10092):424–428.
485. Innes KE, Byers TE, Marshall JA, et al. Association of a woman's own birth weight with subsequent risk for gestational diabetes. *JAMA*. 2002;287(19):2534–2541.
486. Aiken CE, Ozanne SE. Transgenerational developmental programming. *Hum Reprod Update*. 2014;20(1):63–75.
487. Cheong JN, Wlodek ME, Moritz KM, et al. Programming of maternal and offspring disease: impact of growth restriction, fetal sex and transmission across generations. *J Physiol*. 2016;594(17):4727–4740.
488. Harrison M, Langley-Evans SC. Intergenerational programming of impaired nephrogenesis and hypertension in rats following maternal protein restriction during pregnancy. *Br J Nutr*. 2009;101(7):1020–1030.
489. Mueller TF, Luyckx VA. The natural history of residual renal function in transplant donors. *J Am Soc Nephrol*. 2012;23(9):1462–1466.
490. Muzaale AD, Massie AB, Wang MC, et al. Risk of end-stage renal disease following live kidney donation. *JAMA*. 2014;311(6):579–586.
491. Mjoen G, Hallan S, Hartmann A, et al. Long-term risks for kidney donors. *Kidney Int*. 2014;86(1):162–167.
492. Al-Sehli R, Grebe S, Jacaj Z, et al. What should the serum creatinine be after transplantation? An approach to integrate donor and recipient information to assess posttransplant kidney function. *Transplantation*. 2015;99(9):1960–1967.
493. Hakim RM, Goldsizer RC, Brenner BM. Hypertension and proteinuria: long-term sequelae of uninephrectomy in humans. *Kidney Int*. 1984;25(6):930–936.

494. Ibrahim HN, Foley RN, Reule SA, et al. Renal function profile in white kidney donors: the first 4 decades. *J Am Soc Nephrol*. 2016;27(9):2885–2893.
495. Fehrman-Ekholm I, Kvarnstrom N, Softeland JM, et al. Post-nephrectomy development of renal function in living kidney donors: a cross-sectional retrospective study. *Nephrol Dial Transplant*. 2011;26(7):2377–2381.
496. Locke JE, Reed RD, Massie A, et al. Obesity increases the risk of end-stage renal disease among living kidney donors. *Kidney Int*. 2017;91(3):699–703.
497. Massie AB, Muzaale AD, Luo X, et al. Quantifying postdonation risk of ESRD in living kidney donors. *J Am Soc Nephrol*. 2017;28(9):2749–2755.
498. Rogers NM, Lawton PD, Jose MD. Indigenous Australians and living kidney donation. *N Engl J Med*. 2009;361(15):1513–1516.
499. Gibney EM, Parikh CR, Garg AX. Age, gender, race, and associations with kidney failure following living kidney donation. *Transplant Proc*. 2008;40(5):1337–1340.
500. Lentine KL, Schnitzler MA, Xiao H, et al. Racial variation in medical outcomes among living kidney donors. *N Engl J Med*. 2010;363(8):724–732.
501. Storsley LJ, Young A, Rush DN, et al. Long-term medical outcomes among Aboriginal living kidney donors. *Transplantation*. 2010;90(4):401–406.
502. UNICEF, WHO. *Low Birthweight: Country, Regional and Global Estimates*. New York: United Nations Children's Fund and World Health Organization; 2004.
503. Schachtner T, Reinke P. Estimated nephron number of the remaining donor kidney: impact on living kidney donor outcomes. *Nephrol Dial Transplant*. 2016;31(9):1523–1530.
504. Vikse BE, Irgens LM, Leivestad T, et al. Preeclampsia and the risk of end-stage renal disease. *N Engl J Med*. 2008;359(8):800–809.
505. Garg AX, Nevis IF, McArthur E, et al. Gestational hypertension and preeclampsia in living kidney donors. *N Engl J Med*. 2015;372(2):124–133.
506. Azuma H, Nadeau K, Mackenzie HS, et al. Nephron mass modulates the hemodynamic, cellular and molecular response of the rat renal allograft. *Transplantation*. 1997;63(4):519–528.
507. Heeman UW, Azuma H, Tullius SG, et al. The contribution of reduced functioning mass to chronic kidney allograft dysfunction in rats. *Transplantation*. 1994;58(12):1317–1321.
508. Mackenzie HS, Azuma H, Rennke HG, et al. Renal mass as a determinant of late allograft outcome: insights from experimental studies in rats. *Kidney Int*. 1995;48(suppl 52):S38–S42.
509. Mackenzie HS, Azuma H, Troy JL, et al. Augmenting kidney mass at transplantation abrogates chronic renal allograft injury in rats. *Proc Assoc Am Phys*. 1996;108(2):127–133.
510. Szabo AJ, Muller V, Chen GF, et al. Nephron number determines susceptibility to renal mass reduction-induced CKD in Lewis and Fisher 344 rats: implications for development of experimentally induced chronic allograft nephropathy. *Nephrol Dial Transplant*. 2008;23(8):2492–2495.
511. Brenner BM, Milford EL. Nephron underdosing: a programmed cause of chronic renal allograft failure. *Am J Kidney Dis*. 1993;21(5, suppl 2):66–72.
512. Chertow GM, Brenner BM, Mackenzie HS, et al. Non-immunologic predictors of chronic renal allograft failure: data from the United Network of Organ Sharing. *Kidney Int*. 1995;48(suppl 52):S48–S51.
513. Chertow GM, Brenner BM, Mori M, et al. Antigen-independent determinants of graft survival in living-related kidney transplantation. *Kidney Int*. 1997;52(suppl 63):S84–S86.
514. Chertow GM, Milford EL, Mackenzie HS, et al. Antigen-independent determinants of cadaveric kidney transplant failure. *JAMA*. 1996;276(21):1732–1736.
515. Giral M, Nguyen JM, Karam G, et al. Impact of graft mass on the clinical outcome of kidney transplants. *J Am Soc Nephrol*. 2005;16(1):261–268.
516. Douvorny JB, Baptista-Silva JC, Pestana JO, et al. Importance of renal mass on graft function outcome after 12 months of living donor kidney transplantation. *Nephrol Dial Transplant*. 2007;22(12):3646–3651.
517. el Agroudy AE, Hassan NA, Bakr MA, et al. Effect of donor/recipient body weight mismatch on patient and graft outcome in living-donor kidney transplantation. *Am J Nephrol*. 2003;23(5):294–299.
518. Gaston RS, Hudson SL, Julian BA, et al. Impact of donor/recipient size matching on outcomes in renal transplantation. *Transplantation*. 1996;61(3):383–388.
519. Kasiske BL, Snyder JJ, Gilbertson D. Inadequate donor size in cadaver kidney transplantation. *J Am Soc Nephrol*. 2002;13:2152–2159.
520. Kim YS, Moon JI, Kim DK, et al. Ratio of donor kidney weight to recipient bodyweight as an index of graft function. *Lancet*. 2001;357(9263):1180–1181.
521. Goldberg RJ, Smits G, Wiseman AC. Long-term impact of donor-recipient size mismatching in deceased donor kidney transplantation and in expanded criteria donor recipients. *Transplantation*. 2010;90(8):867–874.
522. Nakatani T, Sugimura K, Kawashima H, et al. The influence of recipient body mass on the outcome of cadaver kidney transplants. *Clin Exp Nephrol*. 2002;6:158–162.
523. Taal MW, Tilney NL, Brenner BM, et al. Renal mass: an important determinant of late allograft outcome. *Transpl Rev*. 1998;12(2):74–84.
524. Kim YS, Kim MS, Han DS, et al. Evidence that the ratio of donor kidney weight to recipient body weight, donor age, and episodes of acute rejection correlate independently with live-donor graft function. *Transplantation*. 2002;72(2):280–283.
525. Giral M, Foucher Y, Karam G, et al. Kidney and recipient weight incompatibility reduces long-term graft survival. *J Am Soc Nephrol*. 2010;21(6):1022–1029.
526. Nicholson ML, Windmill DC, Horsburgh T, et al. Influence of allograft size to recipient body-weight ratio on the long-term outcome of renal transplantation. *Br J Surg*. 2000;87:314–319.
527. Choi JY, Kwon OJ. Is the graft function of living donor renal transplants associated with renal mass matching by computed tomography angiographic volumetry? *Transplant Proc*. 2013;45(8):2919–2924.
528. Luyckx VA, Brenner BM. Low birth weight, nephron number, and kidney disease. *Kidney Int Suppl*. 2005;97:S68–S77.
529. Vazquez MA, Jeyarajah DR, Kielar ML, et al. Long-term outcomes of renal transplantation: a result of the original endowment of the donor kidney and the inflammatory response to both alloantigens and injury. *Curr Opin Nephrol Hypertens*. 2000;9(6):643–648.
530. Luyckx VA, Brenner BM. The clinical importance of nephron mass. *J Am Soc Nephrol*. 2010;21(6):898–910.
531. Franco MC, Kawamoto EM, Gorjao R, et al. Biomarkers of oxidative stress and antioxidant status in children born small for gestational age: evidence of lipid peroxidation. *Pediatr Res*. 2007;62(2):204–208.
532. Jennings BJ, Ozanne SE, Durling MW, et al. Early growth determines longevity in male rats and may be related to telomere shortening in the kidney. *FEBS Lett*. 1999;448(1):4–8.
533. Stewart T, Jung FF, Manning J, et al. Kidney immune cell infiltration and oxidative stress contribute to prenatally programmed hypertension. *Kidney Int*. 2005;68(5):2180–2188.
534. Merlet-Benichou C, Gilbert T, Vilar J, et al. Nephron number: variability is the rule. Causes and consequences. *Lab Invest*. 1999;79(5):515–527.
535. McNamara BJ, Diouf B, Hughson MD, et al. Renal pathology, glomerular number and volume in a West African urban community. *Nephrol Dial Transplant*. 2008;23(8):2576–2585.
536. Hoy WE, Ingelfinger JR, Hallan S, et al. The early development of the kidney and implications for future health. *J Dev Origins Health Dis*. 2010;1:216–233.
537. Almeida JR, Mandarim-de-Lacerda CA. Maternal gestational protein-calorie restriction decreases the number of glomeruli and causes glomerular hypertrophy in adult hypertensive rats. *Am J Obstet Gynecol*. 2005;192(3):945–951.
538. Lucas SR, Costa Silva VL, Miraglia SM, et al. Functional and morphometric evaluation of offspring kidney after intrauterine undernutrition. *Pediatr Nephrol*. 1997;11(6):719–723.
539. Langley-Evans A, Phillips CJ, Jackson AA. In utero exposure to maternal low protein diets induces hypertension in weanling rats, independently of maternal blood pressure changes. *Clin Nutr*. 1994;13:319–324.
540. Nwagwu MO, Cook A, Langley-Evans SC. Evidence of progressive deterioration of renal function in rats exposed to a maternal low-protein diet in utero. *Br J Nutr*. 2000;83(1):79–85.
541. Hoppe CC, Evans RG, Bertram JF, et al. Effects of dietary protein restriction on nephron number in the mouse. *Am J Physiol Regul Integr Comp Physiol*. 2007;292(5):R1768–R1774.
542. Gonzalez-Rodriguez P Jr, Tong W, Xue Q, et al. Fetal hypoxia results in programming of aberrant angiotensin II receptor expression patterns and kidney development. *Int J Med Sci*. 2013;10(5):532–538.

543. Jagadapillai R, Chen J, Canales L, et al. Developmental cigarette smoke exposure: kidney proteome profile alterations in low birth weight pups. *Toxicology*. 2012;299(2-3):80–89.
544. Ortiz LA, Quan A, Zarzar F, et al. Prenatal dexamethasone programs hypertension and renal injury in the rat. *Hypertension*. 2003;41(2):328–334.
545. Tendron A, Decramer S, Justabo E, et al. Cyclosporin A administration during pregnancy induces a permanent nephron deficit in young rabbits. *J Am Soc Nephrol*. 2003;14(12):3188–3196.
546. Fassi A, Sangalli F, Maffi R, et al. Progressive glomerular injury in the MWF rat is predicted by inborn nephron deficit. *J Am Soc Nephrol*. 1998;9(8):1399–1406.
547. Cullen-McEwen LA, Kett MM, Dowling J, et al. Nephron number, renal function, and arterial pressure in aged GDNF heterozygous mice. *Hypertension*. 2003;41(2):335–340.
548. Norwood VF, Morham SG, Smithies O. Postnatal development and progression of renal dysplasia in cyclooxygenase-2 null mice. *Kidney Int*. 2000;58(6):2291–2300.
549. Jackson AA, Dunn RL, Marchand MC, et al. Increased systolic blood pressure in rats induced by a maternal low-protein diet is reversed by dietary supplementation with glycine. *Clin Sci*. 2002;103(6):633–639.
550. McNamara BJ, Diouf B, Hughson MD, et al. Associations between age, body size and nephron number with individual glomerular volumes in urban West African males. *Nephrol Dial Transplant*. 2009;24(5):1500–1506.
551. Kaczmarczyk M, Loniewska B, Kuprijanowicz A, et al. An insertion/deletion ACE polymorphism and kidney size in Polish full-term newborns. *J Renin Angiotensin Aldosterone Syst*. 2013;14(4):369–374.
552. Huxley RR, Shiell AW, Law CM. The role of size at birth and postnatal catch-up growth in determining systolic blood pressure: a systematic review of the literature. *J Hypertens*. 2000;18(7):815–831.
553. Stanner SA, Bulmer K, Andres C, et al. Does malnutrition in utero determine diabetes and coronary heart disease in adulthood? Results from the leningrad siege study, a cross sectional study. *BMJ*. 1997;315(7119):1342–1348.
554. Abalos E, Cuesta C, Carroli G, et al. Pre-eclampsia, eclampsia and adverse maternal and perinatal outcomes: a secondary analysis of the world health organization multicountry survey on maternal and newborn health. *BJOG*. 2014;121(suppl 1):14–24.
555. Abou-Jaoude P, Dubourg L, Bessenay L, et al. What about the renal function during childhood of children born from dialysed mothers? *Nephrol Dial Transplant*. 2012;27(6):2365–2369.
556. Geelhoed JJ, Fraser A, Tilling K, et al. Preeclampsia and gestational hypertension are associated with childhood blood pressure independently of family adiposity measures: the Avon Longitudinal Study of Parents and Children. *Circulation*. 2010;122(12):1192–1199.
557. Luyckx VA, Goodyer P, Bertram JF. Nephron endowment and developmental programming of blood pressure and renal function. In: Taal MW, Chertow GM, Marsden PA, et al, eds. *Brenner and Rector's the Kidney*. 10th ed. Philadelphia: Elsevier; 2015.
558. Taal HR, Geelhoed JJ, Steegers EA, et al. Maternal smoking during pregnancy and kidney volume in the offspring: the Generation R study. *Pediatr Nephrol*. 2011;26(8):1275–1283.
559. Cochat P, Decramer S, Robert-Gnansia E, et al. Renal outcome of children exposed to cyclosporine in utero. *Transplant Proc*. 2004;36(2 suppl):208S–210S.
560. Seckl JR. Prenatal glucocorticoids and long-term programming. *Eur J Endocrinol*. 2004;151(suppl 3):U49–U62.
561. Gubhaju L, Sutherland MR, Yoder BA, et al. Is nephrogenesis affected by preterm birth? Studies in a non-human primate model. *Am J Physiol Renal Physiol*. 2009;297(6):F1668–F1677.
562. Rolnik DL, Wright D, Poon LC, et al. Aspirin versus placebo in pregnancies at high risk for preterm preeclampsia. *N Engl J Med*. 2017;377(17):613–622.
563. Ott JJ, Stevens GA, Groeger J, et al. Global epidemiology of hepatitis B virus infection: new estimates of age-specific HBsAg seroprevalence and endemicity. *Vaccine*. 2012;30(12):2212–2219.
564. Abalos E, Cuesta C, Grosso AL, et al. Global and regional estimates of preeclampsia and eclampsia: a systematic review. *Eur J Obstet Gynecol Reprod Biol*. 2013;170(1):1–7.
565. Guariguata L, Whiting DR, Hambleton I, et al. Global estimates of diabetes prevalence for 2013 and projections for 2035. *Diabetes Res Clin Pract*. 2014;103(2):137–149.
566. Cresswell JA, Campbell OM, De Silva MJ, et al. Effect of maternal obesity on neonatal death in sub-Saharan Africa: multivariable analysis of 27 national datasets. *Lancet*. 2012;380(9850):1325–1330.
567. Hanson M, Barker M, Dodd JM, et al. Interventions to prevent maternal obesity before conception, during pregnancy, and post partum. *Lancet Diabetes Endocrinol*. 2017;5(1):65–76.
568. Levy D, Mohlman MK, Zhang Y. Estimating the potential impact of tobacco control policies on adverse maternal and child health outcomes in the United States using the SimSmoke tobacco control policy simulation model. *Nicotine Tob Res*. 2016;18(5):1240–1249.
569. Caleyachetty R, Tait CA, Kengne AP, et al. Tobacco use in pregnant women: analysis of data from demographic and health surveys from 54 low-income and middle-income countries. *Lancet Glob Health*. 2014;2(9):e513–e520.
570. Qin JB, Sheng XQ, Wu D, et al. Worldwide prevalence of adverse pregnancy outcomes among singleton pregnancies after in vitro fertilization/intracytoplasmic sperm injection: a systematic review and meta-analysis. *Arch Gynecol Obstet*. 2016.
571. von Ehr J, von Versen-Hoyneck F. Implications of maternal conditions and pregnancy course on offspring's medical problems in adult life. *Arch Gynecol Obstet*. 2016;294(4):673–679.
572. Alhabe F, Moore JL, Gibbons L, et al. Adverse maternal and perinatal outcomes in adolescent pregnancies: the Global Network's Maternal Newborn Health Registry Study. *Reprod Health*. 2015;12(suppl 2):S8.
573. Ganchimeg T, Mori R, Ota E, et al. Maternal and perinatal outcomes among nulliparous adolescents in low- and middle-income countries: a multi-country study. *BJOG*. 2013;120(13):1622–1630, discussion 30.
574. Blencowe H, Cousens S, Oestergaard MZ, et al. National, regional, and worldwide estimates of preterm birth rates in the year 2010 with time trends since 1990 for selected countries: a systematic analysis and implications. *Lancet*. 2012;379(9832):2162–2172.
575. World Health Organization. *Joint Child Malnutrition Estimates—Levels and Trends 2017*. <http://apps.who.int/gho/data/node.wrapper.nutrition-2016&showonly=nutrition>. 2017. Accessed July 14, 2017.
576. Jamshidi F, Kelishadi R. A systematic review on the effects of maternal calcium supplementation on offspring's blood pressure. *J Res Med Sci*. 2015;20(10):994–999.
577. Haider BA, Bhutta ZA. Multiple-micronutrient supplementation for women during pregnancy. *Cochrane Database Syst Rev*. 2015;(11):CD004905.
578. Kinra S, Rameshwar Sarma KV, Ghafoorunnissa, et al. Effect of integration of supplemental nutrition with public health programmes in pregnancy and early childhood on cardiovascular risk in rural Indian adolescents: long term follow-up of hyderabad nutrition trial. *BMJ*. 2008;337:a605.
579. Hawkesworth S, Prentice AM, Fulford AJ, et al. Maternal protein-energy supplementation does not affect adolescent blood pressure in the Gambia. *Int J Epidemiol*. 2009;38(1):119–127.
580. Hawkesworth S, Sawo Y, Fulford AJ, et al. Effect of maternal calcium supplementation on offspring blood pressure in 5- to 10-year-old rural Gambian children. *Am J Clin Nutr*. 2010;92(4):741–747.
581. Bergel E, Barros AJD. Effect of maternal calcium intake during pregnancy on children's blood pressure: a systematic review of the literature. *BMC Pediatr*. 2007;7:15.
582. Devakumar D, Fall CH, Sachdev HS, et al. Maternal antenatal multiple micronutrient supplementation for long-term health benefits in children: a systematic review and meta-analysis. *BMC Med*. 2016;14:90.
583. Mispireta ML, Caulfield LE, Zavaleta N, et al. Effect of maternal zinc supplementation on the cardiometabolic profile of Peruvian children: results from a randomized clinical trial. *J Dev Orig Health Dis*. 2017;8(1):56–64.

BOARD REVIEW QUESTIONS

1. A mother brings her 6-month-old daughter for a routine health check. The pregnancy was complicated by preeclampsia. The child was born at 32 weeks of gestation and weighed 1.5 kg (appropriate weight for gestation) but now looks happy and healthy. The mother is worried about long-term consequences for her child. Which of the following is incorrect?
 - a. The child herself is at risk of preeclampsia.
 - b. The child is at increased risk of hypertension and cardiovascular disease.
 - c. The child likely has small kidneys, but her glomerular filtration rate (GFR) is normal.
 - d. The child had an appropriate weight for gestational age and therefore is not at increased risk of kidney disease.
 - e. Perinatal steroid use to accelerate lung maturation is not associated with increased risk of hypertension.

Answer: d

Rationale: Preterm birth is associated with a mother's own risk for preeclampsia and gestational hypertension as well as delivering a preterm infant. Exposure to preeclampsia in utero is also associated with higher blood pressures in children. Such findings illustrate the intergenerational impact of developmental programming. Babies born preterm are at increased risk of high blood pressure, cardiovascular, and kidney disease, the risk of which appear exacerbated if the child becomes overweight or obese. In preterm infants, kidney size is smaller, but GFR is generally well preserved, which may reflect some degree of hyperfiltration in the small kidney, which may predispose to subsequent risk. There is no evidence of hypertension after having received short-term steroids prenatally.

2. Which statements about developmental programming in the kidney are true?
 - a. Environmental stress experienced during development may result in permanent structural changes in major organ systems.
 - b. Programming effects can be transmitted across generations.
 - c. Males and females are always equally affected.
 - d. Attempts to reverse effects of developmental programming (e.g., accelerating catch-up growth) can contribute to additional risk of long-term consequences.
 - e. Only low birth weight and preterm infants should be considered at risk and require follow-up.

Answer: a, b, d

Rationale: Programming occurs as a result of a variety of factors that may occur during gestation and impact fetal development. Offspring of a woman who has experienced the effects of developmental programming are at higher risk of being exposed to hypertension, diabetes, and preeclampsia during gestation and thereby themselves experience developmental programming. The impact on males and females is different in both experimental animals and humans, but both should be considered at risk. Rapid crossing of weight centiles (catch-up growth) is associated with increased risk of higher blood pressure and kidney dysfunction, especially

if the child becomes obese. Not all infants who are at risk of developmental programming are low birth weight (birth weight <2.5 kg) or preterm. Being born small for gestational age, of high birth weight, or being exposed to a diabetic, hypertensive, or preeclamptic pregnancy are also known factors impacting offspring risk of hypertension and kidney disease.

3. It is important to identify individuals at risk of developmental programming of hypertension and kidney disease. Which of the following maternal factors may increase the risk of a preterm birth or a low-birth-weight infant?
 - a. Short maternal stature
 - b. Maternal anemia
 - c. Refugee status
 - d. Maternal obesity
 - e. In vitro fertilization

Answer: All of the above

Rationale: All of the factors listed are recognized risk factors for low birth weight or preterm birth, and therefore, such women need be counseled to optimize nutrition before and during pregnancy, plan pregnancies, and attend antenatal clinics to permit monitoring of fetal growth.

4. A 30-year-old Caucasian male wishes to donate a kidney to his mother. The mother has end-stage kidney disease because of hypertension and kidney damage that was first diagnosed during a preeclamptic pregnancy with the son. The son was born at a birth weight of 2.4 kg at 37 weeks of gestation. The son has a blood pressure around 135/80 measured on three occasions by his family doctor, normal renal function, and no microalbuminuria. He is a non-smoker and has a body mass index (BMI) of 29.8. Which of the following could impact your decision on whether he should donate a kidney?
 - a. Low birth weight
 - b. Birth in a preeclamptic pregnancy
 - c. Ethnicity
 - d. Current BMI
 - e. The mother's 56-year-old brother with mild hypertension is willing to donate, although HLA match is less good.

Answer: a, b, d, e

Rationale: Low birth weight and being born in a preeclamptic pregnancy are known risk factors for later-life hypertension and kidney disease. The man is relatively young, and therefore, his risk of future hypertension and kidney dysfunction is real, especially given his borderline high BMI. His ethnicity is not a risk factor for worse outcomes after living donation. In this case, given the possibility that this young donor may have a reduced nephron endowment, one may be more hesitant to accept the son. In this case, a likely potential donor does exist; however, ideally this fact per se should not impact on whether to decline the young man as a donor or not. He should be counseled to lose weight, and given his borderline blood pressure, this requires follow-up and early treatment should his pressures remain in this range.